

Dipanjan_Das_MSc_project_v3

June 8, 2026

```
[1]: # Package verification
import importlib, subprocess, sys

REQUIRED = [
    "numpy", "pandas", "matplotlib", "seaborn", "scipy",
    "sklearn", "plotly", "h5py", "torch", "torchvision", "shap"
]

missing = [p for p in REQUIRED if importlib.util.find_spec(p) is None]
if missing:
    print(f"Installing missing packages: {missing}")
    subprocess.check_call([sys.executable, "-m", "pip", "install", *missing,
        ↪ "-q"])
else:
    print("All required packages are available.")

# Set non-interactive backend BEFORE importing pyplot to prevent GUI crashes on
↪ Windows
import matplotlib
matplotlib.use('Agg')

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import os, pathlib, random, warnings
warnings.filterwarnings("ignore")
print("Core imports OK.")
```

All required packages are available.

Core imports OK.

```
[2]: # Global random seed for full reproducibility
SEED = 42
random.seed(SEED)
np.random.seed(SEED)
try:
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import torch
torch.manual_seed(SEED)
if torch.cuda.is_available():
    torch.cuda.manual_seed_all(SEED)
DEVICE = torch.device("cuda" if torch.cuda.is_available() else "cpu")
print("PyTorch seed set. Device:", DEVICE)
except ImportError:
    DEVICE = None
    print("PyTorch not available, skipping torch seed.")

# Path constants
BASE = pathlib.Path(r"e:\learnings\University Of Hull\
    ↪AI\assignment-september\MSc\
    ↪project\multimodal_agriculture_prediction-notebook_only")
CROPNET      = BASE / "data" / "CropNet"
USDA_ROOT    = CROPNET / "USDA" / "data" / "Soybean"
SENTINEL_ROOT = CROPNET / "Sentinel" / "data"
HRRR_ROOT    = CROPNET / "HRRR"

# Verify every root exists
paths = {"USDA_ROOT": USDA_ROOT, "SENTINEL_ROOT": SENTINEL_ROOT, "HRRR_ROOT":\
    ↪HRRR_ROOT}
for name, p in paths.items():
    status = "OK" if p.exists() else "MISSING"
    print(f"{name}: {status} ({p})")

```

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PyTorch seed set. Device: cpu
USDA_ROOT: OK (e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
notebook_only\data\CropNet\USDA\data\Soybean)
SENTINEL_ROOT: OK (e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
notebook_only\data\CropNet\Sentinel\data)
HRRR_ROOT: OK (e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-notebook_only\data\CropNet\HRRR)

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[3]: # Build usda_index: full coverage across all available years and counties
#
# USDA data spans 2016-2022 (7 years) and covers all US soybean counties\
    ↪(~500+).
# This is the primary label source for the weather-only Bi-LSTM pre-training\
    ↪phase.
# The full 7-year, multi-county dataset (~3,030 county-year records) is loaded\
    ↪here.

USDA_YEARS = ["2016", "2017", "2018", "2019", "2020", "2021", "2022"]
YIELD_COL  = "YIELD, MEASURED IN BU / ACRE"

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PROD_COL    = "PRODUCTION, MEASURED IN BU"

frames = []
for yr in USDA_YEARS:
    csv_path = USDA_ROOT / yr / f"USDA_Soybean_County_{yr}.csv"
    if not csv_path.exists():
        print(f"WARNING: {csv_path} not found, skipping.")
        continue
    df = pd.read_csv(csv_path)
    if df.empty or YIELD_COL not in df.columns:
        print(f"WARNING: {yr} CSV is empty or missing yield column, skipping.")
        continue
    n_rows = len(df.dropna(subset=[YIELD_COL]))
    print(f"Loaded {yr}: {n_rows} valid county records")
    frames.append(df)

usda_raw = pd.concat(frames, ignore_index=True)

# Drop rows with null FIPS components before constructing FIPS string.
# With large CSVs pandas may infer float64 for integer columns if any nulls
↪exist,
# causing astype(str) to produce "5.0" instead of "5". Explicit int conversion
# via fillna + astype(int) prevents malformed FIPS codes like "5.01.0".
usda_raw = usda_raw.dropna(subset=["state_ansi", "county_ansi"])
usda_raw["fips"] = (
    usda_raw["state_ansi"].astype(float).astype(int).astype(str).str.zfill(2) +
    usda_raw["county_ansi"].astype(float).astype(int).astype(str).str.zfill(3)
)

usda_index = (
    usda_raw[["fips", "year", "state_name", "county_name", YIELD_COL, PROD_COL]]
    .rename(columns={YIELD_COL: "yield_bu_acre", PROD_COL: "production_bu"})
    .dropna(subset=["yield_bu_acre"])
    .reset_index(drop=True)
)
usda_index["year"] = usda_index["year"].astype(str)

# Coverage summary
year_counts = usda_index.groupby("year").size()
fips_years  = usda_index.groupby("fips")["year"].nunique()

print()

print("USDA INDEX SUMMARY")

print(f"Total county-year records : {len(usda_index)}")
print(f"Unique FIPS codes          : {usda_index['fips'].nunique()}")

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print(f"Years covered          : {sorted(usda_index['year'].unique())}")
print(f"States covered         : {usda_index['state_name'].nunique()}")
print()
print("Records per year:")
print(year_counts.to_string())
print()
print(f"Counties with all 7 years : {(fips_years == 7).sum()}")
print(f"Counties with >= 5 years : {(fips_years >= 5).sum()}")
print(f"Counties with >= 3 years : {(fips_years >= 3).sum()}")
print()
# NOTE: usda_index is the full label universe for the weather-only Bi-LSTM
↳branch.
# The multimodal branch will use the intersection with Sentinel coverage only

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Loaded 2016: 541 valid county records
Loaded 2017: 543 valid county records
Loaded 2018: 473 valid county records
Loaded 2019: 429 valid county records
Loaded 2020: 546 valid county records
Loaded 2021: 483 valid county records
Loaded 2022: 520 valid county records

```

USDA INDEX SUMMARY

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Total county-year records : 3535
Unique FIPS codes         : 649
Years covered             : ['2016', '2017', '2018', '2019', '2020', '2021',
'2022']
States covered            : 15

```

Records per year:

```

year
2016    541
2017    543
2018    473
2019    429
2020    546
2021    483
2022    520

```

```

Counties with all 7 years : 260
Counties with >= 5 years  : 493
Counties with >= 3 years  : 570

```

```

[4]: # Build sentinel_index by scanning all HDF5 files in the Sentinel tree
import h5py

```



```

sentinel_records = []
SENTINEL_AG_ROOT = SENTINEL_ROOT / "AG"
SENTINEL_NDVI_ROOT = SENTINEL_ROOT / "NDVI"

for year_dir in sorted(SENTINEL_AG_ROOT.iterdir()):
    if not year_dir.is_dir():
        continue
    year = year_dir.name
    for state_dir in sorted(year_dir.iterdir()):
        if not state_dir.is_dir():
            continue
        state = state_dir.name
        for h5_file in sorted(state_dir.glob("*.h5")):
            # Parse quarter dates from filename:
            ↪ Agriculture_NN_ST_YYYY-MM-DD_YYYY-MM-DD.h5
            parts = h5_file.stem.split("_")
            q_start = parts[3] if len(parts) > 4 else "unknown"
            q_end = parts[4] if len(parts) > 4 else "unknown"

            # Open file to get FIPS codes, dates, and tile counts
            try:
                with h5py.File(h5_file, "r") as f:
                    for fips_key in f.keys():
                        date_keys = list(f[fips_key].keys())
                        n_dates = len(date_keys)
                        # Sample tile count from first available date
                        n_tiles = 0
                        if n_dates > 0:
                            first_date = date_keys[0]
                            if "data" in f[fips_key][first_date]:
                                n_tiles = f[fips_key][first_date]["data"].
                                ↪ shape[0]

                        sentinel_records.append({
                            "fips" : fips_key,
                            "year" : year,
                            "state" : state,
                            "q_start" : q_start,
                            "q_end" : q_end,
                            "h5_path" : str(h5_file),
                            "n_dates" : n_dates,
                            "n_tiles" : n_tiles,
                        })
            except Exception as e:
                print(f"WARNING: Could not open {h5_file.name}: {e}")

sentinel_index = pd.DataFrame(sentinel_records)

```

```

print("sentinel_index shape:", sentinel_index.shape)
print("Years:", sorted(sentinel_index["year"].unique()))
print("States:", sorted(sentinel_index["state"].unique()))
print("Unique FIPS codes:", sentinel_index["fips"].nunique())
print("Total tiles across all files (approx):", sentinel_index["n_tiles"].sum())
print()
print(sentinel_index.groupby(["year", "state"]).agg(
    files=("h5_path", "count"),
    fips_codes=("fips", "nunique"),
    total_dates=("n_dates", "sum"),
    total_tiles=("n_tiles", "sum")
).to_string())

```

```

sentinel_index shape: (116, 8)
Years: ['2021', '2022']
States: ['AL', 'AR', 'AZ', 'CA', 'CO', 'DE', 'FL', 'GA', 'IA', 'ID', 'IL', 'IN',
'KS', 'KY', 'LA']
Unique FIPS codes: 15
Total tiles across all files (approx): 3884

```

		files	fips_codes	total_dates	total_tiles
year	state				
2021	AL	4	1	24	60
	AR	4	1	24	148
	CA	4	1	24	120
	CO	4	1	24	120
	DE	4	1	24	80
	FL	4	1	24	116
	GA	4	1	24	76
	IA	4	1	24	48
	ID	4	1	24	132
	IL	4	1	24	96
	IN	4	1	24	24
	KS	4	1	24	32
	KY	4	1	24	52
	LA	4	1	24	76
2022	AL	4	1	24	60
	AR	4	1	24	148
	AZ	4	1	19	1524
	CA	4	1	24	120
	CO	4	1	24	120
	DE	4	1	24	80
	FL	4	1	24	116
	GA	4	1	24	76
	IA	4	1	24	48
	ID	4	1	24	132
	IL	4	1	24	96
	IN	4	1	24	24

KS	4	1	24	32
KY	4	1	24	52
LA	4	1	24	76

```
[5]: # Identify which Sentinel counties have matching USDA yield labels
#
# This cell programmatically looks up each Sentinel FIPS code in usda_index.
# No values are hardcoded. Whatever is found on disk is used; the rest is left
↳out.

sentinel_fips_list = sorted(sentinel_index["fips"].unique())
print("Sentinel FIPS codes (from imagery):", sentinel_fips_list)
print()

# Programmatic join: keep only Sentinel FIPS codes that exist in usda_index
sentinel_labels = (
    usda_index[usda_index["fips"].isin(sentinel_fips_list)]
    .copy()
    .reset_index(drop=True)
)

print("SENTINEL-USDA LABEL INTERSECTION")

print(f"Sentinel FIPS codes queried : {len(sentinel_fips_list)}")
print(f"Matching USDA records found : {len(sentinel_labels)}")
print()

if len(sentinel_labels) == 0:
    print("No USDA yield records found for any Sentinel FIPS code.")
else:
    print("Matching records:")
    print(sentinel_labels[["fips", "year", "state_name", "county_name",
                           "yield_bu_acre"]].to_string(index=False))
```

```
Sentinel FIPS codes (from imagery): ['01001', '04001', '05001', '06001',
'08001', '10001', '12001', '13001', '16001', '17001', '18001', '19001', '20001',
'21001', '22001']
```

```
SENTINEL-USDA LABEL INTERSECTION
Sentinel FIPS codes queried : 15
Matching USDA records found : 105
```

```
Matching records:
  fips year state_name county_name  yield_bu_acre
13001 2016    GEORGIA    APPLING         31.4
08001 2016   COLORADO    ADAMS          32.8
21001 2016   KENTUCKY    ADAIR          43.9
```

18001	2016	INDIANA	ADAMS	59.3
19001	2016	IOWA	ADAIR	60.8
10001	2016	DELAWARE	KENT	44.5
04001	2016	ARIZONA	APACHE	30.1
05001	2016	ARKANSAS	ARKANSAS	48.4
06001	2016	CALIFORNIA	ALAMEDA	32.7
16001	2016	IDAHO	ADA	38.7
12001	2016	FLORIDA	ALACHUA	29.1
22001	2016	LOUISIANA	ACADIA	37.6
20001	2016	KANSAS	ANDERSON	45.4
17001	2016	ILLINOIS	ADAMS	55.5
01001	2016	ALABAMA	AUTAGA	61.8
13001	2017	GEORGIA	APPLING	32.2
08001	2017	COLORADO	ADAMS	35.1
21001	2017	KENTUCKY	ADAIR	44.5
18001	2017	INDIANA	ADAMS	54.1
19001	2017	IOWA	ADAIR	57.7
10001	2017	DELAWARE	KENT	43.2
04001	2017	ARIZONA	APACHE	29.8
05001	2017	ARKANSAS	ARKANSAS	45.8
06001	2017	CALIFORNIA	ALAMEDA	29.4
16001	2017	IDAHO	ADA	34.4
12001	2017	FLORIDA	ALACHUA	26.9
22001	2017	LOUISIANA	ACADIA	36.9
20001	2017	KANSAS	ANDERSON	45.0
17001	2017	ILLINOIS	ADAMS	56.1
01001	2017	ALABAMA	AUTAGA	48.0
13001	2018	GEORGIA	APPLING	27.3
08001	2018	COLORADO	ADAMS	29.9
21001	2018	KENTUCKY	ADAIR	38.2
18001	2018	INDIANA	ADAMS	52.9
19001	2018	IOWA	ADAIR	52.0
10001	2018	DELAWARE	KENT	41.4
04001	2018	ARIZONA	APACHE	25.3
05001	2018	ARKANSAS	ARKANSAS	43.5
06001	2018	CALIFORNIA	ALAMEDA	28.6
16001	2018	IDAHO	ADA	31.6
12001	2018	FLORIDA	ALACHUA	27.7
22001	2018	LOUISIANA	ACADIA	37.7
20001	2018	KANSAS	ANDERSON	43.3
17001	2018	ILLINOIS	ADAMS	51.0
01001	2018	ALABAMA	AUTAGA	65.4
13001	2019	GEORGIA	APPLING	31.7
08001	2019	COLORADO	ADAMS	27.0
21001	2019	KENTUCKY	ADAIR	36.0
18001	2019	INDIANA	ADAMS	47.6
19001	2019	IOWA	ADAIR	55.0
10001	2019	DELAWARE	KENT	45.0

04001	2019	ARIZONA	APACHE	23.9
05001	2019	ARKANSAS	ARKANSAS	38.1
06001	2019	CALIFORNIA	ALAMEDA	28.7
16001	2019	IDAHO	ADA	34.0
12001	2019	FLORIDA	ALACHUA	28.1
22001	2019	LOUISIANA	ACADIA	39.2
20001	2019	KANSAS	ANDERSON	43.1
17001	2019	ILLINOIS	ADAMS	48.6
01001	2019	ALABAMA	AUTAGA	39.3
13001	2020	GEORGIA	APPLING	34.2
08001	2020	COLORADO	ADAMS	33.5
21001	2020	KENTUCKY	ADAIR	42.5
18001	2020	INDIANA	ADAMS	55.3
19001	2020	IOWA	ADAIR	54.1
10001	2020	DELAWARE	KENT	44.2
04001	2020	ARIZONA	APACHE	28.8
05001	2020	ARKANSAS	ARKANSAS	45.8
06001	2020	CALIFORNIA	ALAMEDA	30.5
16001	2020	IDAHO	ADA	35.5
12001	2020	FLORIDA	ALACHUA	29.4
22001	2020	LOUISIANA	ACADIA	37.3
20001	2020	KANSAS	ANDERSON	46.5
17001	2020	ILLINOIS	ADAMS	52.8
01001	2020	ALABAMA	AUTAGA	47.2
13001	2021	GEORGIA	APPLING	33.6
08001	2021	COLORADO	ADAMS	31.0
21001	2021	KENTUCKY	ADAIR	44.2
18001	2021	INDIANA	ADAMS	58.3
19001	2021	IOWA	ADAIR	58.5
10001	2021	DELAWARE	KENT	47.6
04001	2021	ARIZONA	APACHE	28.4
05001	2021	ARKANSAS	ARKANSAS	47.2
06001	2021	CALIFORNIA	ALAMEDA	32.6
16001	2021	IDAHO	ADA	37.5
12001	2021	FLORIDA	ALACHUA	27.9
22001	2021	LOUISIANA	ACADIA	34.8
20001	2021	KANSAS	ANDERSON	46.2
17001	2021	ILLINOIS	ADAMS	57.5
01001	2021	ALABAMA	AUTAGA	54.3
13001	2022	GEORGIA	APPLING	31.6
08001	2022	COLORADO	ADAMS	32.6
21001	2022	KENTUCKY	ADAIR	40.6
18001	2022	INDIANA	ADAMS	54.5
19001	2022	IOWA	ADAIR	57.3
10001	2022	DELAWARE	KENT	41.6
04001	2022	ARIZONA	APACHE	26.6
05001	2022	ARKANSAS	ARKANSAS	45.2
06001	2022	CALIFORNIA	ALAMEDA	30.9

16001	2022	IDAHO	ADA	38.0
12001	2022	FLORIDA	ALACHUA	27.0
22001	2022	LOUISIANA	ACADIA	36.5
20001	2022	KANSAS	ANDERSON	43.4
17001	2022	ILLINOIS	ADAMS	53.2
01001	2022	ALABAMA	AUTAGA	47.1

```
[6]: # Define training universe splits for each model phase
#
# Phase 1 - Weather encoder (Bi-LSTM)
# Source : usda_index - all USDA soybean records 2016-2022 (3,430 records,
#         ↪ 634 counties)
# Split : 80/20 stratified by 5-bin yield quintile
#
# Phase 2 - Satellite encoder (ResNet18 + LSTM)
# Phase 3 - GMU fine-tuning (gated dual-stream)
# Source : sentinel_labels - USDA records that match Sentinel FIPS codes
# Split : 80/20 stratified by 2-bin yield median (constrained by small n)
# Note : If sentinel_labels is empty (no USDA-Sentinel FIPS overlap on
#         ↪ disk),
#         Phase 2 and 3 are skipped and documented as a data gap.

from sklearn.model_selection import train_test_split

# Phase 1: Weather branch
weather_df = usda_index.copy().reset_index(drop=True)
weather_df["yield_quintile"] = pd.qcut(
    weather_df["yield_bu_acre"], q=5, labels=False, duplicates="drop"
)
w_train_idx, w_test_idx = train_test_split(
    weather_df.index,
    test_size=0.20,
    random_state=SEED,
    stratify=weather_df["yield_quintile"]
)
weather_df["split"] = "test"
weather_df.loc[w_train_idx, "split"] = "train"
weather_train = weather_df[weather_df["split"] == "train"].
    ↪ drop(columns=["yield_quintile", "split"]).reset_index(drop=True)
weather_test = weather_df[weather_df["split"] == "test"].
    ↪ drop(columns=["yield_quintile", "split"]).reset_index(drop=True)

print("PHASE 1 - WEATHER BRANCH (full USDA x HRRR, 2016-2022)")

print(f"Total records          : {len(weather_df)}")
print(f"Train (80%)               : {len(weather_train)}")
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print(f"Test (20%) : {len(weather_test)}")
print(f"Unique counties - train : {weather_train['fips'].nunique()}")
print(f"Years in train : {sorted(weather_train['year'].unique())}")
print(f"Yield range - train : {weather_train['yield_bu_acre'].min():.1f} to_
↳ {weather_train['yield_bu_acre'].max():.1f} BU/ACRE")
print(f"Yield range - test : {weather_test['yield_bu_acre'].min():.1f} to_
↳ {weather_test['yield_bu_acre'].max():.1f} BU/ACRE")
print()

# Phase 2 & 3: Satellite + GMU
# Attach Sentinel tile counts to whatever sentinel_labels contains
sentinel_summary = (
    sentinel_index.groupby(["fips", "year"])
    .agg(total_dates=("n_dates", "sum"), total_tiles=("n_tiles", "sum"))
    .reset_index()
)

print("PHASE 2 & 3 - SATELLITE / GMU")

if len(sentinel_labels) == 0:
    # No USDA yield labels match the Sentinel FIPS codes on disk.
    multimodal_index = pd.DataFrame(columns=["fips", "year", "state_name",
                                             "county_name", "yield_bu_acre",
                                             "production_bu", "total_dates",
                                             "total_tiles", "split"])

    train_set = multimodal_index.copy()
    test_set = multimodal_index.copy()
    print("Sentinel-USDA label intersection is empty.")
    print("codes that appear in the Sentinel imagery:")
    print(sorted(sentinel_index["fips"].unique()))
else:
    mm_df = sentinel_labels.copy().reset_index(drop=True)
    # Normalise types so fips/year match usda_index (sentinel_index keys are_
    ↳ strings)
    sentinel_summary["fips"] = sentinel_summary["fips"].astype(str).str.zfill(5)
    sentinel_summary["year"] = sentinel_summary["year"].astype(int)
    mm_df["fips"] = mm_df["fips"].astype(str).str.zfill(5)
    mm_df["year"] = mm_df["year"].astype(int)
    mm_df = mm_df.merge(sentinel_summary, on=["fips", "year"], how="left")

    n_mm = len(mm_df)
    # Use 2-bin stratification when test set is too small for more bins
    mm_df["yield_bin"] = (mm_df["yield_bu_acre"] >= mm_df["yield_bu_acre"].
    ↳ median()).astype(int)

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if n_mm < 5:
    # Too few records to split - use all as train, none as test
    mm_df["split"] = "train"
    print(f"WARNING: Only {n_mm} records in sentinel_labels. Too few for an
↪80/20 split.")
    print("All records assigned to train. Test set is empty.")
else:
    mm_train_idx, mm_test_idx = train_test_split(
        mm_df.index,
        test_size=0.20,
        random_state=SEED,
        stratify=mm_df["yield_bin"]
    )
    mm_df["split"] = "test"
    mm_df.loc[mm_train_idx, "split"] = "train"

    multimodal_index = mm_df.drop(columns=["yield_bin"]).reset_index(drop=True)
    train_set = multimodal_index[multimodal_index["split"] == "train"].
↪reset_index(drop=True)
    test_set = multimodal_index[multimodal_index["split"] == "test"].
↪reset_index(drop=True)

    print(f"Total records          : {len(multimodal_index)}")
    print(f"Train (80%)              : {len(train_set)}")
    print(f"Test (20%)                  : {len(test_set)}")
    print(f"Approx train tiles          : {int(train_set['total_tiles'].sum())}")
    print(f"Approx test tiles           : {int(test_set['total_tiles'].sum())}")
    print()
    print("Train set:")
    print(train_set[["fips", "state_name", "county_name", "year",
                    "yield_bu_acre", "total_tiles"]].to_string(index=False))

    print()
    print("Test set:")
    print(test_set[["fips", "state_name", "county_name", "year",
                   "yield_bu_acre", "total_tiles"]].to_string(index=False))

print()
print("COVERAGE SUMMARY")
print(f"Weather branch : {len(weather_train)} train records, "
      f"{weather_train['fips'].nunique()} counties, "
      f"{weather_train['state_name'].nunique()} states, "
      f"years 2016-2022")
if len(train_set) > 0:
    print(f"Satellite branch: {len(train_set)} train county-years, "
          f"~{int(train_set['total_tiles'].sum())} tiles, "
          f"years {sorted(train_set['year'].unique())}")

```



```

else:

    print("Satellite branch: 0 records - data gap (no USDA-Sentinel FIPS_
    ↳overlap on disk)")
print(f"Split seed: {SEED}, method: 80/20 stratified random")

```

PHASE 1 - WEATHER BRANCH (full USDA x HRRR, 2016-2022)

```

Total records      : 3535
Train (80%)        : 2828
Test (20%)         : 707
Unique counties - train : 638
Years in train      : ['2016', '2017', '2018', '2019', '2020', '2021',
'2022']
Yield range - train   : 9.3 to 80.4 BU/ACRE
Yield range - test    : 9.4 to 75.2 BU/ACRE

```

PHASE 2 & 3 - SATELLITE / GMU

```

Total records      : 105
Train (80%)        : 84
Test (20%)         : 21
Approx train tiles  : 3420
Approx test tiles   : 464

```

Train set:

fips	state_name	county_name	year	yield_bu_acre	total_tiles
13001	GEORGIA	APPLING	2016	31.4	NaN
08001	COLORADO	ADAMS	2016	32.8	NaN
21001	KENTUCKY	ADAIR	2016	43.9	NaN
19001	IOWA	ADAIR	2016	60.8	NaN
10001	DELAWARE	KENT	2016	44.5	NaN
04001	ARIZONA	APACHE	2016	30.1	NaN
05001	ARKANSAS	ARKANSAS	2016	48.4	NaN
06001	CALIFORNIA	ALAMEDA	2016	32.7	NaN
16001	IDAHO	ADA	2016	38.7	NaN
12001	FLORIDA	ALACHUA	2016	29.1	NaN
22001	LOUISIANA	ACADIA	2016	37.6	NaN
20001	KANSAS	ANDERSON	2016	45.4	NaN
08001	COLORADO	ADAMS	2017	35.1	NaN
21001	KENTUCKY	ADAIR	2017	44.5	NaN
18001	INDIANA	ADAMS	2017	54.1	NaN
19001	IOWA	ADAIR	2017	57.7	NaN
10001	DELAWARE	KENT	2017	43.2	NaN
04001	ARIZONA	APACHE	2017	29.8	NaN
05001	ARKANSAS	ARKANSAS	2017	45.8	NaN
16001	IDAHO	ADA	2017	34.4	NaN
12001	FLORIDA	ALACHUA	2017	26.9	NaN
22001	LOUISIANA	ACADIA	2017	36.9	NaN
17001	ILLINOIS	ADAMS	2017	56.1	NaN

01001	ALABAMA	AUTAGA	2017	48.0	NaN
08001	COLORADO	ADAMS	2018	29.9	NaN
21001	KENTUCKY	ADAIR	2018	38.2	NaN
19001	IOWA	ADAIR	2018	52.0	NaN
10001	DELAWARE	KENT	2018	41.4	NaN
04001	ARIZONA	APACHE	2018	25.3	NaN
16001	IDAHO	ADA	2018	31.6	NaN
22001	LOUISIANA	ACADIA	2018	37.7	NaN
20001	KANSAS	ANDERSON	2018	43.3	NaN
17001	ILLINOIS	ADAMS	2018	51.0	NaN
01001	ALABAMA	AUTAGA	2018	65.4	NaN
08001	COLORADO	ADAMS	2019	27.0	NaN
21001	KENTUCKY	ADAIR	2019	36.0	NaN
18001	INDIANA	ADAMS	2019	47.6	NaN
19001	IOWA	ADAIR	2019	55.0	NaN
10001	DELAWARE	KENT	2019	45.0	NaN
04001	ARIZONA	APACHE	2019	23.9	NaN
05001	ARKANSAS	ARKANSAS	2019	38.1	NaN
06001	CALIFORNIA	ALAMEDA	2019	28.7	NaN
12001	FLORIDA	ALACHUA	2019	28.1	NaN
22001	LOUISIANA	ACADIA	2019	39.2	NaN
20001	KANSAS	ANDERSON	2019	43.1	NaN
01001	ALABAMA	AUTAGA	2019	39.3	NaN
13001	GEORGIA	APPLING	2020	34.2	NaN
08001	COLORADO	ADAMS	2020	33.5	NaN
21001	KENTUCKY	ADAIR	2020	42.5	NaN
18001	INDIANA	ADAMS	2020	55.3	NaN
19001	IOWA	ADAIR	2020	54.1	NaN
10001	DELAWARE	KENT	2020	44.2	NaN
04001	ARIZONA	APACHE	2020	28.8	NaN
05001	ARKANSAS	ARKANSAS	2020	45.8	NaN
06001	CALIFORNIA	ALAMEDA	2020	30.5	NaN
16001	IDAHO	ADA	2020	35.5	NaN
12001	FLORIDA	ALACHUA	2020	29.4	NaN
22001	LOUISIANA	ACADIA	2020	37.3	NaN
17001	ILLINOIS	ADAMS	2020	52.8	NaN
01001	ALABAMA	AUTAGA	2020	47.2	NaN
08001	COLORADO	ADAMS	2021	31.0	120.0
21001	KENTUCKY	ADAIR	2021	44.2	52.0
18001	INDIANA	ADAMS	2021	58.3	24.0
19001	IOWA	ADAIR	2021	58.5	48.0
10001	DELAWARE	KENT	2021	47.6	80.0
04001	ARIZONA	APACHE	2021	28.4	NaN
05001	ARKANSAS	ARKANSAS	2021	47.2	148.0
06001	CALIFORNIA	ALAMEDA	2021	32.6	120.0
12001	FLORIDA	ALACHUA	2021	27.9	116.0
22001	LOUISIANA	ACADIA	2021	34.8	76.0
20001	KANSAS	ANDERSON	2021	46.2	32.0

01001	ALABAMA	AUTAGA	2021	54.3	60.0
13001	GEORGIA	APPLING	2022	31.6	76.0
08001	COLORADO	ADAMS	2022	32.6	120.0
21001	KENTUCKY	ADAIR	2022	40.6	52.0
19001	IOWA	ADAIR	2022	57.3	48.0
10001	DELAWARE	KENT	2022	41.6	80.0
04001	ARIZONA	APACHE	2022	26.6	1524.0
05001	ARKANSAS	ARKANSAS	2022	45.2	148.0
06001	CALIFORNIA	ALAMEDA	2022	30.9	120.0
16001	IDAHO	ADA	2022	38.0	132.0
12001	FLORIDA	ALACHUA	2022	27.0	116.0
20001	KANSAS	ANDERSON	2022	43.4	32.0
17001	ILLINOIS	ADAMS	2022	53.2	96.0

Test set:

fips	state_name	county_name	year	yield_bu_acre	total_tiles
18001	INDIANA	ADAMS	2016	59.3	NaN
17001	ILLINOIS	ADAMS	2016	55.5	NaN
01001	ALABAMA	AUTAGA	2016	61.8	NaN
13001	GEORGIA	APPLING	2017	32.2	NaN
06001	CALIFORNIA	ALAMEDA	2017	29.4	NaN
20001	KANSAS	ANDERSON	2017	45.0	NaN
13001	GEORGIA	APPLING	2018	27.3	NaN
18001	INDIANA	ADAMS	2018	52.9	NaN
05001	ARKANSAS	ARKANSAS	2018	43.5	NaN
06001	CALIFORNIA	ALAMEDA	2018	28.6	NaN
12001	FLORIDA	ALACHUA	2018	27.7	NaN
13001	GEORGIA	APPLING	2019	31.7	NaN
16001	IDAHO	ADA	2019	34.0	NaN
17001	ILLINOIS	ADAMS	2019	48.6	NaN
20001	KANSAS	ANDERSON	2020	46.5	NaN
13001	GEORGIA	APPLING	2021	33.6	76.0
16001	IDAHO	ADA	2021	37.5	132.0
17001	ILLINOIS	ADAMS	2021	57.5	96.0
18001	INDIANA	ADAMS	2022	54.5	24.0
22001	LOUISIANA	ACADIA	2022	36.5	76.0
01001	ALABAMA	AUTAGA	2022	47.1	60.0

COVERAGE SUMMARY

Weather branch : 2828 train records, 638 counties, 15 states, years 2016-2022
 Satellite branch: 84 train county-years, ~3420 tiles, years [np.int64(2016), np.int64(2017), np.int64(2018), np.int64(2019), np.int64(2020), np.int64(2021), np.int64(2022)]
 Split seed: 42, method: 80/20 stratified random

1 USDA Soybean EDA (Target Variable)

It fully characterises the regression target - county-level soybean yield in BU/ACRE. All analysis uses the usda_index and multimodal_index tables built previously. - Full county inventory, data quality check, FIPS resolution - Target distribution: histogram, boxplot, normality test - Year-over-year yield change per county - Geographic choropleth map with Sentinel coverage overlay - Freeze and export final label tables

```
[7]: # Full county inventory, quality check, FIPS resolution
print("USDA INDEX - FULL COVERAGE SUMMARY")
print(f"Total records (all years, all counties): {len(usda_index)}")
print(f"Unique FIPS codes: {usda_index['fips'].
    ↪nunique()}")
print(f"Years with data: {sorted(usda_index['year'].
    ↪astype(str).unique())}")
print(f"States with data: {sorted(usda_index['state_name'].unique())}")
print()

# Completeness check: every FIPS must have both years
pivot = usda_index.pivot_table(index="fips", columns="year",
    ↪values="yield_bu_acre", aggfunc="first")
print("County x Year yield matrix (BU/ACRE):")
print(pivot.to_string())
print()

missing_either_year = pivot[pivot.isnull().any(axis=1)]
if len(missing_either_year) > 0:
    print("WARNING: Counties missing yield for at least one year:")
    print(missing_either_year.to_string())
else:
    print("All counties with USDA data have at least one year of yield records.
    ↪")
print()

# FIPS resolution and intersection report
usda_fips = set(usda_index["fips"].unique())
sentinel_fips = set(sentinel_index["fips"].unique())
overlap_fips = usda_fips & sentinel_fips

print("INTERSECTION REPORT")
print(f"USDA counties: {len(usda_fips)}")
print(f"Sentinel counties: {len(sentinel_fips)}")
print(f"Multimodal intersection: {len(overlap_fips)}")
print(f"Sentinel-only (no USDA): {len(sentinel_fips - usda_fips)}")
print(f" -> {sorted(sentinel_fips - usda_fips)}")
```

```

print()

# Show the full multimodal_index with all metadata
print("Full multimodal_index:")
print(multimodal_index[["fips","state_name","county_name","year","yield_bu_acre",
                        "production_bu","total_tiles","split"]].
      ↪to_string(index=False))

```

USDA INDEX - FULL COVERAGE SUMMARY

Total records (all years, all counties): 3535

Unique FIPS codes: 649

Years with data: ['2016', '2017', '2018', '2019',
'2020', '2021', '2022']

States with data: ['ALABAMA', 'ARIZONA', 'ARKANSAS',
'CALIFORNIA', 'COLORADO', 'DELAWARE', 'FLORIDA', 'GEORGIA', 'IDAHO', 'ILLINOIS',
'INDIANA', 'IOWA', 'KANSAS', 'KENTUCKY', 'LOUISIANA']

County x Year yield matrix (BU/ACRE):

year	2016	2017	2018	2019	2020	2021	2022
fips							
01001	61.8	48.0	65.4	39.3	47.2	54.3	47.1
01003	41.8	42.0	45.4	NaN	37.2	44.3	37.1
01009	25.0	48.6	39.3	35.3	47.0	42.5	37.4
01015	NaN	NaN	NaN	NaN	35.1	44.7	45.5
01019	20.8	42.1	45.3	30.1	36.3	45.1	39.2
01031	43.6	NaN	NaN	NaN	NaN	31.0	NaN
01033	31.2	48.2	36.1	38.7	42.2	NaN	35.6
01035	33.8	NaN	NaN	NaN	NaN	NaN	NaN
01039	32.6	NaN	NaN	NaN	35.1	NaN	NaN
01043	30.3	55.3	44.6	42.8	51.1	50.1	47.4
01047	32.4	43.2	42.0	26.1	42.7	58.6	41.3
01049	29.2	46.1	49.9	30.6	39.9	44.9	48.8
01051	20.3	39.9	42.8	NaN	NaN	50.4	NaN
01053	42.4	46.2	NaN	22.3	NaN	50.7	NaN
01055	25.6	52.7	NaN	30.4	41.9	34.7	46.4
01057	28.7	25.2	26.4	24.9	38.8	40.6	42.0
01059	39.3	41.9	37.5	37.7	38.8	NaN	NaN
01061	33.8	48.4	NaN	NaN	NaN	NaN	NaN
01065	23.0	NaN	NaN	NaN	NaN	NaN	NaN
01069	23.3	47.2	28.4	46.2	48.7	NaN	42.6
01071	21.5	40.3	39.0	29.1	40.1	42.2	45.5
01075	26.2	27.9	NaN	NaN	NaN	NaN	NaN
01077	29.9	46.3	35.4	38.6	39.6	47.2	42.5
01079	33.0	54.3	39.4	39.3	43.4	51.5	42.9
01083	31.8	50.3	43.3	40.5	36.9	45.4	37.1
01085	29.3	NaN	NaN	NaN	NaN	NaN	NaN
01087	NaN	32.1	NaN	NaN	44.3	30.8	42.5
01089	37.6	42.4	38.2	39.5	42.1	49.6	47.1

01091	30.2	31.4	NaN	NaN	30.0	NaN	NaN
01093	NaN	NaN	NaN	NaN	48.0	NaN	46.2
01095	NaN	43.2	36.1	31.0	42.4	43.8	43.1
01099	38.3	NaN	NaN	NaN	NaN	49.1	NaN
01103	35.3	47.0	NaN	39.5	47.4	44.2	33.4
01105	36.5	39.9	37.6	NaN	37.4	39.1	39.3
01109	40.1	NaN	NaN	51.6	NaN	NaN	33.8
01117	39.7	NaN	NaN	NaN	NaN	NaN	NaN
01119	30.7	40.7	43.1	38.0	NaN	45.3	NaN
01121	NaN	41.9	38.2	NaN	46.7	NaN	46.3
01125	38.5	51.3	46.3	31.8	41.2	53.0	39.2
01127	NaN	NaN	38.5	32.4	NaN	NaN	NaN
04001	30.1	29.8	25.3	23.9	28.8	28.4	26.6
05001	48.4	45.8	43.5	38.1	45.8	47.2	45.2
05003	58.2	56.8	54.9	50.6	55.4	63.7	59.7
05007	NaN	NaN	NaN	NaN	NaN	41.2	NaN
05017	52.7	51.2	52.0	54.1	53.9	55.2	54.1
05021	46.8	54.8	50.8	48.0	53.3	44.0	54.7
05029	NaN	39.4	37.0	NaN	33.2	32.9	32.0
05031	48.8	53.1	50.0	46.3	48.7	NaN	46.5
05035	43.8	52.7	NaN	46.1	50.4	52.3	53.2
05037	47.7	51.1	53.3	52.0	50.4	53.3	51.9
05041	55.7	56.7	59.5	59.8	57.3	52.1	58.8
05043	53.5	54.5	56.9	57.5	56.7	58.6	61.9
05045	37.2	42.0	43.2	NaN	33.6	33.2	39.9
05047	NaN	35.5	NaN	NaN	37.3	NaN	NaN
05055	43.7	51.0	41.5	45.8	46.9	NaN	52.0
05063	38.5	43.4	NaN	40.9	43.1	NaN	42.9
05067	39.2	42.2	40.1	40.3	40.6	46.8	38.4
05069	52.2	53.1	56.5	54.3	55.8	57.0	57.6
05071	NaN	NaN	NaN	NaN	34.7	NaN	NaN
05073	NaN	NaN	NaN	NaN	53.1	NaN	NaN
05075	35.7	42.0	40.4	39.9	42.9	NaN	48.7
05077	43.5	48.5	44.4	NaN	55.6	53.5	57.7
05079	56.2	57.8	55.4	57.2	54.6	53.2	60.8
05081	NaN	NaN	NaN	30.2	NaN	NaN	31.3
05083	NaN	NaN	NaN	NaN	37.5	35.3	NaN
05085	46.3	42.5	NaN	48.2	47.6	47.5	49.6
05093	48.9	56.7	50.5	47.6	53.4	NaN	59.3
05095	43.3	47.2	47.3	47.7	50.9	45.1	44.2
05107	48.9	52.9	50.5	NaN	55.5	58.8	55.4
05111	51.0	55.5	56.4	51.0	51.9	54.2	49.2
05115	NaN	NaN	NaN	NaN	NaN	NaN	43.8
05117	50.0	50.9	53.6	44.0	50.0	55.5	50.6
05119	NaN	46.7	44.3	NaN	35.9	38.8	35.6
05121	37.9	48.2	NaN	42.5	45.3	NaN	NaN
05123	44.5	49.7	46.2	50.3	50.2	51.9	52.0
05131	NaN	NaN	NaN	NaN	33.3	NaN	NaN

05145	37.9	42.6	NaN	43.0	47.9	45.8	36.4
05147	35.7	44.6	45.8	43.8	50.8	48.4	40.1
05149	NaN	42.6	NaN	NaN	42.5	38.2	34.6
06001	32.7	29.4	28.6	28.7	30.5	32.6	30.9
08001	32.8	35.1	29.9	27.0	33.5	31.0	32.6
10001	44.5	43.2	41.4	45.0	44.2	47.6	41.6
10003	47.0	NaN	45.8	47.2	48.4	46.9	36.9
10005	38.3	49.5	41.3	44.8	47.8	49.7	42.5
12001	29.1	26.9	27.7	28.1	29.4	27.9	27.0
13001	31.4	32.2	27.3	31.7	34.2	33.6	31.6
13005	NaN	NaN	NaN	NaN	37.7	36.8	NaN
13015	17.0	52.3	NaN	12.6	NaN	NaN	NaN
13027	49.4	45.7	NaN	36.2	NaN	48.7	47.8
13031	20.7	30.8	NaN	NaN	NaN	NaN	NaN
13033	32.4	34.0	NaN	NaN	51.3	41.1	38.3
13043	21.6	NaN	NaN	NaN	NaN	NaN	NaN
13055	21.4	41.5	37.2	40.4	42.0	47.1	NaN
13069	36.4	36.9	NaN	25.2	NaN	NaN	46.0
13071	38.8	44.8	43.5	NaN	NaN	NaN	NaN
13075	NaN	NaN	NaN	NaN	NaN	48.8	48.4
13087	43.3	NaN	NaN	NaN	NaN	NaN	NaN
13091	29.3	NaN	NaN	NaN	NaN	NaN	NaN
13093	NaN	42.3	NaN	NaN	42.0	NaN	NaN
13099	22.7	NaN	NaN	NaN	NaN	NaN	NaN
13105	NaN	39.4	45.5	24.7	48.7	47.9	31.8
13107	34.4	51.0	29.9	23.8	39.5	36.3	31.4
13115	15.3	53.0	NaN	NaN	NaN	42.7	NaN
13119	9.8	37.3	NaN	NaN	NaN	NaN	NaN
13129	NaN	NaN	NaN	NaN	NaN	NaN	44.1
13131	36.6	40.8	NaN	NaN	51.3	NaN	52.3
13147	16.0	36.4	41.2	23.2	52.3	NaN	NaN
13151	NaN	42.1	NaN	NaN	NaN	NaN	NaN
13153	40.7	33.9	NaN	NaN	NaN	NaN	49.9
13161	NaN	NaN	NaN	NaN	NaN	NaN	47.4
13163	28.7	NaN	38.1	34.0	46.2	36.1	NaN
13165	29.6	53.1	32.9	NaN	NaN	NaN	NaN
13167	21.3	NaN	NaN	NaN	NaN	58.2	NaN
13175	NaN	NaN	NaN	NaN	37.1	40.9	35.8
13177	NaN	NaN	35.6	NaN	NaN	40.2	NaN
13185	37.5	39.8	NaN	NaN	36.7	36.9	NaN
13193	37.1	36.7	NaN	28.4	59.7	47.1	NaN
13195	16.1	NaN	NaN	NaN	30.8	NaN	NaN
13197	16.3	35.3	NaN	NaN	NaN	40.1	41.3
13205	41.2	47.9	41.6	NaN	NaN	NaN	NaN
13209	35.3	45.4	NaN	NaN	NaN	NaN	NaN
13211	36.3	NaN	NaN	NaN	NaN	NaN	NaN
13213	23.2	NaN	NaN	NaN	NaN	NaN	NaN
13221	NaN	NaN	35.3	NaN	NaN	NaN	NaN

13225	36.8	NaN	26.9	NaN	34.8	NaN	38.9
13229	23.9	NaN	NaN	NaN	NaN	NaN	NaN
13233	21.8	NaN	NaN	NaN	NaN	NaN	NaN
13243	34.1	53.5	49.1	NaN	57.1	NaN	41.9
13251	29.7	42.7	NaN	NaN	36.1	NaN	NaN
13253	49.1	50.8	38.7	NaN	NaN	NaN	NaN
13267	24.1	NaN	NaN	NaN	NaN	NaN	NaN
13269	NaN	40.8	NaN	NaN	NaN	NaN	37.4
13273	12.3	24.0	NaN	NaN	NaN	45.2	49.1
13275	NaN	44.7	46.1	NaN	NaN	53.8	52.1
13279	34.0	41.8	NaN	NaN	NaN	42.7	NaN
13295	NaN	47.2	NaN	NaN	NaN	42.3	NaN
13299	20.3	NaN	NaN	NaN	44.4	NaN	NaN
13303	NaN	NaN	NaN	NaN	48.8	NaN	NaN
16001	38.7	34.4	31.6	34.0	35.5	37.5	38.0
17001	55.5	56.1	51.0	48.6	52.8	57.5	53.2
17003	38.4	42.3	NaN	NaN	48.7	58.5	51.3
17005	NaN	54.5	67.0	51.7	59.0	65.0	58.6
17007	62.3	47.4	60.3	50.0	53.0	69.2	62.2
17009	52.0	57.1	60.3	56.2	61.4	59.4	64.4
17011	64.2	59.1	68.5	57.4	64.8	66.4	67.5
17013	NaN	53.6	58.6	46.9	54.1	NaN	57.8
17015	69.0	64.3	68.9	64.7	68.1	68.6	68.9
17017	62.4	62.9	71.9	NaN	63.0	68.4	64.6
17019	67.7	64.3	67.9	52.8	66.1	75.3	69.4
17021	62.9	65.6	73.6	NaN	65.9	71.2	69.5
17023	NaN	56.5	64.1	NaN	58.4	65.5	61.3
17025	52.5	44.3	58.5	40.5	52.4	59.2	57.3
17027	54.1	50.8	60.1	51.9	52.5	63.6	54.6
17029	62.6	64.3	74.9	63.7	66.6	NaN	NaN
17031	NaN	NaN	NaN	NaN	NaN	45.8	NaN
17033	56.4	52.7	63.0	49.5	56.3	65.0	61.8
17035	59.0	54.1	67.2	50.6	56.5	NaN	67.2
17037	67.5	53.6	61.1	55.6	59.0	63.9	64.9
17039	62.1	65.9	75.0	NaN	66.4	70.7	68.0
17041	67.4	63.7	78.1	62.4	67.1	NaN	69.0
17045	65.0	64.7	69.5	59.5	61.0	70.8	67.9
17047	47.7	55.6	44.1	NaN	54.7	63.1	53.5
17049	58.2	55.3	72.1	51.5	57.9	64.7	59.9
17051	55.6	53.5	59.5	48.9	54.4	63.5	NaN
17053	60.5	62.4	60.7	52.6	56.3	66.7	66.7
17055	45.2	46.4	47.5	38.9	51.2	53.3	47.0
17057	56.4	60.1	62.4	53.9	58.4	65.5	62.5
17059	48.4	54.5	50.0	47.5	57.6	62.5	55.3
17061	62.5	63.9	70.1	60.7	65.2	NaN	66.6
17063	58.0	57.3	58.5	49.0	56.2	59.5	63.1
17065	46.1	51.3	51.2	NaN	51.8	NaN	50.2
17067	63.7	66.1	69.8	NaN	66.6	67.3	67.6

17069	39.8	NaN	43.3	44.8	50.8	50.6	NaN
17071	57.8	61.6	60.2	NaN	62.1	66.6	67.2
17073	61.4	58.9	63.2	52.6	62.3	68.1	66.8
17075	59.1	57.1	55.3	45.3	60.0	57.9	63.8
17077	NaN	50.1	48.9	39.6	52.2	49.9	51.1
17079	58.1	54.6	60.7	53.9	54.5	NaN	55.8
17081	48.4	40.9	51.4	37.4	47.6	47.7	48.1
17083	55.3	56.9	NaN	60.1	65.3	NaN	63.2
17085	60.6	53.4	NaN	55.6	59.1	58.7	62.6
17087	NaN	48.0	NaN	NaN	45.8	43.8	45.1
17089	64.5	53.6	58.9	50.3	56.0	67.7	64.4
17091	55.8	52.2	54.2	52.3	54.8	61.7	59.7
17093	NaN	58.9	60.8	57.4	57.5	66.3	NaN
17095	63.6	63.7	64.1	55.9	63.9	68.4	67.0
17097	52.5	NaN	41.3	48.2	51.8	50.3	57.3
17099	61.2	60.8	67.1	55.4	59.3	NaN	65.7
17101	NaN	51.0	54.1	47.8	57.9	60.4	58.0
17103	60.8	59.4	63.7	59.7	64.7	67.6	66.8
17105	60.6	58.8	61.0	56.9	56.0	66.0	63.5
17107	66.6	64.7	70.1	NaN	68.7	NaN	71.8
17109	62.5	64.7	69.3	60.2	65.1	NaN	66.3
17111	59.1	47.2	58.9	NaN	52.5	NaN	NaN
17113	63.8	66.8	67.5	61.0	63.9	67.4	68.8
17115	68.2	66.2	75.2	63.8	66.2	74.4	73.8
17117	63.4	61.3	69.7	58.3	67.4	70.8	65.1
17119	54.7	56.7	62.5	56.6	58.5	61.0	59.6
17121	53.3	45.9	60.5	46.5	54.9	60.1	54.3
17123	63.3	60.1	65.5	53.8	63.1	NaN	64.1
17125	59.2	62.1	NaN	55.4	61.2	64.7	62.6
17127	47.5	NaN	NaN	43.5	51.6	52.5	50.0
17129	63.6	63.9	68.6	62.0	67.3	66.4	67.9
17131	59.9	60.0	63.2	51.9	59.4	NaN	65.0
17133	48.4	43.4	54.1	47.2	55.7	56.3	53.1
17135	61.5	61.9	73.0	61.5	63.2	NaN	68.6
17137	60.4	60.2	79.8	64.1	62.9	NaN	70.0
17139	68.5	66.4	75.9	59.3	65.0	72.4	65.7
17141	65.6	55.0	64.3	59.9	61.9	61.8	67.6
17143	60.0	63.9	66.6	55.4	65.2	69.7	64.0
17145	48.0	42.7	52.5	35.6	48.2	53.2	43.4
17147	71.3	69.4	77.4	NaN	71.7	77.3	74.2
17149	58.1	57.7	61.2	NaN	62.1	NaN	NaN
17151	35.1	44.6	NaN	NaN	53.1	NaN	NaN
17153	44.4	44.5	48.2	43.2	54.8	56.8	53.5
17155	63.3	57.2	NaN	NaN	61.8	62.2	64.6
17157	50.0	44.4	52.1	41.6	53.4	55.3	50.3
17159	51.8	48.8	56.3	NaN	54.0	NaN	56.1
17161	58.0	58.6	60.5	NaN	61.3	64.6	66.3
17163	52.4	50.9	56.0	53.2	52.7	59.6	53.6

17165	46.8	49.9	49.4	NaN	59.0	59.2	52.8
17167	66.2	70.3	80.4	65.5	68.8	72.1	72.7
17169	56.9	58.6	59.9	NaN	58.1	63.2	63.1
17171	57.9	62.9	NaN	NaN	65.1	64.9	72.1
17173	57.5	61.0	70.4	56.2	60.4	67.3	67.1
17175	65.2	64.6	NaN	58.8	63.0	70.1	70.8
17177	66.3	59.2	63.3	56.4	64.1	64.5	62.3
17179	65.0	66.8	72.1	63.8	63.8	70.8	71.0
17181	43.8	51.6	NaN	34.0	50.2	44.5	48.0
17183	64.6	64.9	71.7	NaN	61.8	67.4	67.6
17185	55.2	54.4	NaN	NaN	54.5	60.8	58.1
17187	62.6	63.3	67.5	58.5	66.6	NaN	64.5
17189	48.8	48.3	57.4	43.9	51.8	58.7	50.5
17191	48.3	47.5	52.9	43.4	52.0	NaN	56.4
17193	47.0	52.8	53.0	46.6	55.0	63.4	54.5
17195	60.8	58.7	62.7	NaN	NaN	67.0	66.4
17197	55.2	50.9	57.8	47.8	54.5	58.4	57.2
17199	42.5	47.2	45.6	NaN	52.8	NaN	45.9
17201	58.7	47.2	56.4	55.8	56.7	56.9	62.8
17203	67.0	60.4	64.7	61.4	63.3	68.1	69.1
18001	59.3	54.1	52.9	47.6	55.3	58.3	54.5
18003	54.8	49.9	58.2	52.3	53.8	59.5	54.7
18005	54.8	54.2	55.5	NaN	59.9	NaN	57.3
18007	61.9	59.7	66.5	54.5	64.1	60.5	61.4
18009	63.5	54.7	57.4	53.8	NaN	62.4	56.1
18011	62.2	54.5	61.2	52.4	65.0	62.7	62.4
18013	51.7	49.8	NaN	NaN	NaN	54.4	53.8
18015	65.1	58.5	63.9	58.7	67.9	NaN	NaN
18017	60.0	53.5	61.8	NaN	59.4	NaN	57.1
18019	45.0	47.8	49.8	40.6	54.4	50.6	52.6
18021	NaN	55.6	52.8	48.4	56.4	58.7	58.1
18023	65.6	58.0	65.2	NaN	67.4	61.5	61.0
18025	NaN	NaN	47.9	46.3	53.4	NaN	NaN
18027	55.4	58.3	54.3	53.5	60.9	68.2	58.6
18029	NaN	47.9	54.1	30.9	51.4	47.5	47.4
18031	61.1	59.5	62.9	53.9	68.4	61.6	68.3
18033	49.2	48.2	49.3	52.4	45.6	55.6	54.2
18035	59.4	50.8	57.0	48.8	59.3	60.8	57.9
18037	53.2	63.6	59.0	51.4	61.7	63.6	58.4
18039	54.2	54.9	57.6	48.3	55.7	58.0	58.6
18041	57.8	52.1	57.0	51.5	60.6	58.0	57.3
18043	NaN	50.3	NaN	37.1	54.8	NaN	NaN
18045	59.4	56.5	NaN	NaN	64.4	NaN	NaN
18047	56.7	53.3	59.0	41.6	56.5	53.5	57.0
18049	56.3	55.8	57.5	51.4	54.3	56.3	NaN
18051	55.3	58.5	56.3	55.0	62.1	64.7	62.0
18053	59.3	52.9	59.4	NaN	59.9	62.1	56.3
18055	50.8	46.3	53.2	NaN	55.2	61.4	52.3

18057	60.0	58.5	62.8	48.8	65.2	61.6	59.7
18059	56.4	48.9	57.1	42.3	59.3	57.5	56.2
18061	48.1	50.5	51.6	43.6	59.3	55.0	56.1
18063	61.7	53.7	60.2	53.1	64.4	62.2	59.8
18065	60.8	49.2	56.9	45.8	61.5	61.4	55.8
18067	62.8	54.8	62.5	55.4	66.0	66.4	64.2
18069	57.6	51.4	58.5	54.3	57.9	62.6	57.5
18071	55.0	50.5	51.2	NaN	58.9	NaN	59.3
18073	54.7	52.6	56.9	NaN	55.1	58.2	54.6
18075	58.9	50.0	59.5	49.3	55.7	NaN	55.3
18077	43.8	45.0	48.3	32.1	50.3	49.8	49.4
18079	51.9	52.3	55.0	45.7	57.7	52.9	52.8
18081	NaN	59.1	NaN	53.6	60.9	58.0	62.2
18083	56.6	56.6	52.5	NaN	62.5	67.2	57.4
18085	53.3	51.1	60.6	49.0	52.1	56.8	54.9
18087	54.9	50.2	52.9	47.5	52.7	58.5	56.0
18089	54.8	50.5	55.7	48.3	52.0	NaN	51.8
18091	56.0	50.8	58.3	50.1	49.7	55.0	52.0
18093	47.5	46.9	52.5	NaN	54.7	52.0	50.0
18095	62.0	50.2	60.5	47.6	64.5	65.4	56.7
18097	NaN	44.0	NaN	47.6	NaN	47.0	57.8
18099	54.4	52.0	53.3	NaN	50.0	57.8	NaN
18101	55.3	NaN	NaN	43.8	60.6	65.0	57.6
18103	55.8	48.9	55.0	49.1	54.5	NaN	57.1
18105	52.2	45.8	44.7	NaN	51.3	48.9	55.1
18107	64.0	62.2	63.8	58.5	63.0	65.0	64.5
18109	54.4	49.0	56.9	NaN	61.3	60.0	59.3
18111	61.5	54.4	59.0	52.0	60.8	55.3	59.3
18113	54.0	49.7	51.6	54.3	52.2	57.0	59.1
18115	56.1	52.0	50.0	30.8	52.7	42.2	52.6
18117	55.0	63.2	55.8	56.6	64.9	65.1	63.0
18119	48.9	44.8	NaN	NaN	54.0	55.4	50.9
18121	60.6	55.8	56.5	52.3	61.7	58.9	57.4
18123	NaN	NaN	45.0	47.2	57.0	NaN	51.6
18125	51.0	NaN	NaN	51.8	56.7	NaN	50.4
18127	57.1	53.0	59.7	52.1	50.5	60.5	54.8
18129	53.9	56.6	54.4	56.3	66.5	NaN	60.7
18131	50.1	51.0	54.5	44.3	49.9	NaN	51.0
18133	58.8	55.5	57.4	58.8	61.0	58.8	60.3
18135	58.5	51.6	58.2	50.9	62.5	62.2	55.8
18137	52.1	57.4	51.1	42.2	53.9	52.3	51.2
18139	62.2	56.5	57.9	56.9	63.2	60.4	61.8
18141	53.4	49.4	54.4	47.3	50.4	55.5	53.4
18143	NaN	50.6	49.6	NaN	NaN	49.1	NaN
18145	61.9	57.6	56.0	51.8	61.8	56.8	63.6
18147	52.4	59.0	51.1	53.3	58.5	60.0	NaN
18149	50.4	48.9	53.8	NaN	46.9	NaN	50.6
18151	50.4	48.2	46.9	NaN	44.7	55.9	54.2

18153	52.9	53.7	55.5	50.4	55.6	61.9	54.9
18155	52.1	48.5	47.6	NaN	57.4	49.0	55.2
18157	NaN	56.1	63.2	54.7	66.6	58.8	58.4
18159	66.8	61.3	NaN	56.4	69.8	69.2	62.2
18161	60.0	59.5	59.8	51.0	65.5	60.6	59.5
18163	57.9	62.9	55.5	60.9	69.4	73.3	59.1
18165	62.8	54.2	60.3	51.4	57.1	64.4	NaN
18167	NaN	55.2	51.5	49.3	50.6	58.4	55.7
18169	57.4	49.5	59.2	53.1	51.5	55.7	58.6
18171	64.5	64.9	66.5	60.7	67.5	63.4	64.7
18173	50.2	60.9	49.8	53.6	59.4	NaN	57.2
18175	52.5	51.1	52.3	44.1	56.7	NaN	54.7
18177	58.7	55.4	59.2	50.1	61.7	60.4	55.9
18179	61.8	50.9	61.0	NaN	58.7	66.0	57.4
18181	61.1	58.8	60.4	52.8	59.3	NaN	56.2
18183	58.0	51.7	54.7	53.2	51.3	61.7	54.6
19001	60.8	57.7	52.0	55.0	54.1	58.5	57.3
19003	56.5	52.6	53.2	NaN	NaN	NaN	NaN
19005	56.8	52.6	52.8	NaN	58.8	61.9	61.2
19007	55.9	50.7	45.1	NaN	39.1	57.5	40.4
19009	62.0	57.2	52.3	58.9	48.3	67.3	57.9
19011	61.1	57.6	61.0	57.9	56.7	67.1	66.1
19013	61.8	59.5	58.4	51.9	54.2	60.1	64.5
19015	60.2	56.7	54.6	53.2	50.9	64.4	60.4
19017	62.1	56.8	57.3	52.5	NaN	62.0	66.7
19019	60.2	55.9	57.2	55.2	56.7	63.4	65.0
19021	60.2	56.5	55.5	53.5	NaN	63.8	57.2
19023	59.9	55.9	56.1	55.0	54.2	60.8	61.6
19025	62.3	57.5	53.1	55.1	47.7	63.5	61.2
19027	58.7	60.4	59.0	58.8	54.8	65.3	58.9
19029	59.6	56.9	52.6	56.9	49.3	63.3	51.2
19031	60.5	59.6	63.3	NaN	56.4	65.3	67.0
19033	58.6	55.2	51.7	52.7	59.8	61.0	60.8
19035	69.6	62.4	64.3	58.8	57.2	NaN	59.6
19037	56.0	50.4	53.2	54.5	55.9	NaN	63.6
19039	49.9	NaN	47.9	NaN	45.9	57.7	40.2
19041	61.9	57.6	52.6	50.9	50.3	62.4	56.9
19043	63.2	59.0	58.8	56.4	60.1	NaN	64.5
19045	59.8	59.7	58.3	59.5	57.0	63.9	66.0
19047	64.3	61.3	61.7	61.9	54.2	63.5	62.8
19049	58.7	54.6	52.8	54.4	47.1	63.6	59.8
19051	57.4	37.6	50.2	54.6	42.0	56.7	44.6
19053	57.2	NaN	48.0	NaN	47.4	57.0	37.9
19055	63.8	63.0	63.9	NaN	61.1	66.4	67.4
19057	60.4	59.6	63.8	55.1	60.7	NaN	66.9
19059	56.1	57.7	49.8	51.3	55.5	56.0	58.8
19061	57.8	59.6	62.0	60.2	NaN	70.5	68.6
19063	58.5	58.9	46.7	NaN	56.0	59.6	62.8

19065	62.3	56.3	56.5	55.5	55.9	64.4	63.4
19067	57.5	54.1	49.8	54.3	54.0	NaN	61.9
19069	60.5	57.0	58.5	54.1	55.4	59.7	62.8
19071	55.5	60.1	50.6	NaN	54.8	63.1	53.4
19073	63.5	56.7	55.5	54.2	49.0	64.1	59.0
19075	64.8	63.6	61.9	56.3	60.9	66.5	64.9
19077	58.4	59.2	52.0	54.2	45.5	NaN	58.6
19079	61.0	57.2	49.9	51.4	51.5	62.2	61.3
19081	62.4	57.4	53.2	49.6	60.0	63.5	63.0
19083	59.9	59.7	59.3	53.9	52.8	65.0	63.4
19085	58.2	53.7	52.6	NaN	52.1	NaN	52.4
19087	57.2	60.5	62.3	56.9	54.9	58.3	60.3
19089	56.4	51.6	53.7	52.1	54.5	60.3	60.9
19091	61.0	58.0	50.4	54.9	56.9	62.4	59.0
19093	67.6	62.1	63.8	NaN	55.5	65.6	63.1
19095	57.4	56.7	56.0	52.3	49.8	NaN	61.6
19097	61.1	59.2	58.1	NaN	54.7	NaN	66.3
19099	60.3	60.5	60.7	56.5	NaN	67.3	60.1
19101	55.1	46.1	62.0	53.3	53.8	53.9	50.7
19103	56.1	56.4	59.9	55.5	51.8	NaN	62.1
19105	60.4	54.9	60.1	57.2	54.0	64.5	61.7
19107	57.1	51.6	57.7	55.8	51.6	NaN	53.1
19109	63.4	60.4	54.2	57.4	60.2	60.4	56.2
19111	61.4	53.5	64.7	54.8	52.1	59.7	54.6
19113	60.0	56.0	57.7	56.5	NaN	NaN	64.6
19115	56.4	59.0	58.1	54.3	53.3	NaN	61.4
19117	54.4	41.4	41.5	NaN	43.8	54.1	43.1
19119	66.1	63.8	60.6	54.8	60.8	63.6	62.2
19121	57.1	52.5	50.9	52.3	51.4	58.8	55.4
19123	59.1	54.4	59.7	53.7	NaN	62.0	59.3
19125	56.6	52.9	58.3	54.0	NaN	60.6	57.8
19127	62.6	63.5	61.9	60.6	NaN	67.1	65.0
19129	56.3	55.7	NaN	NaN	49.3	NaN	52.1
19131	61.0	54.4	55.6	49.8	59.4	62.5	64.6
19133	56.1	57.1	55.3	55.8	56.0	61.8	48.6
19135	57.3	51.0	50.4	56.1	45.4	59.3	47.4
19137	57.8	56.2	51.8	58.3	49.2	63.0	54.8
19139	60.8	60.3	61.7	55.2	55.6	62.6	65.5
19141	63.9	62.3	59.9	56.0	62.0	66.4	59.4
19143	56.1	59.7	57.2	50.5	NaN	62.9	56.3
19145	57.5	55.4	50.5	55.9	52.4	62.6	54.0
19147	59.7	57.8	48.8	55.6	NaN	61.6	57.9
19149	66.1	59.6	61.3	NaN	NaN	NaN	47.8
19151	60.6	58.2	49.7	NaN	54.6	65.0	56.5
19153	55.2	57.1	53.4	51.0	50.2	64.6	54.1
19155	60.6	57.0	NaN	58.5	48.0	64.4	51.0
19157	57.5	56.6	58.2	53.4	54.8	NaN	60.4
19159	51.7	48.8	47.2	NaN	50.6	54.0	41.8

19161	63.3	57.8	60.7	59.3	47.1	NaN	61.3
19163	64.4	63.4	63.7	60.5	61.0	68.1	69.1
19165	60.5	58.2	57.8	58.7	52.8	NaN	58.0
19167	65.8	65.2	65.8	60.8	NaN	70.5	62.9
19169	58.4	57.9	52.8	50.0	52.2	63.6	60.0
19171	63.5	59.5	60.1	53.4	56.2	66.3	61.4
19173	51.6	52.8	NaN	49.9	47.0	61.3	52.9
19175	56.0	44.5	48.6	53.7	52.1	61.9	41.4
19177	56.6	36.6	54.2	53.9	44.2	53.0	43.6
19179	58.1	45.0	60.2	55.5	52.0	59.0	53.7
19181	54.3	49.9	52.0	52.3	49.5	NaN	50.2
19183	55.8	61.7	61.5	54.7	53.2	NaN	57.2
19185	54.2	NaN	49.0	52.5	44.6	57.1	NaN
19187	58.4	57.9	51.1	49.1	50.4	64.6	57.5
19189	62.5	59.9	51.8	54.2	60.8	61.6	64.4
19191	58.4	54.6	51.7	55.6	53.6	61.8	66.3
19193	61.2	58.1	59.5	56.8	56.6	NaN	56.4
19195	58.2	54.8	53.7	51.0	58.2	60.1	62.0
19197	58.8	57.9	52.6	54.1	55.9	61.0	60.6
20001	45.4	45.0	43.3	43.1	46.5	46.2	43.4
20003	46.9	47.1	48.0	34.2	37.4	41.8	23.2
20005	55.2	51.9	47.4	51.9	49.4	53.7	44.3
20007	NaN	NaN	NaN	26.7	NaN	28.6	20.2
20009	NaN	31.9	NaN	29.9	43.2	37.7	19.1
20011	NaN	40.8	38.2	35.9	35.0	33.1	17.0
20013	62.5	49.4	48.6	56.5	57.2	64.8	53.3
20015	46.1	35.4	47.6	37.6	34.3	35.9	18.3
20017	47.9	43.7	NaN	NaN	39.7	41.8	33.6
20019	31.4	33.5	33.8	40.3	NaN	18.8	12.2
20021	NaN	NaN	NaN	38.9	39.6	32.2	16.0
20023	NaN	56.2	NaN	NaN	62.1	NaN	62.4
20027	48.9	39.7	46.2	45.3	51.9	NaN	28.5
20029	52.5	37.1	NaN	NaN	44.7	39.8	26.3
20031	46.7	39.2	38.7	33.9	38.2	36.2	27.2
20035	42.4	30.9	NaN	37.5	37.4	26.6	9.5
20037	39.2	NaN	38.7	NaN	32.4	34.2	NaN
20041	47.7	28.8	43.6	41.8	40.9	40.1	28.0
20043	62.3	58.5	53.3	NaN	58.3	64.3	NaN
20045	46.1	46.3	33.4	41.0	43.4	36.1	29.8
20047	NaN	53.4	NaN	NaN	63.9	55.9	36.0
20049	41.9	34.3	43.6	39.9	NaN	33.7	21.0
20053	NaN	NaN	48.1	NaN	41.3	34.3	14.5
20055	68.2	NaN	NaN	56.2	NaN	NaN	NaN
20057	NaN	NaN	NaN	NaN	61.6	NaN	NaN
20059	49.9	41.7	43.4	NaN	40.1	36.9	23.5
20061	55.5	41.8	44.6	46.3	50.2	40.6	41.1
20063	NaN	NaN	50.9	NaN	NaN	NaN	NaN
20065	NaN	50.2	54.5	NaN	NaN	NaN	NaN

20067	NaN	NaN	NaN	NaN	NaN	53.1	NaN
20069	68.1	62.8	NaN	53.4	NaN	59.7	NaN
20073	NaN	36.4	34.8	NaN	32.0	30.8	19.6
20077	NaN	22.0	29.4	31.0	32.4	24.0	9.4
20079	48.7	30.8	48.8	37.3	35.5	42.4	25.9
20081	64.5	59.7	64.5	65.0	63.8	62.6	30.5
20083	NaN	55.7	57.1	NaN	NaN	NaN	49.2
20085	47.7	42.9	39.7	46.2	47.1	45.0	NaN
20087	49.2	52.8	NaN	49.2	46.2	52.0	36.7
20089	55.5	39.3	42.9	49.4	47.0	41.2	34.2
20091	NaN	NaN	34.1	NaN	NaN	38.8	32.8
20095	45.9	26.7	46.0	26.7	34.5	35.4	22.5
20097	57.8	NaN	63.3	NaN	53.3	50.0	41.0
20099	NaN	39.4	33.7	NaN	36.7	24.4	9.3
20103	NaN	NaN	NaN	46.0	47.3	42.0	38.2
20105	NaN	NaN	NaN	NaN	42.2	NaN	12.3
20107	41.8	NaN	NaN	NaN	40.0	33.5	20.9
20111	41.7	38.6	NaN	34.0	35.9	36.1	23.2
20113	45.1	29.3	45.5	35.1	30.2	34.4	16.9
20115	52.0	25.7	42.4	35.1	29.7	38.3	17.9
20117	45.3	28.3	38.6	48.0	42.7	47.9	38.8
20119	NaN	70.9	NaN	NaN	73.1	69.1	67.6
20121	47.8	49.7	42.2	43.3	48.9	46.1	29.5
20123	53.4	NaN	43.0	43.3	44.0	29.1	24.7
20125	35.9	37.0	39.6	35.6	40.7	27.9	10.7
20127	42.8	38.8	32.8	44.9	40.2	36.7	32.7
20131	56.1	33.3	41.5	NaN	45.7	57.6	39.8
20133	43.6	35.7	38.9	NaN	31.4	29.5	11.0
20137	NaN	NaN	47.8	NaN	32.5	33.4	NaN
20139	45.1	42.8	35.5	41.0	37.5	39.5	25.1
20141	50.3	27.5	NaN	NaN	38.5	NaN	13.6
20143	45.0	29.9	NaN	NaN	43.9	NaN	20.5
20145	49.5	43.1	NaN	NaN	56.7	NaN	44.3
20147	54.0	NaN	NaN	43.8	27.4	38.7	25.7
20149	54.5	48.1	40.0	49.5	45.8	40.7	NaN
20151	57.4	51.8	54.0	NaN	54.7	44.9	37.8
20153	NaN	NaN	NaN	NaN	NaN	36.5	NaN
20155	45.9	23.1	NaN	32.8	38.0	NaN	21.4
20157	55.7	44.8	51.0	45.1	50.5	54.6	43.3
20159	43.3	29.3	NaN	NaN	35.7	37.6	16.4
20161	50.9	37.0	38.7	46.1	48.4	47.8	35.0
20163	NaN	25.5	NaN	NaN	29.7	30.3	13.9
20165	44.1	NaN	37.3	NaN	NaN	NaN	NaN
20167	41.7	25.2	NaN	30.3	41.9	33.5	NaN
20169	42.4	22.6	NaN	43.0	39.4	34.4	19.7
20171	NaN	NaN	51.5	NaN	NaN	NaN	NaN
20173	45.9	32.0	43.5	36.3	35.3	32.2	25.2
20175	59.3	72.5	65.8	64.4	60.6	58.9	57.6

20177	48.9	39.0	37.7	NaN	41.5	42.1	35.2
20179	NaN	NaN	NaN	NaN	59.5	59.1	NaN
20181	NaN	44.4	NaN	NaN	NaN	NaN	NaN
20183	NaN	39.6	45.8	NaN	37.5	43.8	24.7
20185	48.7	NaN	NaN	NaN	43.8	NaN	NaN
20189	61.3	58.1	NaN	NaN	62.0	57.9	64.1
20191	39.6	26.5	35.7	31.0	30.4	21.4	12.5
20193	NaN	NaN	NaN	NaN	45.6	NaN	40.2
20195	NaN	NaN	NaN	34.8	NaN	NaN	NaN
20197	NaN	36.8	31.0	NaN	41.9	39.8	34.2
20201	43.4	33.2	NaN	NaN	45.2	48.8	42.0
20203	NaN	NaN	52.5	NaN	NaN	NaN	NaN
20205	44.1	37.7	NaN	39.4	29.2	32.3	15.8
20207	NaN	36.0	NaN	NaN	33.1	31.7	NaN
21001	43.9	44.5	38.2	36.0	42.5	44.2	40.6
21003	NaN	NaN	NaN	NaN	54.2	41.3	NaN
21005	NaN	NaN	NaN	NaN	NaN	53.4	NaN
21007	50.1	49.4	NaN	51.5	56.9	NaN	49.2
21009	53.3	56.7	NaN	44.0	60.9	54.5	52.8
21011	46.4	NaN	NaN	48.1	43.4	47.3	52.1
21015	52.3	60.3	55.9	53.4	53.1	51.3	54.8
21017	47.5	50.9	47.4	NaN	50.4	61.6	NaN
21021	54.3	59.7	56.0	40.0	56.1	58.8	57.9
21023	45.4	51.3	51.9	NaN	NaN	51.3	NaN
21027	53.2	53.9	49.0	36.0	55.7	50.8	53.6
21029	47.8	51.8	52.4	30.1	52.6	47.5	NaN
21031	40.8	50.5	50.8	51.0	54.0	57.8	55.2
21033	50.6	50.3	48.1	46.5	58.3	56.6	41.2
21035	41.0	43.7	43.0	NaN	52.4	57.7	42.0
21039	48.1	NaN	51.2	NaN	56.0	54.2	45.3
21041	39.6	NaN	NaN	NaN	50.2	47.6	53.2
21045	NaN	54.0	NaN	47.0	55.1	56.8	51.0
21047	48.1	54.4	51.6	44.5	54.9	55.3	44.0
21049	47.9	58.1	49.3	58.0	55.0	59.6	54.4
21051	NaN	NaN	NaN	NaN	49.2	46.1	44.8
21053	49.4	NaN	NaN	NaN	54.8	NaN	53.8
21055	50.0	49.7	51.5	NaN	54.1	55.1	43.4
21057	NaN	NaN	46.3	53.9	59.9	NaN	50.5
21059	54.2	60.3	56.2	54.5	60.0	62.7	59.2
21061	46.9	49.9	52.2	48.0	54.9	50.1	53.8
21065	42.9	42.5	35.2	35.0	50.3	NaN	NaN
21067	50.6	54.1	52.9	44.5	51.4	53.4	50.0
21069	45.7	44.9	NaN	40.5	48.5	49.8	50.5
21073	NaN	NaN	NaN	NaN	48.2	51.9	NaN
21075	49.4	52.1	NaN	48.0	54.7	57.1	45.5
21077	42.0	52.0	NaN	NaN	NaN	NaN	NaN
21079	NaN	49.0	NaN	NaN	NaN	56.7	NaN
21081	NaN	NaN	NaN	49.5	51.7	48.2	50.0

21083	50.2	51.4	45.4	NaN	52.0	53.1	42.8
21085	51.8	51.7	50.4	NaN	50.5	53.7	51.3
21087	51.5	55.5	55.0	NaN	55.8	56.2	59.7
21089	46.5	NaN	46.7	NaN	42.3	41.9	48.0
21091	50.6	59.0	NaN	54.0	58.6	66.5	NaN
21093	53.4	52.4	52.9	36.5	54.3	54.0	52.7
21097	46.9	NaN	NaN	NaN	50.7	NaN	NaN
21099	54.8	51.3	55.7	46.0	54.5	55.9	53.2
21101	NaN	58.8	52.0	53.0	59.0	NaN	58.8
21103	40.0	45.0	48.5	42.5	51.3	55.6	NaN
21105	47.4	53.6	50.8	52.0	53.8	56.7	46.6
21107	NaN	NaN	NaN	NaN	53.2	NaN	NaN
21111	NaN	NaN	NaN	25.0	50.7	56.6	NaN
21113	42.5	47.6	51.8	31.0	45.9	48.7	47.1
21117	NaN	NaN	NaN	NaN	NaN	51.3	NaN
21123	NaN	58.0	NaN	47.0	58.5	57.6	56.9
21135	50.2	NaN	NaN	44.0	46.7	45.8	50.7
21137	53.4	NaN	49.9	37.0	54.1	56.9	51.6
21139	48.4	46.0	NaN	47.0	54.0	49.3	41.3
21141	48.7	49.1	47.8	43.0	52.5	50.8	45.3
21143	NaN	NaN	46.0	43.0	51.6	NaN	45.0
21145	NaN	NaN	NaN	NaN	51.4	49.9	NaN
21149	NaN	55.2	58.3	55.0	58.1	60.4	54.7
21151	47.5	53.9	48.6	30.0	54.2	57.9	NaN
21155	52.9	59.2	57.2	38.5	59.6	58.5	52.7
21157	NaN	46.7	43.0	34.0	52.2	52.8	NaN
21161	53.8	55.0	56.4	44.0	54.4	55.6	57.6
21163	56.5	58.8	55.0	43.0	59.9	58.4	54.8
21167	54.8	54.0	NaN	42.0	NaN	NaN	59.7
21169	53.0	54.0	NaN	NaN	53.2	58.9	58.3
21171	NaN	NaN	49.9	NaN	NaN	NaN	57.8
21173	NaN	NaN	NaN	NaN	44.9	NaN	NaN
21177	NaN	NaN	47.3	NaN	51.0	54.5	50.4
21179	52.1	50.3	NaN	33.5	50.3	53.4	NaN
21181	44.7	49.3	NaN	36.0	NaN	NaN	49.0
21183	NaN	56.4	58.5	53.0	53.6	57.6	60.7
21185	45.5	46.4	46.1	39.5	51.7	50.0	NaN
21187	NaN	NaN	34.7	NaN	53.1	53.4	NaN
21191	46.0	60.9	46.8	NaN	46.1	NaN	50.5
21197	NaN	NaN	NaN	NaN	NaN	NaN	46.7
21199	54.9	58.9	45.1	43.0	49.1	53.0	51.9
21203	45.4	NaN	NaN	44.0	46.2	43.4	NaN
21205	36.7	46.8	NaN	NaN	NaN	NaN	NaN
21207	50.8	51.3	50.4	47.5	52.0	51.4	58.6
21209	50.8	57.5	NaN	41.0	50.0	NaN	56.1
21211	48.2	53.3	56.8	40.5	56.9	55.2	56.7
21213	51.5	53.3	51.2	45.5	54.5	51.4	46.8
21215	53.7	50.3	57.9	38.5	53.5	56.8	52.6

21217	48.7	54.8	45.8	48.0	54.2	55.8	55.9
21219	52.1	50.7	49.5	43.5	51.9	54.8	45.5
21221	46.2	51.5	46.7	41.0	51.1	55.8	40.6
21223	43.8	46.4	45.1	35.0	52.6	50.9	46.6
21225	53.9	60.4	57.4	56.5	63.9	67.0	60.6
21227	51.5	53.3	53.0	47.5	56.8	53.0	56.9
21229	54.0	52.7	NaN	30.5	52.7	56.5	53.2
21231	52.2	60.4	48.1	48.0	52.9	54.2	55.7
21233	50.5	53.4	54.1	51.5	60.1	60.4	57.4
21239	50.9	54.6	56.9	50.5	56.4	57.2	55.7
22001	37.6	36.9	37.7	39.2	37.3	34.8	36.5
22003	25.9	NaN	NaN	NaN	NaN	NaN	NaN
22007	46.2	39.7	NaN	38.1	NaN	42.6	50.0
22009	46.0	49.0	44.2	40.8	46.4	43.4	48.3
22011	NaN	37.1	NaN	NaN	NaN	NaN	NaN
22015	29.7	56.0	NaN	NaN	NaN	37.6	NaN
22017	37.9	48.9	40.4	42.3	46.9	33.8	33.5
22019	NaN	34.1	NaN	NaN	NaN	NaN	NaN
22025	46.8	51.4	44.3	45.2	52.6	49.1	44.2
22029	55.3	55.3	54.7	55.1	61.8	51.9	48.3
22035	59.8	65.6	60.6	60.5	65.0	61.9	57.4
22039	29.5	38.2	29.3	31.9	34.9	39.0	32.4
22041	57.9	59.0	61.7	54.6	50.3	56.1	54.7
22043	45.4	NaN	39.8	38.4	38.7	46.4	49.3
22045	34.4	42.3	32.0	38.2	36.5	42.4	40.4
22047	51.9	58.0	57.6	47.3	51.5	NaN	NaN
22053	24.1	31.3	36.9	NaN	NaN	NaN	27.9
22055	33.6	37.6	NaN	NaN	NaN	NaN	NaN
22065	51.8	56.8	59.4	50.1	55.0	57.4	48.5
22067	54.0	61.5	54.1	NaN	53.6	60.0	50.7
22069	37.6	47.0	37.7	42.0	37.7	40.4	32.0
22073	38.8	56.9	NaN	NaN	44.5	NaN	35.1
22077	48.6	54.0	52.1	51.3	54.1	47.3	43.1
22079	49.7	49.4	47.8	41.7	40.4	44.2	39.9
22081	NaN	NaN	36.2	36.6	39.6	NaN	NaN
22083	48.7	57.0	56.5	NaN	57.6	61.5	50.6
22093	NaN	NaN	NaN	NaN	NaN	38.3	NaN
22097	38.3	43.2	38.3	40.1	42.2	41.1	36.7
22099	36.3	41.1	30.1	37.4	40.0	33.6	36.5
22101	42.9	45.8	NaN	NaN	38.7	38.4	42.7
22107	54.6	63.8	61.8	54.2	58.0	58.8	46.1
22113	27.1	35.6	26.3	41.8	46.6	40.0	43.8
22117	55.8	NaN	NaN	35.4	50.8	38.0	45.0
22121	NaN	57.4	NaN	NaN	50.5	58.7	52.1
22123	51.4	61.5	60.5	54.3	59.5	57.0	56.0

year	2016	2017	2018	2019	2020	2021	2022
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01001	61.8	48.0	65.4	39.3	47.2	54.3	47.1
01003	41.8	42.0	45.4	NaN	37.2	44.3	37.1
01009	25.0	48.6	39.3	35.3	47.0	42.5	37.4
01015	NaN	NaN	NaN	NaN	35.1	44.7	45.5
01019	20.8	42.1	45.3	30.1	36.3	45.1	39.2
01031	43.6	NaN	NaN	NaN	NaN	31.0	NaN
01033	31.2	48.2	36.1	38.7	42.2	NaN	35.6
01035	33.8	NaN	NaN	NaN	NaN	NaN	NaN
01039	32.6	NaN	NaN	NaN	35.1	NaN	NaN
01043	30.3	55.3	44.6	42.8	51.1	50.1	47.4
01047	32.4	43.2	42.0	26.1	42.7	58.6	41.3
01049	29.2	46.1	49.9	30.6	39.9	44.9	48.8
01051	20.3	39.9	42.8	NaN	NaN	50.4	NaN
01053	42.4	46.2	NaN	22.3	NaN	50.7	NaN
01055	25.6	52.7	NaN	30.4	41.9	34.7	46.4
01057	28.7	25.2	26.4	24.9	38.8	40.6	42.0
01059	39.3	41.9	37.5	37.7	38.8	NaN	NaN
01061	33.8	48.4	NaN	NaN	NaN	NaN	NaN
01065	23.0	NaN	NaN	NaN	NaN	NaN	NaN
01069	23.3	47.2	28.4	46.2	48.7	NaN	42.6
01071	21.5	40.3	39.0	29.1	40.1	42.2	45.5
01075	26.2	27.9	NaN	NaN	NaN	NaN	NaN
01077	29.9	46.3	35.4	38.6	39.6	47.2	42.5
01079	33.0	54.3	39.4	39.3	43.4	51.5	42.9
01083	31.8	50.3	43.3	40.5	36.9	45.4	37.1
01085	29.3	NaN	NaN	NaN	NaN	NaN	NaN
01087	NaN	32.1	NaN	NaN	44.3	30.8	42.5
01089	37.6	42.4	38.2	39.5	42.1	49.6	47.1
01091	30.2	31.4	NaN	NaN	30.0	NaN	NaN
01093	NaN	NaN	NaN	NaN	48.0	NaN	46.2
01095	NaN	43.2	36.1	31.0	42.4	43.8	43.1
01099	38.3	NaN	NaN	NaN	NaN	49.1	NaN
01103	35.3	47.0	NaN	39.5	47.4	44.2	33.4
01105	36.5	39.9	37.6	NaN	37.4	39.1	39.3
01109	40.1	NaN	NaN	51.6	NaN	NaN	33.8
01117	39.7	NaN	NaN	NaN	NaN	NaN	NaN
01119	30.7	40.7	43.1	38.0	NaN	45.3	NaN
01121	NaN	41.9	38.2	NaN	46.7	NaN	46.3
01125	38.5	51.3	46.3	31.8	41.2	53.0	39.2
01127	NaN	NaN	38.5	32.4	NaN	NaN	NaN
04001	30.1	29.8	25.3	23.9	28.8	28.4	26.6
05001	48.4	45.8	43.5	38.1	45.8	47.2	45.2
05003	58.2	56.8	54.9	50.6	55.4	63.7	59.7
05007	NaN	NaN	NaN	NaN	NaN	41.2	NaN
05017	52.7	51.2	52.0	54.1	53.9	55.2	54.1
05021	46.8	54.8	50.8	48.0	53.3	44.0	54.7
05029	NaN	39.4	37.0	NaN	33.2	32.9	32.0
05031	48.8	53.1	50.0	46.3	48.7	NaN	46.5

05035	43.8	52.7	NaN	46.1	50.4	52.3	53.2
05037	47.7	51.1	53.3	52.0	50.4	53.3	51.9
05041	55.7	56.7	59.5	59.8	57.3	52.1	58.8
05043	53.5	54.5	56.9	57.5	56.7	58.6	61.9
05045	37.2	42.0	43.2	NaN	33.6	33.2	39.9
05047	NaN	35.5	NaN	NaN	37.3	NaN	NaN
05055	43.7	51.0	41.5	45.8	46.9	NaN	52.0
05063	38.5	43.4	NaN	40.9	43.1	NaN	42.9
05067	39.2	42.2	40.1	40.3	40.6	46.8	38.4
05069	52.2	53.1	56.5	54.3	55.8	57.0	57.6
05071	NaN	NaN	NaN	NaN	34.7	NaN	NaN
05073	NaN	NaN	NaN	NaN	53.1	NaN	NaN
05075	35.7	42.0	40.4	39.9	42.9	NaN	48.7
05077	43.5	48.5	44.4	NaN	55.6	53.5	57.7
05079	56.2	57.8	55.4	57.2	54.6	53.2	60.8
05081	NaN	NaN	NaN	30.2	NaN	NaN	31.3
05083	NaN	NaN	NaN	NaN	37.5	35.3	NaN
05085	46.3	42.5	NaN	48.2	47.6	47.5	49.6
05093	48.9	56.7	50.5	47.6	53.4	NaN	59.3
05095	43.3	47.2	47.3	47.7	50.9	45.1	44.2
05107	48.9	52.9	50.5	NaN	55.5	58.8	55.4
05111	51.0	55.5	56.4	51.0	51.9	54.2	49.2
05115	NaN	NaN	NaN	NaN	NaN	NaN	43.8
05117	50.0	50.9	53.6	44.0	50.0	55.5	50.6
05119	NaN	46.7	44.3	NaN	35.9	38.8	35.6
05121	37.9	48.2	NaN	42.5	45.3	NaN	NaN
05123	44.5	49.7	46.2	50.3	50.2	51.9	52.0
05131	NaN	NaN	NaN	NaN	33.3	NaN	NaN
05145	37.9	42.6	NaN	43.0	47.9	45.8	36.4
05147	35.7	44.6	45.8	43.8	50.8	48.4	40.1
05149	NaN	42.6	NaN	NaN	42.5	38.2	34.6
06001	32.7	29.4	28.6	28.7	30.5	32.6	30.9
08001	32.8	35.1	29.9	27.0	33.5	31.0	32.6
10001	44.5	43.2	41.4	45.0	44.2	47.6	41.6
10003	47.0	NaN	45.8	47.2	48.4	46.9	36.9
10005	38.3	49.5	41.3	44.8	47.8	49.7	42.5
12001	29.1	26.9	27.7	28.1	29.4	27.9	27.0
13001	31.4	32.2	27.3	31.7	34.2	33.6	31.6
13005	NaN	NaN	NaN	NaN	37.7	36.8	NaN
13015	17.0	52.3	NaN	12.6	NaN	NaN	NaN
13027	49.4	45.7	NaN	36.2	NaN	48.7	47.8
13031	20.7	30.8	NaN	NaN	NaN	NaN	NaN
13033	32.4	34.0	NaN	NaN	51.3	41.1	38.3
13043	21.6	NaN	NaN	NaN	NaN	NaN	NaN
13055	21.4	41.5	37.2	40.4	42.0	47.1	NaN
13069	36.4	36.9	NaN	25.2	NaN	NaN	46.0
13071	38.8	44.8	43.5	NaN	NaN	NaN	NaN
13075	NaN	NaN	NaN	NaN	NaN	48.8	48.4

13087	43.3	NaN	NaN	NaN	NaN	NaN	NaN
13091	29.3	NaN	NaN	NaN	NaN	NaN	NaN
13093	NaN	42.3	NaN	NaN	42.0	NaN	NaN
13099	22.7	NaN	NaN	NaN	NaN	NaN	NaN
13105	NaN	39.4	45.5	24.7	48.7	47.9	31.8
13107	34.4	51.0	29.9	23.8	39.5	36.3	31.4
13115	15.3	53.0	NaN	NaN	NaN	42.7	NaN
13119	9.8	37.3	NaN	NaN	NaN	NaN	NaN
13129	NaN	NaN	NaN	NaN	NaN	NaN	44.1
13131	36.6	40.8	NaN	NaN	51.3	NaN	52.3
13147	16.0	36.4	41.2	23.2	52.3	NaN	NaN
13151	NaN	42.1	NaN	NaN	NaN	NaN	NaN
13153	40.7	33.9	NaN	NaN	NaN	NaN	49.9
13161	NaN	NaN	NaN	NaN	NaN	NaN	47.4
13163	28.7	NaN	38.1	34.0	46.2	36.1	NaN
13165	29.6	53.1	32.9	NaN	NaN	NaN	NaN
13167	21.3	NaN	NaN	NaN	NaN	58.2	NaN
13175	NaN	NaN	NaN	NaN	37.1	40.9	35.8
13177	NaN	NaN	35.6	NaN	NaN	40.2	NaN
13185	37.5	39.8	NaN	NaN	36.7	36.9	NaN
13193	37.1	36.7	NaN	28.4	59.7	47.1	NaN
13195	16.1	NaN	NaN	NaN	30.8	NaN	NaN
13197	16.3	35.3	NaN	NaN	NaN	40.1	41.3
13205	41.2	47.9	41.6	NaN	NaN	NaN	NaN
13209	35.3	45.4	NaN	NaN	NaN	NaN	NaN
13211	36.3	NaN	NaN	NaN	NaN	NaN	NaN
13213	23.2	NaN	NaN	NaN	NaN	NaN	NaN
13221	NaN	NaN	35.3	NaN	NaN	NaN	NaN
13225	36.8	NaN	26.9	NaN	34.8	NaN	38.9
13229	23.9	NaN	NaN	NaN	NaN	NaN	NaN
13233	21.8	NaN	NaN	NaN	NaN	NaN	NaN
13243	34.1	53.5	49.1	NaN	57.1	NaN	41.9
13251	29.7	42.7	NaN	NaN	36.1	NaN	NaN
13253	49.1	50.8	38.7	NaN	NaN	NaN	NaN
13267	24.1	NaN	NaN	NaN	NaN	NaN	NaN
13269	NaN	40.8	NaN	NaN	NaN	NaN	37.4
13273	12.3	24.0	NaN	NaN	NaN	45.2	49.1
13275	NaN	44.7	46.1	NaN	NaN	53.8	52.1
13279	34.0	41.8	NaN	NaN	NaN	42.7	NaN
13295	NaN	47.2	NaN	NaN	NaN	42.3	NaN
13299	20.3	NaN	NaN	NaN	44.4	NaN	NaN
13303	NaN	NaN	NaN	NaN	48.8	NaN	NaN
16001	38.7	34.4	31.6	34.0	35.5	37.5	38.0
17001	55.5	56.1	51.0	48.6	52.8	57.5	53.2
17003	38.4	42.3	NaN	NaN	48.7	58.5	51.3
17005	NaN	54.5	67.0	51.7	59.0	65.0	58.6
17007	62.3	47.4	60.3	50.0	53.0	69.2	62.2
17009	52.0	57.1	60.3	56.2	61.4	59.4	64.4

17011	64.2	59.1	68.5	57.4	64.8	66.4	67.5
17013	NaN	53.6	58.6	46.9	54.1	NaN	57.8
17015	69.0	64.3	68.9	64.7	68.1	68.6	68.9
17017	62.4	62.9	71.9	NaN	63.0	68.4	64.6
17019	67.7	64.3	67.9	52.8	66.1	75.3	69.4
17021	62.9	65.6	73.6	NaN	65.9	71.2	69.5
17023	NaN	56.5	64.1	NaN	58.4	65.5	61.3
17025	52.5	44.3	58.5	40.5	52.4	59.2	57.3
17027	54.1	50.8	60.1	51.9	52.5	63.6	54.6
17029	62.6	64.3	74.9	63.7	66.6	NaN	NaN
17031	NaN	NaN	NaN	NaN	NaN	45.8	NaN
17033	56.4	52.7	63.0	49.5	56.3	65.0	61.8
17035	59.0	54.1	67.2	50.6	56.5	NaN	67.2
17037	67.5	53.6	61.1	55.6	59.0	63.9	64.9
17039	62.1	65.9	75.0	NaN	66.4	70.7	68.0
17041	67.4	63.7	78.1	62.4	67.1	NaN	69.0
17045	65.0	64.7	69.5	59.5	61.0	70.8	67.9
17047	47.7	55.6	44.1	NaN	54.7	63.1	53.5
17049	58.2	55.3	72.1	51.5	57.9	64.7	59.9
17051	55.6	53.5	59.5	48.9	54.4	63.5	NaN
17053	60.5	62.4	60.7	52.6	56.3	66.7	66.7
17055	45.2	46.4	47.5	38.9	51.2	53.3	47.0
17057	56.4	60.1	62.4	53.9	58.4	65.5	62.5
17059	48.4	54.5	50.0	47.5	57.6	62.5	55.3
17061	62.5	63.9	70.1	60.7	65.2	NaN	66.6
17063	58.0	57.3	58.5	49.0	56.2	59.5	63.1
17065	46.1	51.3	51.2	NaN	51.8	NaN	50.2
17067	63.7	66.1	69.8	NaN	66.6	67.3	67.6
17069	39.8	NaN	43.3	44.8	50.8	50.6	NaN
17071	57.8	61.6	60.2	NaN	62.1	66.6	67.2
17073	61.4	58.9	63.2	52.6	62.3	68.1	66.8
17075	59.1	57.1	55.3	45.3	60.0	57.9	63.8
17077	NaN	50.1	48.9	39.6	52.2	49.9	51.1
17079	58.1	54.6	60.7	53.9	54.5	NaN	55.8
17081	48.4	40.9	51.4	37.4	47.6	47.7	48.1
17083	55.3	56.9	NaN	60.1	65.3	NaN	63.2
17085	60.6	53.4	NaN	55.6	59.1	58.7	62.6
17087	NaN	48.0	NaN	NaN	45.8	43.8	45.1
17089	64.5	53.6	58.9	50.3	56.0	67.7	64.4
17091	55.8	52.2	54.2	52.3	54.8	61.7	59.7
17093	NaN	58.9	60.8	57.4	57.5	66.3	NaN
17095	63.6	63.7	64.1	55.9	63.9	68.4	67.0
17097	52.5	NaN	41.3	48.2	51.8	50.3	57.3
17099	61.2	60.8	67.1	55.4	59.3	NaN	65.7
17101	NaN	51.0	54.1	47.8	57.9	60.4	58.0
17103	60.8	59.4	63.7	59.7	64.7	67.6	66.8
17105	60.6	58.8	61.0	56.9	56.0	66.0	63.5
17107	66.6	64.7	70.1	NaN	68.7	NaN	71.8

17109	62.5	64.7	69.3	60.2	65.1	NaN	66.3
17111	59.1	47.2	58.9	NaN	52.5	NaN	NaN
17113	63.8	66.8	67.5	61.0	63.9	67.4	68.8
17115	68.2	66.2	75.2	63.8	66.2	74.4	73.8
17117	63.4	61.3	69.7	58.3	67.4	70.8	65.1
17119	54.7	56.7	62.5	56.6	58.5	61.0	59.6
17121	53.3	45.9	60.5	46.5	54.9	60.1	54.3
17123	63.3	60.1	65.5	53.8	63.1	NaN	64.1
17125	59.2	62.1	NaN	55.4	61.2	64.7	62.6
17127	47.5	NaN	NaN	43.5	51.6	52.5	50.0
17129	63.6	63.9	68.6	62.0	67.3	66.4	67.9
17131	59.9	60.0	63.2	51.9	59.4	NaN	65.0
17133	48.4	43.4	54.1	47.2	55.7	56.3	53.1
17135	61.5	61.9	73.0	61.5	63.2	NaN	68.6
17137	60.4	60.2	79.8	64.1	62.9	NaN	70.0
17139	68.5	66.4	75.9	59.3	65.0	72.4	65.7
17141	65.6	55.0	64.3	59.9	61.9	61.8	67.6
17143	60.0	63.9	66.6	55.4	65.2	69.7	64.0
17145	48.0	42.7	52.5	35.6	48.2	53.2	43.4
17147	71.3	69.4	77.4	NaN	71.7	77.3	74.2
17149	58.1	57.7	61.2	NaN	62.1	NaN	NaN
17151	35.1	44.6	NaN	NaN	53.1	NaN	NaN
17153	44.4	44.5	48.2	43.2	54.8	56.8	53.5
17155	63.3	57.2	NaN	NaN	61.8	62.2	64.6
17157	50.0	44.4	52.1	41.6	53.4	55.3	50.3
17159	51.8	48.8	56.3	NaN	54.0	NaN	56.1
17161	58.0	58.6	60.5	NaN	61.3	64.6	66.3
17163	52.4	50.9	56.0	53.2	52.7	59.6	53.6
17165	46.8	49.9	49.4	NaN	59.0	59.2	52.8
17167	66.2	70.3	80.4	65.5	68.8	72.1	72.7
17169	56.9	58.6	59.9	NaN	58.1	63.2	63.1
17171	57.9	62.9	NaN	NaN	65.1	64.9	72.1
17173	57.5	61.0	70.4	56.2	60.4	67.3	67.1
17175	65.2	64.6	NaN	58.8	63.0	70.1	70.8
17177	66.3	59.2	63.3	56.4	64.1	64.5	62.3
17179	65.0	66.8	72.1	63.8	63.8	70.8	71.0
17181	43.8	51.6	NaN	34.0	50.2	44.5	48.0
17183	64.6	64.9	71.7	NaN	61.8	67.4	67.6
17185	55.2	54.4	NaN	NaN	54.5	60.8	58.1
17187	62.6	63.3	67.5	58.5	66.6	NaN	64.5
17189	48.8	48.3	57.4	43.9	51.8	58.7	50.5
17191	48.3	47.5	52.9	43.4	52.0	NaN	56.4
17193	47.0	52.8	53.0	46.6	55.0	63.4	54.5
17195	60.8	58.7	62.7	NaN	NaN	67.0	66.4
17197	55.2	50.9	57.8	47.8	54.5	58.4	57.2
17199	42.5	47.2	45.6	NaN	52.8	NaN	45.9
17201	58.7	47.2	56.4	55.8	56.7	56.9	62.8
17203	67.0	60.4	64.7	61.4	63.3	68.1	69.1

18001	59.3	54.1	52.9	47.6	55.3	58.3	54.5
18003	54.8	49.9	58.2	52.3	53.8	59.5	54.7
18005	54.8	54.2	55.5	NaN	59.9	NaN	57.3
18007	61.9	59.7	66.5	54.5	64.1	60.5	61.4
18009	63.5	54.7	57.4	53.8	NaN	62.4	56.1
18011	62.2	54.5	61.2	52.4	65.0	62.7	62.4
18013	51.7	49.8	NaN	NaN	NaN	54.4	53.8
18015	65.1	58.5	63.9	58.7	67.9	NaN	NaN
18017	60.0	53.5	61.8	NaN	59.4	NaN	57.1
18019	45.0	47.8	49.8	40.6	54.4	50.6	52.6
18021	NaN	55.6	52.8	48.4	56.4	58.7	58.1
18023	65.6	58.0	65.2	NaN	67.4	61.5	61.0
18025	NaN	NaN	47.9	46.3	53.4	NaN	NaN
18027	55.4	58.3	54.3	53.5	60.9	68.2	58.6
18029	NaN	47.9	54.1	30.9	51.4	47.5	47.4
18031	61.1	59.5	62.9	53.9	68.4	61.6	68.3
18033	49.2	48.2	49.3	52.4	45.6	55.6	54.2
18035	59.4	50.8	57.0	48.8	59.3	60.8	57.9
18037	53.2	63.6	59.0	51.4	61.7	63.6	58.4
18039	54.2	54.9	57.6	48.3	55.7	58.0	58.6
18041	57.8	52.1	57.0	51.5	60.6	58.0	57.3
18043	NaN	50.3	NaN	37.1	54.8	NaN	NaN
18045	59.4	56.5	NaN	NaN	64.4	NaN	NaN
18047	56.7	53.3	59.0	41.6	56.5	53.5	57.0
18049	56.3	55.8	57.5	51.4	54.3	56.3	NaN
18051	55.3	58.5	56.3	55.0	62.1	64.7	62.0
18053	59.3	52.9	59.4	NaN	59.9	62.1	56.3
18055	50.8	46.3	53.2	NaN	55.2	61.4	52.3
18057	60.0	58.5	62.8	48.8	65.2	61.6	59.7
18059	56.4	48.9	57.1	42.3	59.3	57.5	56.2
18061	48.1	50.5	51.6	43.6	59.3	55.0	56.1
18063	61.7	53.7	60.2	53.1	64.4	62.2	59.8
18065	60.8	49.2	56.9	45.8	61.5	61.4	55.8
18067	62.8	54.8	62.5	55.4	66.0	66.4	64.2
18069	57.6	51.4	58.5	54.3	57.9	62.6	57.5
18071	55.0	50.5	51.2	NaN	58.9	NaN	59.3
18073	54.7	52.6	56.9	NaN	55.1	58.2	54.6
18075	58.9	50.0	59.5	49.3	55.7	NaN	55.3
18077	43.8	45.0	48.3	32.1	50.3	49.8	49.4
18079	51.9	52.3	55.0	45.7	57.7	52.9	52.8
18081	NaN	59.1	NaN	53.6	60.9	58.0	62.2
18083	56.6	56.6	52.5	NaN	62.5	67.2	57.4
18085	53.3	51.1	60.6	49.0	52.1	56.8	54.9
18087	54.9	50.2	52.9	47.5	52.7	58.5	56.0
18089	54.8	50.5	55.7	48.3	52.0	NaN	51.8
18091	56.0	50.8	58.3	50.1	49.7	55.0	52.0
18093	47.5	46.9	52.5	NaN	54.7	52.0	50.0
18095	62.0	50.2	60.5	47.6	64.5	65.4	56.7

18097	NaN	44.0	NaN	47.6	NaN	47.0	57.8
18099	54.4	52.0	53.3	NaN	50.0	57.8	NaN
18101	55.3	NaN	NaN	43.8	60.6	65.0	57.6
18103	55.8	48.9	55.0	49.1	54.5	NaN	57.1
18105	52.2	45.8	44.7	NaN	51.3	48.9	55.1
18107	64.0	62.2	63.8	58.5	63.0	65.0	64.5
18109	54.4	49.0	56.9	NaN	61.3	60.0	59.3
18111	61.5	54.4	59.0	52.0	60.8	55.3	59.3
18113	54.0	49.7	51.6	54.3	52.2	57.0	59.1
18115	56.1	52.0	50.0	30.8	52.7	42.2	52.6
18117	55.0	63.2	55.8	56.6	64.9	65.1	63.0
18119	48.9	44.8	NaN	NaN	54.0	55.4	50.9
18121	60.6	55.8	56.5	52.3	61.7	58.9	57.4
18123	NaN	NaN	45.0	47.2	57.0	NaN	51.6
18125	51.0	NaN	NaN	51.8	56.7	NaN	50.4
18127	57.1	53.0	59.7	52.1	50.5	60.5	54.8
18129	53.9	56.6	54.4	56.3	66.5	NaN	60.7
18131	50.1	51.0	54.5	44.3	49.9	NaN	51.0
18133	58.8	55.5	57.4	58.8	61.0	58.8	60.3
18135	58.5	51.6	58.2	50.9	62.5	62.2	55.8
18137	52.1	57.4	51.1	42.2	53.9	52.3	51.2
18139	62.2	56.5	57.9	56.9	63.2	60.4	61.8
18141	53.4	49.4	54.4	47.3	50.4	55.5	53.4
18143	NaN	50.6	49.6	NaN	NaN	49.1	NaN
18145	61.9	57.6	56.0	51.8	61.8	56.8	63.6
18147	52.4	59.0	51.1	53.3	58.5	60.0	NaN
18149	50.4	48.9	53.8	NaN	46.9	NaN	50.6
18151	50.4	48.2	46.9	NaN	44.7	55.9	54.2
18153	52.9	53.7	55.5	50.4	55.6	61.9	54.9
18155	52.1	48.5	47.6	NaN	57.4	49.0	55.2
18157	NaN	56.1	63.2	54.7	66.6	58.8	58.4
18159	66.8	61.3	NaN	56.4	69.8	69.2	62.2
18161	60.0	59.5	59.8	51.0	65.5	60.6	59.5
18163	57.9	62.9	55.5	60.9	69.4	73.3	59.1
18165	62.8	54.2	60.3	51.4	57.1	64.4	NaN
18167	NaN	55.2	51.5	49.3	50.6	58.4	55.7
18169	57.4	49.5	59.2	53.1	51.5	55.7	58.6
18171	64.5	64.9	66.5	60.7	67.5	63.4	64.7
18173	50.2	60.9	49.8	53.6	59.4	NaN	57.2
18175	52.5	51.1	52.3	44.1	56.7	NaN	54.7
18177	58.7	55.4	59.2	50.1	61.7	60.4	55.9
18179	61.8	50.9	61.0	NaN	58.7	66.0	57.4
18181	61.1	58.8	60.4	52.8	59.3	NaN	56.2
18183	58.0	51.7	54.7	53.2	51.3	61.7	54.6
19001	60.8	57.7	52.0	55.0	54.1	58.5	57.3
19003	56.5	52.6	53.2	NaN	NaN	NaN	NaN
19005	56.8	52.6	52.8	NaN	58.8	61.9	61.2
19007	55.9	50.7	45.1	NaN	39.1	57.5	40.4

19009	62.0	57.2	52.3	58.9	48.3	67.3	57.9
19011	61.1	57.6	61.0	57.9	56.7	67.1	66.1
19013	61.8	59.5	58.4	51.9	54.2	60.1	64.5
19015	60.2	56.7	54.6	53.2	50.9	64.4	60.4
19017	62.1	56.8	57.3	52.5	NaN	62.0	66.7
19019	60.2	55.9	57.2	55.2	56.7	63.4	65.0
19021	60.2	56.5	55.5	53.5	NaN	63.8	57.2
19023	59.9	55.9	56.1	55.0	54.2	60.8	61.6
19025	62.3	57.5	53.1	55.1	47.7	63.5	61.2
19027	58.7	60.4	59.0	58.8	54.8	65.3	58.9
19029	59.6	56.9	52.6	56.9	49.3	63.3	51.2
19031	60.5	59.6	63.3	NaN	56.4	65.3	67.0
19033	58.6	55.2	51.7	52.7	59.8	61.0	60.8
19035	69.6	62.4	64.3	58.8	57.2	NaN	59.6
19037	56.0	50.4	53.2	54.5	55.9	NaN	63.6
19039	49.9	NaN	47.9	NaN	45.9	57.7	40.2
19041	61.9	57.6	52.6	50.9	50.3	62.4	56.9
19043	63.2	59.0	58.8	56.4	60.1	NaN	64.5
19045	59.8	59.7	58.3	59.5	57.0	63.9	66.0
19047	64.3	61.3	61.7	61.9	54.2	63.5	62.8
19049	58.7	54.6	52.8	54.4	47.1	63.6	59.8
19051	57.4	37.6	50.2	54.6	42.0	56.7	44.6
19053	57.2	NaN	48.0	NaN	47.4	57.0	37.9
19055	63.8	63.0	63.9	NaN	61.1	66.4	67.4
19057	60.4	59.6	63.8	55.1	60.7	NaN	66.9
19059	56.1	57.7	49.8	51.3	55.5	56.0	58.8
19061	57.8	59.6	62.0	60.2	NaN	70.5	68.6
19063	58.5	58.9	46.7	NaN	56.0	59.6	62.8
19065	62.3	56.3	56.5	55.5	55.9	64.4	63.4
19067	57.5	54.1	49.8	54.3	54.0	NaN	61.9
19069	60.5	57.0	58.5	54.1	55.4	59.7	62.8
19071	55.5	60.1	50.6	NaN	54.8	63.1	53.4
19073	63.5	56.7	55.5	54.2	49.0	64.1	59.0
19075	64.8	63.6	61.9	56.3	60.9	66.5	64.9
19077	58.4	59.2	52.0	54.2	45.5	NaN	58.6
19079	61.0	57.2	49.9	51.4	51.5	62.2	61.3
19081	62.4	57.4	53.2	49.6	60.0	63.5	63.0
19083	59.9	59.7	59.3	53.9	52.8	65.0	63.4
19085	58.2	53.7	52.6	NaN	52.1	NaN	52.4
19087	57.2	60.5	62.3	56.9	54.9	58.3	60.3
19089	56.4	51.6	53.7	52.1	54.5	60.3	60.9
19091	61.0	58.0	50.4	54.9	56.9	62.4	59.0
19093	67.6	62.1	63.8	NaN	55.5	65.6	63.1
19095	57.4	56.7	56.0	52.3	49.8	NaN	61.6
19097	61.1	59.2	58.1	NaN	54.7	NaN	66.3
19099	60.3	60.5	60.7	56.5	NaN	67.3	60.1
19101	55.1	46.1	62.0	53.3	53.8	53.9	50.7
19103	56.1	56.4	59.9	55.5	51.8	NaN	62.1

19105	60.4	54.9	60.1	57.2	54.0	64.5	61.7
19107	57.1	51.6	57.7	55.8	51.6	NaN	53.1
19109	63.4	60.4	54.2	57.4	60.2	60.4	56.2
19111	61.4	53.5	64.7	54.8	52.1	59.7	54.6
19113	60.0	56.0	57.7	56.5	NaN	NaN	64.6
19115	56.4	59.0	58.1	54.3	53.3	NaN	61.4
19117	54.4	41.4	41.5	NaN	43.8	54.1	43.1
19119	66.1	63.8	60.6	54.8	60.8	63.6	62.2
19121	57.1	52.5	50.9	52.3	51.4	58.8	55.4
19123	59.1	54.4	59.7	53.7	NaN	62.0	59.3
19125	56.6	52.9	58.3	54.0	NaN	60.6	57.8
19127	62.6	63.5	61.9	60.6	NaN	67.1	65.0
19129	56.3	55.7	NaN	NaN	49.3	NaN	52.1
19131	61.0	54.4	55.6	49.8	59.4	62.5	64.6
19133	56.1	57.1	55.3	55.8	56.0	61.8	48.6
19135	57.3	51.0	50.4	56.1	45.4	59.3	47.4
19137	57.8	56.2	51.8	58.3	49.2	63.0	54.8
19139	60.8	60.3	61.7	55.2	55.6	62.6	65.5
19141	63.9	62.3	59.9	56.0	62.0	66.4	59.4
19143	56.1	59.7	57.2	50.5	NaN	62.9	56.3
19145	57.5	55.4	50.5	55.9	52.4	62.6	54.0
19147	59.7	57.8	48.8	55.6	NaN	61.6	57.9
19149	66.1	59.6	61.3	NaN	NaN	NaN	47.8
19151	60.6	58.2	49.7	NaN	54.6	65.0	56.5
19153	55.2	57.1	53.4	51.0	50.2	64.6	54.1
19155	60.6	57.0	NaN	58.5	48.0	64.4	51.0
19157	57.5	56.6	58.2	53.4	54.8	NaN	60.4
19159	51.7	48.8	47.2	NaN	50.6	54.0	41.8
19161	63.3	57.8	60.7	59.3	47.1	NaN	61.3
19163	64.4	63.4	63.7	60.5	61.0	68.1	69.1
19165	60.5	58.2	57.8	58.7	52.8	NaN	58.0
19167	65.8	65.2	65.8	60.8	NaN	70.5	62.9
19169	58.4	57.9	52.8	50.0	52.2	63.6	60.0
19171	63.5	59.5	60.1	53.4	56.2	66.3	61.4
19173	51.6	52.8	NaN	49.9	47.0	61.3	52.9
19175	56.0	44.5	48.6	53.7	52.1	61.9	41.4
19177	56.6	36.6	54.2	53.9	44.2	53.0	43.6
19179	58.1	45.0	60.2	55.5	52.0	59.0	53.7
19181	54.3	49.9	52.0	52.3	49.5	NaN	50.2
19183	55.8	61.7	61.5	54.7	53.2	NaN	57.2
19185	54.2	NaN	49.0	52.5	44.6	57.1	NaN
19187	58.4	57.9	51.1	49.1	50.4	64.6	57.5
19189	62.5	59.9	51.8	54.2	60.8	61.6	64.4
19191	58.4	54.6	51.7	55.6	53.6	61.8	66.3
19193	61.2	58.1	59.5	56.8	56.6	NaN	56.4
19195	58.2	54.8	53.7	51.0	58.2	60.1	62.0
19197	58.8	57.9	52.6	54.1	55.9	61.0	60.6
20001	45.4	45.0	43.3	43.1	46.5	46.2	43.4

20003	46.9	47.1	48.0	34.2	37.4	41.8	23.2
20005	55.2	51.9	47.4	51.9	49.4	53.7	44.3
20007	NaN	NaN	NaN	26.7	NaN	28.6	20.2
20009	NaN	31.9	NaN	29.9	43.2	37.7	19.1
20011	NaN	40.8	38.2	35.9	35.0	33.1	17.0
20013	62.5	49.4	48.6	56.5	57.2	64.8	53.3
20015	46.1	35.4	47.6	37.6	34.3	35.9	18.3
20017	47.9	43.7	NaN	NaN	39.7	41.8	33.6
20019	31.4	33.5	33.8	40.3	NaN	18.8	12.2
20021	NaN	NaN	NaN	38.9	39.6	32.2	16.0
20023	NaN	56.2	NaN	NaN	62.1	NaN	62.4
20027	48.9	39.7	46.2	45.3	51.9	NaN	28.5
20029	52.5	37.1	NaN	NaN	44.7	39.8	26.3
20031	46.7	39.2	38.7	33.9	38.2	36.2	27.2
20035	42.4	30.9	NaN	37.5	37.4	26.6	9.5
20037	39.2	NaN	38.7	NaN	32.4	34.2	NaN
20041	47.7	28.8	43.6	41.8	40.9	40.1	28.0
20043	62.3	58.5	53.3	NaN	58.3	64.3	NaN
20045	46.1	46.3	33.4	41.0	43.4	36.1	29.8
20047	NaN	53.4	NaN	NaN	63.9	55.9	36.0
20049	41.9	34.3	43.6	39.9	NaN	33.7	21.0
20053	NaN	NaN	48.1	NaN	41.3	34.3	14.5
20055	68.2	NaN	NaN	56.2	NaN	NaN	NaN
20057	NaN	NaN	NaN	NaN	61.6	NaN	NaN
20059	49.9	41.7	43.4	NaN	40.1	36.9	23.5
20061	55.5	41.8	44.6	46.3	50.2	40.6	41.1
20063	NaN	NaN	50.9	NaN	NaN	NaN	NaN
20065	NaN	50.2	54.5	NaN	NaN	NaN	NaN
20067	NaN	NaN	NaN	NaN	NaN	53.1	NaN
20069	68.1	62.8	NaN	53.4	NaN	59.7	NaN
20073	NaN	36.4	34.8	NaN	32.0	30.8	19.6
20077	NaN	22.0	29.4	31.0	32.4	24.0	9.4
20079	48.7	30.8	48.8	37.3	35.5	42.4	25.9
20081	64.5	59.7	64.5	65.0	63.8	62.6	30.5
20083	NaN	55.7	57.1	NaN	NaN	NaN	49.2
20085	47.7	42.9	39.7	46.2	47.1	45.0	NaN
20087	49.2	52.8	NaN	49.2	46.2	52.0	36.7
20089	55.5	39.3	42.9	49.4	47.0	41.2	34.2
20091	NaN	NaN	34.1	NaN	NaN	38.8	32.8
20095	45.9	26.7	46.0	26.7	34.5	35.4	22.5
20097	57.8	NaN	63.3	NaN	53.3	50.0	41.0
20099	NaN	39.4	33.7	NaN	36.7	24.4	9.3
20103	NaN	NaN	NaN	46.0	47.3	42.0	38.2
20105	NaN	NaN	NaN	NaN	42.2	NaN	12.3
20107	41.8	NaN	NaN	NaN	40.0	33.5	20.9
20111	41.7	38.6	NaN	34.0	35.9	36.1	23.2
20113	45.1	29.3	45.5	35.1	30.2	34.4	16.9
20115	52.0	25.7	42.4	35.1	29.7	38.3	17.9

20117	45.3	28.3	38.6	48.0	42.7	47.9	38.8
20119	NaN	70.9	NaN	NaN	73.1	69.1	67.6
20121	47.8	49.7	42.2	43.3	48.9	46.1	29.5
20123	53.4	NaN	43.0	43.3	44.0	29.1	24.7
20125	35.9	37.0	39.6	35.6	40.7	27.9	10.7
20127	42.8	38.8	32.8	44.9	40.2	36.7	32.7
20131	56.1	33.3	41.5	NaN	45.7	57.6	39.8
20133	43.6	35.7	38.9	NaN	31.4	29.5	11.0
20137	NaN	NaN	47.8	NaN	32.5	33.4	NaN
20139	45.1	42.8	35.5	41.0	37.5	39.5	25.1
20141	50.3	27.5	NaN	NaN	38.5	NaN	13.6
20143	45.0	29.9	NaN	NaN	43.9	NaN	20.5
20145	49.5	43.1	NaN	NaN	56.7	NaN	44.3
20147	54.0	NaN	NaN	43.8	27.4	38.7	25.7
20149	54.5	48.1	40.0	49.5	45.8	40.7	NaN
20151	57.4	51.8	54.0	NaN	54.7	44.9	37.8
20153	NaN	NaN	NaN	NaN	NaN	36.5	NaN
20155	45.9	23.1	NaN	32.8	38.0	NaN	21.4
20157	55.7	44.8	51.0	45.1	50.5	54.6	43.3
20159	43.3	29.3	NaN	NaN	35.7	37.6	16.4
20161	50.9	37.0	38.7	46.1	48.4	47.8	35.0
20163	NaN	25.5	NaN	NaN	29.7	30.3	13.9
20165	44.1	NaN	37.3	NaN	NaN	NaN	NaN
20167	41.7	25.2	NaN	30.3	41.9	33.5	NaN
20169	42.4	22.6	NaN	43.0	39.4	34.4	19.7
20171	NaN	NaN	51.5	NaN	NaN	NaN	NaN
20173	45.9	32.0	43.5	36.3	35.3	32.2	25.2
20175	59.3	72.5	65.8	64.4	60.6	58.9	57.6
20177	48.9	39.0	37.7	NaN	41.5	42.1	35.2
20179	NaN	NaN	NaN	NaN	59.5	59.1	NaN
20181	NaN	44.4	NaN	NaN	NaN	NaN	NaN
20183	NaN	39.6	45.8	NaN	37.5	43.8	24.7
20185	48.7	NaN	NaN	NaN	43.8	NaN	NaN
20189	61.3	58.1	NaN	NaN	62.0	57.9	64.1
20191	39.6	26.5	35.7	31.0	30.4	21.4	12.5
20193	NaN	NaN	NaN	NaN	45.6	NaN	40.2
20195	NaN	NaN	NaN	34.8	NaN	NaN	NaN
20197	NaN	36.8	31.0	NaN	41.9	39.8	34.2
20201	43.4	33.2	NaN	NaN	45.2	48.8	42.0
20203	NaN	NaN	52.5	NaN	NaN	NaN	NaN
20205	44.1	37.7	NaN	39.4	29.2	32.3	15.8
20207	NaN	36.0	NaN	NaN	33.1	31.7	NaN
21001	43.9	44.5	38.2	36.0	42.5	44.2	40.6
21003	NaN	NaN	NaN	NaN	54.2	41.3	NaN
21005	NaN	NaN	NaN	NaN	NaN	53.4	NaN
21007	50.1	49.4	NaN	51.5	56.9	NaN	49.2
21009	53.3	56.7	NaN	44.0	60.9	54.5	52.8
21011	46.4	NaN	NaN	48.1	43.4	47.3	52.1

21015	52.3	60.3	55.9	53.4	53.1	51.3	54.8
21017	47.5	50.9	47.4	NaN	50.4	61.6	NaN
21021	54.3	59.7	56.0	40.0	56.1	58.8	57.9
21023	45.4	51.3	51.9	NaN	NaN	51.3	NaN
21027	53.2	53.9	49.0	36.0	55.7	50.8	53.6
21029	47.8	51.8	52.4	30.1	52.6	47.5	NaN
21031	40.8	50.5	50.8	51.0	54.0	57.8	55.2
21033	50.6	50.3	48.1	46.5	58.3	56.6	41.2
21035	41.0	43.7	43.0	NaN	52.4	57.7	42.0
21039	48.1	NaN	51.2	NaN	56.0	54.2	45.3
21041	39.6	NaN	NaN	NaN	50.2	47.6	53.2
21045	NaN	54.0	NaN	47.0	55.1	56.8	51.0
21047	48.1	54.4	51.6	44.5	54.9	55.3	44.0
21049	47.9	58.1	49.3	58.0	55.0	59.6	54.4
21051	NaN	NaN	NaN	NaN	49.2	46.1	44.8
21053	49.4	NaN	NaN	NaN	54.8	NaN	53.8
21055	50.0	49.7	51.5	NaN	54.1	55.1	43.4
21057	NaN	NaN	46.3	53.9	59.9	NaN	50.5
21059	54.2	60.3	56.2	54.5	60.0	62.7	59.2
21061	46.9	49.9	52.2	48.0	54.9	50.1	53.8
21065	42.9	42.5	35.2	35.0	50.3	NaN	NaN
21067	50.6	54.1	52.9	44.5	51.4	53.4	50.0
21069	45.7	44.9	NaN	40.5	48.5	49.8	50.5
21073	NaN	NaN	NaN	NaN	48.2	51.9	NaN
21075	49.4	52.1	NaN	48.0	54.7	57.1	45.5
21077	42.0	52.0	NaN	NaN	NaN	NaN	NaN
21079	NaN	49.0	NaN	NaN	NaN	56.7	NaN
21081	NaN	NaN	NaN	49.5	51.7	48.2	50.0
21083	50.2	51.4	45.4	NaN	52.0	53.1	42.8
21085	51.8	51.7	50.4	NaN	50.5	53.7	51.3
21087	51.5	55.5	55.0	NaN	55.8	56.2	59.7
21089	46.5	NaN	46.7	NaN	42.3	41.9	48.0
21091	50.6	59.0	NaN	54.0	58.6	66.5	NaN
21093	53.4	52.4	52.9	36.5	54.3	54.0	52.7
21097	46.9	NaN	NaN	NaN	50.7	NaN	NaN
21099	54.8	51.3	55.7	46.0	54.5	55.9	53.2
21101	NaN	58.8	52.0	53.0	59.0	NaN	58.8
21103	40.0	45.0	48.5	42.5	51.3	55.6	NaN
21105	47.4	53.6	50.8	52.0	53.8	56.7	46.6
21107	NaN	NaN	NaN	NaN	53.2	NaN	NaN
21111	NaN	NaN	NaN	25.0	50.7	56.6	NaN
21113	42.5	47.6	51.8	31.0	45.9	48.7	47.1
21117	NaN	NaN	NaN	NaN	NaN	51.3	NaN
21123	NaN	58.0	NaN	47.0	58.5	57.6	56.9
21135	50.2	NaN	NaN	44.0	46.7	45.8	50.7
21137	53.4	NaN	49.9	37.0	54.1	56.9	51.6
21139	48.4	46.0	NaN	47.0	54.0	49.3	41.3
21141	48.7	49.1	47.8	43.0	52.5	50.8	45.3

21143	NaN	NaN	46.0	43.0	51.6	NaN	45.0
21145	NaN	NaN	NaN	NaN	51.4	49.9	NaN
21149	NaN	55.2	58.3	55.0	58.1	60.4	54.7
21151	47.5	53.9	48.6	30.0	54.2	57.9	NaN
21155	52.9	59.2	57.2	38.5	59.6	58.5	52.7
21157	NaN	46.7	43.0	34.0	52.2	52.8	NaN
21161	53.8	55.0	56.4	44.0	54.4	55.6	57.6
21163	56.5	58.8	55.0	43.0	59.9	58.4	54.8
21167	54.8	54.0	NaN	42.0	NaN	NaN	59.7
21169	53.0	54.0	NaN	NaN	53.2	58.9	58.3
21171	NaN	NaN	49.9	NaN	NaN	NaN	57.8
21173	NaN	NaN	NaN	NaN	44.9	NaN	NaN
21177	NaN	NaN	47.3	NaN	51.0	54.5	50.4
21179	52.1	50.3	NaN	33.5	50.3	53.4	NaN
21181	44.7	49.3	NaN	36.0	NaN	NaN	49.0
21183	NaN	56.4	58.5	53.0	53.6	57.6	60.7
21185	45.5	46.4	46.1	39.5	51.7	50.0	NaN
21187	NaN	NaN	34.7	NaN	53.1	53.4	NaN
21191	46.0	60.9	46.8	NaN	46.1	NaN	50.5
21197	NaN	NaN	NaN	NaN	NaN	NaN	46.7
21199	54.9	58.9	45.1	43.0	49.1	53.0	51.9
21203	45.4	NaN	NaN	44.0	46.2	43.4	NaN
21205	36.7	46.8	NaN	NaN	NaN	NaN	NaN
21207	50.8	51.3	50.4	47.5	52.0	51.4	58.6
21209	50.8	57.5	NaN	41.0	50.0	NaN	56.1
21211	48.2	53.3	56.8	40.5	56.9	55.2	56.7
21213	51.5	53.3	51.2	45.5	54.5	51.4	46.8
21215	53.7	50.3	57.9	38.5	53.5	56.8	52.6
21217	48.7	54.8	45.8	48.0	54.2	55.8	55.9
21219	52.1	50.7	49.5	43.5	51.9	54.8	45.5
21221	46.2	51.5	46.7	41.0	51.1	55.8	40.6
21223	43.8	46.4	45.1	35.0	52.6	50.9	46.6
21225	53.9	60.4	57.4	56.5	63.9	67.0	60.6
21227	51.5	53.3	53.0	47.5	56.8	53.0	56.9
21229	54.0	52.7	NaN	30.5	52.7	56.5	53.2
21231	52.2	60.4	48.1	48.0	52.9	54.2	55.7
21233	50.5	53.4	54.1	51.5	60.1	60.4	57.4
21239	50.9	54.6	56.9	50.5	56.4	57.2	55.7
22001	37.6	36.9	37.7	39.2	37.3	34.8	36.5
22003	25.9	NaN	NaN	NaN	NaN	NaN	NaN
22007	46.2	39.7	NaN	38.1	NaN	42.6	50.0
22009	46.0	49.0	44.2	40.8	46.4	43.4	48.3
22011	NaN	37.1	NaN	NaN	NaN	NaN	NaN
22015	29.7	56.0	NaN	NaN	NaN	37.6	NaN
22017	37.9	48.9	40.4	42.3	46.9	33.8	33.5
22019	NaN	34.1	NaN	NaN	NaN	NaN	NaN
22025	46.8	51.4	44.3	45.2	52.6	49.1	44.2
22029	55.3	55.3	54.7	55.1	61.8	51.9	48.3

22035	59.8	65.6	60.6	60.5	65.0	61.9	57.4
22039	29.5	38.2	29.3	31.9	34.9	39.0	32.4
22041	57.9	59.0	61.7	54.6	50.3	56.1	54.7
22043	45.4	NaN	39.8	38.4	38.7	46.4	49.3
22045	34.4	42.3	32.0	38.2	36.5	42.4	40.4
22047	51.9	58.0	57.6	47.3	51.5	NaN	NaN
22053	24.1	31.3	36.9	NaN	NaN	NaN	27.9
22055	33.6	37.6	NaN	NaN	NaN	NaN	NaN
22065	51.8	56.8	59.4	50.1	55.0	57.4	48.5
22067	54.0	61.5	54.1	NaN	53.6	60.0	50.7
22069	37.6	47.0	37.7	42.0	37.7	40.4	32.0
22073	38.8	56.9	NaN	NaN	44.5	NaN	35.1
22077	48.6	54.0	52.1	51.3	54.1	47.3	43.1
22079	49.7	49.4	47.8	41.7	40.4	44.2	39.9
22081	NaN	NaN	36.2	36.6	39.6	NaN	NaN
22083	48.7	57.0	56.5	NaN	57.6	61.5	50.6
22093	NaN	NaN	NaN	NaN	NaN	38.3	NaN
22097	38.3	43.2	38.3	40.1	42.2	41.1	36.7
22099	36.3	41.1	30.1	37.4	40.0	33.6	36.5
22101	42.9	45.8	NaN	NaN	38.7	38.4	42.7
22107	54.6	63.8	61.8	54.2	58.0	58.8	46.1
22113	27.1	35.6	26.3	41.8	46.6	40.0	43.8
22117	55.8	NaN	NaN	35.4	50.8	38.0	45.0
22121	NaN	57.4	NaN	NaN	50.5	58.7	52.1
22123	51.4	61.5	60.5	54.3	59.5	57.0	56.0

WARNING: Counties missing yield for at least one year:

year	2016	2017	2018	2019	2020	2021	2022
fips							
01003	41.8	42.0	45.4	NaN	37.2	44.3	37.1
01015	NaN	NaN	NaN	NaN	35.1	44.7	45.5
01031	43.6	NaN	NaN	NaN	NaN	31.0	NaN
01033	31.2	48.2	36.1	38.7	42.2	NaN	35.6
01035	33.8	NaN	NaN	NaN	NaN	NaN	NaN
01039	32.6	NaN	NaN	NaN	35.1	NaN	NaN
01051	20.3	39.9	42.8	NaN	NaN	50.4	NaN
01053	42.4	46.2	NaN	22.3	NaN	50.7	NaN
01055	25.6	52.7	NaN	30.4	41.9	34.7	46.4
01059	39.3	41.9	37.5	37.7	38.8	NaN	NaN
01061	33.8	48.4	NaN	NaN	NaN	NaN	NaN
01065	23.0	NaN	NaN	NaN	NaN	NaN	NaN
01069	23.3	47.2	28.4	46.2	48.7	NaN	42.6
01075	26.2	27.9	NaN	NaN	NaN	NaN	NaN
01085	29.3	NaN	NaN	NaN	NaN	NaN	NaN
01087	NaN	32.1	NaN	NaN	44.3	30.8	42.5
01091	30.2	31.4	NaN	NaN	30.0	NaN	NaN
01093	NaN	NaN	NaN	NaN	48.0	NaN	46.2
01095	NaN	43.2	36.1	31.0	42.4	43.8	43.1

01099	38.3	NaN	NaN	NaN	NaN	49.1	NaN
01103	35.3	47.0	NaN	39.5	47.4	44.2	33.4
01105	36.5	39.9	37.6	NaN	37.4	39.1	39.3
01109	40.1	NaN	NaN	51.6	NaN	NaN	33.8
01117	39.7	NaN	NaN	NaN	NaN	NaN	NaN
01119	30.7	40.7	43.1	38.0	NaN	45.3	NaN
01121	NaN	41.9	38.2	NaN	46.7	NaN	46.3
01127	NaN	NaN	38.5	32.4	NaN	NaN	NaN
05007	NaN	NaN	NaN	NaN	NaN	41.2	NaN
05029	NaN	39.4	37.0	NaN	33.2	32.9	32.0
05031	48.8	53.1	50.0	46.3	48.7	NaN	46.5
05035	43.8	52.7	NaN	46.1	50.4	52.3	53.2
05045	37.2	42.0	43.2	NaN	33.6	33.2	39.9
05047	NaN	35.5	NaN	NaN	37.3	NaN	NaN
05055	43.7	51.0	41.5	45.8	46.9	NaN	52.0
05063	38.5	43.4	NaN	40.9	43.1	NaN	42.9
05071	NaN	NaN	NaN	NaN	34.7	NaN	NaN
05073	NaN	NaN	NaN	NaN	53.1	NaN	NaN
05075	35.7	42.0	40.4	39.9	42.9	NaN	48.7
05077	43.5	48.5	44.4	NaN	55.6	53.5	57.7
05081	NaN	NaN	NaN	30.2	NaN	NaN	31.3
05083	NaN	NaN	NaN	NaN	37.5	35.3	NaN
05085	46.3	42.5	NaN	48.2	47.6	47.5	49.6
05093	48.9	56.7	50.5	47.6	53.4	NaN	59.3
05107	48.9	52.9	50.5	NaN	55.5	58.8	55.4
05115	NaN	NaN	NaN	NaN	NaN	NaN	43.8
05119	NaN	46.7	44.3	NaN	35.9	38.8	35.6
05121	37.9	48.2	NaN	42.5	45.3	NaN	NaN
05131	NaN	NaN	NaN	NaN	33.3	NaN	NaN
05145	37.9	42.6	NaN	43.0	47.9	45.8	36.4
05149	NaN	42.6	NaN	NaN	42.5	38.2	34.6
10003	47.0	NaN	45.8	47.2	48.4	46.9	36.9
13005	NaN	NaN	NaN	NaN	37.7	36.8	NaN
13015	17.0	52.3	NaN	12.6	NaN	NaN	NaN
13027	49.4	45.7	NaN	36.2	NaN	48.7	47.8
13031	20.7	30.8	NaN	NaN	NaN	NaN	NaN
13033	32.4	34.0	NaN	NaN	51.3	41.1	38.3
13043	21.6	NaN	NaN	NaN	NaN	NaN	NaN
13055	21.4	41.5	37.2	40.4	42.0	47.1	NaN
13069	36.4	36.9	NaN	25.2	NaN	NaN	46.0
13071	38.8	44.8	43.5	NaN	NaN	NaN	NaN
13075	NaN	NaN	NaN	NaN	NaN	48.8	48.4
13087	43.3	NaN	NaN	NaN	NaN	NaN	NaN
13091	29.3	NaN	NaN	NaN	NaN	NaN	NaN
13093	NaN	42.3	NaN	NaN	42.0	NaN	NaN
13099	22.7	NaN	NaN	NaN	NaN	NaN	NaN
13105	NaN	39.4	45.5	24.7	48.7	47.9	31.8
13115	15.3	53.0	NaN	NaN	NaN	42.7	NaN

13119	9.8	37.3	NaN	NaN	NaN	NaN	NaN
13129	NaN	NaN	NaN	NaN	NaN	NaN	44.1
13131	36.6	40.8	NaN	NaN	51.3	NaN	52.3
13147	16.0	36.4	41.2	23.2	52.3	NaN	NaN
13151	NaN	42.1	NaN	NaN	NaN	NaN	NaN
13153	40.7	33.9	NaN	NaN	NaN	NaN	49.9
13161	NaN	NaN	NaN	NaN	NaN	NaN	47.4
13163	28.7	NaN	38.1	34.0	46.2	36.1	NaN
13165	29.6	53.1	32.9	NaN	NaN	NaN	NaN
13167	21.3	NaN	NaN	NaN	NaN	58.2	NaN
13175	NaN	NaN	NaN	NaN	37.1	40.9	35.8
13177	NaN	NaN	35.6	NaN	NaN	40.2	NaN
13185	37.5	39.8	NaN	NaN	36.7	36.9	NaN
13193	37.1	36.7	NaN	28.4	59.7	47.1	NaN
13195	16.1	NaN	NaN	NaN	30.8	NaN	NaN
13197	16.3	35.3	NaN	NaN	NaN	40.1	41.3
13205	41.2	47.9	41.6	NaN	NaN	NaN	NaN
13209	35.3	45.4	NaN	NaN	NaN	NaN	NaN
13211	36.3	NaN	NaN	NaN	NaN	NaN	NaN
13213	23.2	NaN	NaN	NaN	NaN	NaN	NaN
13221	NaN	NaN	35.3	NaN	NaN	NaN	NaN
13225	36.8	NaN	26.9	NaN	34.8	NaN	38.9
13229	23.9	NaN	NaN	NaN	NaN	NaN	NaN
13233	21.8	NaN	NaN	NaN	NaN	NaN	NaN
13243	34.1	53.5	49.1	NaN	57.1	NaN	41.9
13251	29.7	42.7	NaN	NaN	36.1	NaN	NaN
13253	49.1	50.8	38.7	NaN	NaN	NaN	NaN
13267	24.1	NaN	NaN	NaN	NaN	NaN	NaN
13269	NaN	40.8	NaN	NaN	NaN	NaN	37.4
13273	12.3	24.0	NaN	NaN	NaN	45.2	49.1
13275	NaN	44.7	46.1	NaN	NaN	53.8	52.1
13279	34.0	41.8	NaN	NaN	NaN	42.7	NaN
13295	NaN	47.2	NaN	NaN	NaN	42.3	NaN
13299	20.3	NaN	NaN	NaN	44.4	NaN	NaN
13303	NaN	NaN	NaN	NaN	48.8	NaN	NaN
17003	38.4	42.3	NaN	NaN	48.7	58.5	51.3
17005	NaN	54.5	67.0	51.7	59.0	65.0	58.6
17013	NaN	53.6	58.6	46.9	54.1	NaN	57.8
17017	62.4	62.9	71.9	NaN	63.0	68.4	64.6
17021	62.9	65.6	73.6	NaN	65.9	71.2	69.5
17023	NaN	56.5	64.1	NaN	58.4	65.5	61.3
17029	62.6	64.3	74.9	63.7	66.6	NaN	NaN
17031	NaN	NaN	NaN	NaN	NaN	45.8	NaN
17035	59.0	54.1	67.2	50.6	56.5	NaN	67.2
17039	62.1	65.9	75.0	NaN	66.4	70.7	68.0
17041	67.4	63.7	78.1	62.4	67.1	NaN	69.0
17047	47.7	55.6	44.1	NaN	54.7	63.1	53.5
17051	55.6	53.5	59.5	48.9	54.4	63.5	NaN

17061	62.5	63.9	70.1	60.7	65.2	NaN	66.6
17065	46.1	51.3	51.2	NaN	51.8	NaN	50.2
17067	63.7	66.1	69.8	NaN	66.6	67.3	67.6
17069	39.8	NaN	43.3	44.8	50.8	50.6	NaN
17071	57.8	61.6	60.2	NaN	62.1	66.6	67.2
17077	NaN	50.1	48.9	39.6	52.2	49.9	51.1
17079	58.1	54.6	60.7	53.9	54.5	NaN	55.8
17083	55.3	56.9	NaN	60.1	65.3	NaN	63.2
17085	60.6	53.4	NaN	55.6	59.1	58.7	62.6
17087	NaN	48.0	NaN	NaN	45.8	43.8	45.1
17093	NaN	58.9	60.8	57.4	57.5	66.3	NaN
17097	52.5	NaN	41.3	48.2	51.8	50.3	57.3
17099	61.2	60.8	67.1	55.4	59.3	NaN	65.7
17101	NaN	51.0	54.1	47.8	57.9	60.4	58.0
17107	66.6	64.7	70.1	NaN	68.7	NaN	71.8
17109	62.5	64.7	69.3	60.2	65.1	NaN	66.3
17111	59.1	47.2	58.9	NaN	52.5	NaN	NaN
17123	63.3	60.1	65.5	53.8	63.1	NaN	64.1
17125	59.2	62.1	NaN	55.4	61.2	64.7	62.6
17127	47.5	NaN	NaN	43.5	51.6	52.5	50.0
17131	59.9	60.0	63.2	51.9	59.4	NaN	65.0
17135	61.5	61.9	73.0	61.5	63.2	NaN	68.6
17137	60.4	60.2	79.8	64.1	62.9	NaN	70.0
17147	71.3	69.4	77.4	NaN	71.7	77.3	74.2
17149	58.1	57.7	61.2	NaN	62.1	NaN	NaN
17151	35.1	44.6	NaN	NaN	53.1	NaN	NaN
17155	63.3	57.2	NaN	NaN	61.8	62.2	64.6
17159	51.8	48.8	56.3	NaN	54.0	NaN	56.1
17161	58.0	58.6	60.5	NaN	61.3	64.6	66.3
17165	46.8	49.9	49.4	NaN	59.0	59.2	52.8
17169	56.9	58.6	59.9	NaN	58.1	63.2	63.1
17171	57.9	62.9	NaN	NaN	65.1	64.9	72.1
17175	65.2	64.6	NaN	58.8	63.0	70.1	70.8
17181	43.8	51.6	NaN	34.0	50.2	44.5	48.0
17183	64.6	64.9	71.7	NaN	61.8	67.4	67.6
17185	55.2	54.4	NaN	NaN	54.5	60.8	58.1
17187	62.6	63.3	67.5	58.5	66.6	NaN	64.5
17191	48.3	47.5	52.9	43.4	52.0	NaN	56.4
17195	60.8	58.7	62.7	NaN	NaN	67.0	66.4
17199	42.5	47.2	45.6	NaN	52.8	NaN	45.9
18005	54.8	54.2	55.5	NaN	59.9	NaN	57.3
18009	63.5	54.7	57.4	53.8	NaN	62.4	56.1
18013	51.7	49.8	NaN	NaN	NaN	54.4	53.8
18015	65.1	58.5	63.9	58.7	67.9	NaN	NaN
18017	60.0	53.5	61.8	NaN	59.4	NaN	57.1
18021	NaN	55.6	52.8	48.4	56.4	58.7	58.1
18023	65.6	58.0	65.2	NaN	67.4	61.5	61.0
18025	NaN	NaN	47.9	46.3	53.4	NaN	NaN

18029	NaN	47.9	54.1	30.9	51.4	47.5	47.4
18043	NaN	50.3	NaN	37.1	54.8	NaN	NaN
18045	59.4	56.5	NaN	NaN	64.4	NaN	NaN
18049	56.3	55.8	57.5	51.4	54.3	56.3	NaN
18053	59.3	52.9	59.4	NaN	59.9	62.1	56.3
18055	50.8	46.3	53.2	NaN	55.2	61.4	52.3
18071	55.0	50.5	51.2	NaN	58.9	NaN	59.3
18073	54.7	52.6	56.9	NaN	55.1	58.2	54.6
18075	58.9	50.0	59.5	49.3	55.7	NaN	55.3
18081	NaN	59.1	NaN	53.6	60.9	58.0	62.2
18083	56.6	56.6	52.5	NaN	62.5	67.2	57.4
18089	54.8	50.5	55.7	48.3	52.0	NaN	51.8
18093	47.5	46.9	52.5	NaN	54.7	52.0	50.0
18097	NaN	44.0	NaN	47.6	NaN	47.0	57.8
18099	54.4	52.0	53.3	NaN	50.0	57.8	NaN
18101	55.3	NaN	NaN	43.8	60.6	65.0	57.6
18103	55.8	48.9	55.0	49.1	54.5	NaN	57.1
18105	52.2	45.8	44.7	NaN	51.3	48.9	55.1
18109	54.4	49.0	56.9	NaN	61.3	60.0	59.3
18119	48.9	44.8	NaN	NaN	54.0	55.4	50.9
18123	NaN	NaN	45.0	47.2	57.0	NaN	51.6
18125	51.0	NaN	NaN	51.8	56.7	NaN	50.4
18129	53.9	56.6	54.4	56.3	66.5	NaN	60.7
18131	50.1	51.0	54.5	44.3	49.9	NaN	51.0
18143	NaN	50.6	49.6	NaN	NaN	49.1	NaN
18147	52.4	59.0	51.1	53.3	58.5	60.0	NaN
18149	50.4	48.9	53.8	NaN	46.9	NaN	50.6
18151	50.4	48.2	46.9	NaN	44.7	55.9	54.2
18155	52.1	48.5	47.6	NaN	57.4	49.0	55.2
18157	NaN	56.1	63.2	54.7	66.6	58.8	58.4
18159	66.8	61.3	NaN	56.4	69.8	69.2	62.2
18165	62.8	54.2	60.3	51.4	57.1	64.4	NaN
18167	NaN	55.2	51.5	49.3	50.6	58.4	55.7
18173	50.2	60.9	49.8	53.6	59.4	NaN	57.2
18175	52.5	51.1	52.3	44.1	56.7	NaN	54.7
18179	61.8	50.9	61.0	NaN	58.7	66.0	57.4
18181	61.1	58.8	60.4	52.8	59.3	NaN	56.2
19003	56.5	52.6	53.2	NaN	NaN	NaN	NaN
19005	56.8	52.6	52.8	NaN	58.8	61.9	61.2
19007	55.9	50.7	45.1	NaN	39.1	57.5	40.4
19017	62.1	56.8	57.3	52.5	NaN	62.0	66.7
19021	60.2	56.5	55.5	53.5	NaN	63.8	57.2
19031	60.5	59.6	63.3	NaN	56.4	65.3	67.0
19035	69.6	62.4	64.3	58.8	57.2	NaN	59.6
19037	56.0	50.4	53.2	54.5	55.9	NaN	63.6
19039	49.9	NaN	47.9	NaN	45.9	57.7	40.2
19043	63.2	59.0	58.8	56.4	60.1	NaN	64.5
19053	57.2	NaN	48.0	NaN	47.4	57.0	37.9

19055	63.8	63.0	63.9	NaN	61.1	66.4	67.4
19057	60.4	59.6	63.8	55.1	60.7	NaN	66.9
19061	57.8	59.6	62.0	60.2	NaN	70.5	68.6
19063	58.5	58.9	46.7	NaN	56.0	59.6	62.8
19067	57.5	54.1	49.8	54.3	54.0	NaN	61.9
19071	55.5	60.1	50.6	NaN	54.8	63.1	53.4
19077	58.4	59.2	52.0	54.2	45.5	NaN	58.6
19085	58.2	53.7	52.6	NaN	52.1	NaN	52.4
19093	67.6	62.1	63.8	NaN	55.5	65.6	63.1
19095	57.4	56.7	56.0	52.3	49.8	NaN	61.6
19097	61.1	59.2	58.1	NaN	54.7	NaN	66.3
19099	60.3	60.5	60.7	56.5	NaN	67.3	60.1
19103	56.1	56.4	59.9	55.5	51.8	NaN	62.1
19107	57.1	51.6	57.7	55.8	51.6	NaN	53.1
19113	60.0	56.0	57.7	56.5	NaN	NaN	64.6
19115	56.4	59.0	58.1	54.3	53.3	NaN	61.4
19117	54.4	41.4	41.5	NaN	43.8	54.1	43.1
19123	59.1	54.4	59.7	53.7	NaN	62.0	59.3
19125	56.6	52.9	58.3	54.0	NaN	60.6	57.8
19127	62.6	63.5	61.9	60.6	NaN	67.1	65.0
19129	56.3	55.7	NaN	NaN	49.3	NaN	52.1
19143	56.1	59.7	57.2	50.5	NaN	62.9	56.3
19147	59.7	57.8	48.8	55.6	NaN	61.6	57.9
19149	66.1	59.6	61.3	NaN	NaN	NaN	47.8
19151	60.6	58.2	49.7	NaN	54.6	65.0	56.5
19155	60.6	57.0	NaN	58.5	48.0	64.4	51.0
19157	57.5	56.6	58.2	53.4	54.8	NaN	60.4
19159	51.7	48.8	47.2	NaN	50.6	54.0	41.8
19161	63.3	57.8	60.7	59.3	47.1	NaN	61.3
19165	60.5	58.2	57.8	58.7	52.8	NaN	58.0
19167	65.8	65.2	65.8	60.8	NaN	70.5	62.9
19173	51.6	52.8	NaN	49.9	47.0	61.3	52.9
19181	54.3	49.9	52.0	52.3	49.5	NaN	50.2
19183	55.8	61.7	61.5	54.7	53.2	NaN	57.2
19185	54.2	NaN	49.0	52.5	44.6	57.1	NaN
19193	61.2	58.1	59.5	56.8	56.6	NaN	56.4
20007	NaN	NaN	NaN	26.7	NaN	28.6	20.2
20009	NaN	31.9	NaN	29.9	43.2	37.7	19.1
20011	NaN	40.8	38.2	35.9	35.0	33.1	17.0
20017	47.9	43.7	NaN	NaN	39.7	41.8	33.6
20019	31.4	33.5	33.8	40.3	NaN	18.8	12.2
20021	NaN	NaN	NaN	38.9	39.6	32.2	16.0
20023	NaN	56.2	NaN	NaN	62.1	NaN	62.4
20027	48.9	39.7	46.2	45.3	51.9	NaN	28.5
20029	52.5	37.1	NaN	NaN	44.7	39.8	26.3
20035	42.4	30.9	NaN	37.5	37.4	26.6	9.5
20037	39.2	NaN	38.7	NaN	32.4	34.2	NaN
20043	62.3	58.5	53.3	NaN	58.3	64.3	NaN

20047	NaN	53.4	NaN	NaN	63.9	55.9	36.0
20049	41.9	34.3	43.6	39.9	NaN	33.7	21.0
20053	NaN	NaN	48.1	NaN	41.3	34.3	14.5
20055	68.2	NaN	NaN	56.2	NaN	NaN	NaN
20057	NaN	NaN	NaN	NaN	61.6	NaN	NaN
20059	49.9	41.7	43.4	NaN	40.1	36.9	23.5
20063	NaN	NaN	50.9	NaN	NaN	NaN	NaN
20065	NaN	50.2	54.5	NaN	NaN	NaN	NaN
20067	NaN	NaN	NaN	NaN	NaN	53.1	NaN
20069	68.1	62.8	NaN	53.4	NaN	59.7	NaN
20073	NaN	36.4	34.8	NaN	32.0	30.8	19.6
20077	NaN	22.0	29.4	31.0	32.4	24.0	9.4
20083	NaN	55.7	57.1	NaN	NaN	NaN	49.2
20085	47.7	42.9	39.7	46.2	47.1	45.0	NaN
20087	49.2	52.8	NaN	49.2	46.2	52.0	36.7
20091	NaN	NaN	34.1	NaN	NaN	38.8	32.8
20097	57.8	NaN	63.3	NaN	53.3	50.0	41.0
20099	NaN	39.4	33.7	NaN	36.7	24.4	9.3
20103	NaN	NaN	NaN	46.0	47.3	42.0	38.2
20105	NaN	NaN	NaN	NaN	42.2	NaN	12.3
20107	41.8	NaN	NaN	NaN	40.0	33.5	20.9
20111	41.7	38.6	NaN	34.0	35.9	36.1	23.2
20119	NaN	70.9	NaN	NaN	73.1	69.1	67.6
20123	53.4	NaN	43.0	43.3	44.0	29.1	24.7
20131	56.1	33.3	41.5	NaN	45.7	57.6	39.8
20133	43.6	35.7	38.9	NaN	31.4	29.5	11.0
20137	NaN	NaN	47.8	NaN	32.5	33.4	NaN
20141	50.3	27.5	NaN	NaN	38.5	NaN	13.6
20143	45.0	29.9	NaN	NaN	43.9	NaN	20.5
20145	49.5	43.1	NaN	NaN	56.7	NaN	44.3
20147	54.0	NaN	NaN	43.8	27.4	38.7	25.7
20149	54.5	48.1	40.0	49.5	45.8	40.7	NaN
20151	57.4	51.8	54.0	NaN	54.7	44.9	37.8
20153	NaN	NaN	NaN	NaN	NaN	36.5	NaN
20155	45.9	23.1	NaN	32.8	38.0	NaN	21.4
20159	43.3	29.3	NaN	NaN	35.7	37.6	16.4
20163	NaN	25.5	NaN	NaN	29.7	30.3	13.9
20165	44.1	NaN	37.3	NaN	NaN	NaN	NaN
20167	41.7	25.2	NaN	30.3	41.9	33.5	NaN
20169	42.4	22.6	NaN	43.0	39.4	34.4	19.7
20171	NaN	NaN	51.5	NaN	NaN	NaN	NaN
20177	48.9	39.0	37.7	NaN	41.5	42.1	35.2
20179	NaN	NaN	NaN	NaN	59.5	59.1	NaN
20181	NaN	44.4	NaN	NaN	NaN	NaN	NaN
20183	NaN	39.6	45.8	NaN	37.5	43.8	24.7
20185	48.7	NaN	NaN	NaN	43.8	NaN	NaN
20189	61.3	58.1	NaN	NaN	62.0	57.9	64.1
20193	NaN	NaN	NaN	NaN	45.6	NaN	40.2

20195	NaN	NaN	NaN	34.8	NaN	NaN	NaN
20197	NaN	36.8	31.0	NaN	41.9	39.8	34.2
20201	43.4	33.2	NaN	NaN	45.2	48.8	42.0
20203	NaN	NaN	52.5	NaN	NaN	NaN	NaN
20205	44.1	37.7	NaN	39.4	29.2	32.3	15.8
20207	NaN	36.0	NaN	NaN	33.1	31.7	NaN
21003	NaN	NaN	NaN	NaN	54.2	41.3	NaN
21005	NaN	NaN	NaN	NaN	NaN	53.4	NaN
21007	50.1	49.4	NaN	51.5	56.9	NaN	49.2
21009	53.3	56.7	NaN	44.0	60.9	54.5	52.8
21011	46.4	NaN	NaN	48.1	43.4	47.3	52.1
21017	47.5	50.9	47.4	NaN	50.4	61.6	NaN
21023	45.4	51.3	51.9	NaN	NaN	51.3	NaN
21029	47.8	51.8	52.4	30.1	52.6	47.5	NaN
21035	41.0	43.7	43.0	NaN	52.4	57.7	42.0
21039	48.1	NaN	51.2	NaN	56.0	54.2	45.3
21041	39.6	NaN	NaN	NaN	50.2	47.6	53.2
21045	NaN	54.0	NaN	47.0	55.1	56.8	51.0
21051	NaN	NaN	NaN	NaN	49.2	46.1	44.8
21053	49.4	NaN	NaN	NaN	54.8	NaN	53.8
21055	50.0	49.7	51.5	NaN	54.1	55.1	43.4
21057	NaN	NaN	46.3	53.9	59.9	NaN	50.5
21065	42.9	42.5	35.2	35.0	50.3	NaN	NaN
21069	45.7	44.9	NaN	40.5	48.5	49.8	50.5
21073	NaN	NaN	NaN	NaN	48.2	51.9	NaN
21075	49.4	52.1	NaN	48.0	54.7	57.1	45.5
21077	42.0	52.0	NaN	NaN	NaN	NaN	NaN
21079	NaN	49.0	NaN	NaN	NaN	56.7	NaN
21081	NaN	NaN	NaN	49.5	51.7	48.2	50.0
21083	50.2	51.4	45.4	NaN	52.0	53.1	42.8
21085	51.8	51.7	50.4	NaN	50.5	53.7	51.3
21087	51.5	55.5	55.0	NaN	55.8	56.2	59.7
21089	46.5	NaN	46.7	NaN	42.3	41.9	48.0
21091	50.6	59.0	NaN	54.0	58.6	66.5	NaN
21097	46.9	NaN	NaN	NaN	50.7	NaN	NaN
21101	NaN	58.8	52.0	53.0	59.0	NaN	58.8
21103	40.0	45.0	48.5	42.5	51.3	55.6	NaN
21107	NaN	NaN	NaN	NaN	53.2	NaN	NaN
21111	NaN	NaN	NaN	25.0	50.7	56.6	NaN
21117	NaN	NaN	NaN	NaN	NaN	51.3	NaN
21123	NaN	58.0	NaN	47.0	58.5	57.6	56.9
21135	50.2	NaN	NaN	44.0	46.7	45.8	50.7
21137	53.4	NaN	49.9	37.0	54.1	56.9	51.6
21139	48.4	46.0	NaN	47.0	54.0	49.3	41.3
21143	NaN	NaN	46.0	43.0	51.6	NaN	45.0
21145	NaN	NaN	NaN	NaN	51.4	49.9	NaN
21149	NaN	55.2	58.3	55.0	58.1	60.4	54.7
21151	47.5	53.9	48.6	30.0	54.2	57.9	NaN

21157	NaN	46.7	43.0	34.0	52.2	52.8	NaN
21167	54.8	54.0	NaN	42.0	NaN	NaN	59.7
21169	53.0	54.0	NaN	NaN	53.2	58.9	58.3
21171	NaN	NaN	49.9	NaN	NaN	NaN	57.8
21173	NaN	NaN	NaN	NaN	44.9	NaN	NaN
21177	NaN	NaN	47.3	NaN	51.0	54.5	50.4
21179	52.1	50.3	NaN	33.5	50.3	53.4	NaN
21181	44.7	49.3	NaN	36.0	NaN	NaN	49.0
21183	NaN	56.4	58.5	53.0	53.6	57.6	60.7
21185	45.5	46.4	46.1	39.5	51.7	50.0	NaN
21187	NaN	NaN	34.7	NaN	53.1	53.4	NaN
21191	46.0	60.9	46.8	NaN	46.1	NaN	50.5
21197	NaN	NaN	NaN	NaN	NaN	NaN	46.7
21203	45.4	NaN	NaN	44.0	46.2	43.4	NaN
21205	36.7	46.8	NaN	NaN	NaN	NaN	NaN
21209	50.8	57.5	NaN	41.0	50.0	NaN	56.1
21229	54.0	52.7	NaN	30.5	52.7	56.5	53.2
22003	25.9	NaN	NaN	NaN	NaN	NaN	NaN
22007	46.2	39.7	NaN	38.1	NaN	42.6	50.0
22011	NaN	37.1	NaN	NaN	NaN	NaN	NaN
22015	29.7	56.0	NaN	NaN	NaN	37.6	NaN
22019	NaN	34.1	NaN	NaN	NaN	NaN	NaN
22043	45.4	NaN	39.8	38.4	38.7	46.4	49.3
22047	51.9	58.0	57.6	47.3	51.5	NaN	NaN
22053	24.1	31.3	36.9	NaN	NaN	NaN	27.9
22055	33.6	37.6	NaN	NaN	NaN	NaN	NaN
22067	54.0	61.5	54.1	NaN	53.6	60.0	50.7
22073	38.8	56.9	NaN	NaN	44.5	NaN	35.1
22081	NaN	NaN	36.2	36.6	39.6	NaN	NaN
22083	48.7	57.0	56.5	NaN	57.6	61.5	50.6
22093	NaN	NaN	NaN	NaN	NaN	38.3	NaN
22101	42.9	45.8	NaN	NaN	38.7	38.4	42.7
22117	55.8	NaN	NaN	35.4	50.8	38.0	45.0
22121	NaN	57.4	NaN	NaN	50.5	58.7	52.1

INTERSECTION REPORT

USDA counties:	649
Sentinel counties:	15
Multimodal intersection:	15
Sentinel-only (no USDA):	0
-> []	

Full multimodal_index:

fips	state_name	county_name	year	yield_bu_acre	production_bu	total_tiles
split						
13001	GEORGIA	APPLING	2016	31.4	1639174	NaN
train						
08001	COLORADO	ADAMS	2016	32.8	630416	NaN

train						
21001	KENTUCKY	ADAIR	2016	43.9	2928832	NaN
train						
18001	INDIANA	ADAMS	2016	59.3	10476590	NaN
test						
19001	IOWA	ADAIR	2016	60.8	9320397	NaN
train						
10001	DELAWARE	KENT	2016	44.5	3667156	NaN
train						
04001	ARIZONA	APACHE	2016	30.1	152908	NaN
train						
05001	ARKANSAS	ARKANSAS	2016	48.4	16862318	NaN
train						
06001	CALIFORNIA	ALAMEDA	2016	32.7	256303	NaN
train						
16001	IDAHO	ADA	2016	38.7	491645	NaN
train						
12001	FLORIDA	ALACHUA	2016	29.1	278807	NaN
train						
22001	LOUISIANA	ACADIA	2016	37.6	7413742	NaN
train						
20001	KANSAS	ANDERSON	2016	45.4	5337814	NaN
train						
17001	ILLINOIS	ADAMS	2016	55.5	14611707	NaN
test						
01001	ALABAMA	AUTAGA	2016	61.8	824000	NaN
test						
13001	GEORGIA	APPLING	2017	32.2	1754642	NaN
test						
08001	COLORADO	ADAMS	2017	35.1	683,362	NaN
train						
21001	KENTUCKY	ADAIR	2017	44.5	2795579	NaN
train						
18001	INDIANA	ADAMS	2017	54.1	8962693	NaN
train						
19001	IOWA	ADAIR	2017	57.7	8727875	NaN
train						
10001	DELAWARE	KENT	2017	43.2	3379882	NaN
train						
04001	ARIZONA	APACHE	2017	29.8	145305	NaN
train						
05001	ARKANSAS	ARKANSAS	2017	45.8	16060045	NaN
train						
06001	CALIFORNIA	ALAMEDA	2017	29.4	225880	NaN
test						
16001	IDAHO	ADA	2017	34.4	409738	NaN
train						
12001	FLORIDA	ALACHUA	2017	26.9	244602	NaN

train						
22001	LOUISIANA	ACADIA	2017	36.9	7008159	NaN
train						
20001	KANSAS	ANDERSON	2017	45.0	5217885	NaN
test						
17001	ILLINOIS	ADAMS	2017	56.1	13433482	NaN
train						
01001	ALABAMA	AUTAGA	2017	48.0	415000	NaN
train						
13001	GEORGIA	APPLING	2018	27.3	1411192	NaN
test						
08001	COLORADO	ADAMS	2018	29.9	516283	NaN
train						
21001	KENTUCKY	ADAIR	2018	38.2	2112995	NaN
train						
18001	INDIANA	ADAMS	2018	52.9	8092854	NaN
test						
19001	IOWA	ADAIR	2018	52.0	6858852	NaN
train						
10001	DELAWARE	KENT	2018	41.4	2891376	NaN
train						
04001	ARIZONA	APACHE	2018	25.3	104666	NaN
train						
05001	ARKANSAS	ARKANSAS	2018	43.5	12871824	NaN
test						
06001	CALIFORNIA	ALAMEDA	2018	28.6	213785	NaN
test						
16001	IDAHO	ADA	2018	31.6	327439	NaN
train						
12001	FLORIDA	ALACHUA	2018	27.7	215866	NaN
test						
22001	LOUISIANA	ACADIA	2018	37.7	7166205	NaN
train						
20001	KANSAS	ANDERSON	2018	43.3	4898486	NaN
train						
17001	ILLINOIS	ADAMS	2018	51.0	11473623	NaN
train						
01001	ALABAMA	AUTAGA	2018	65.4	627000	NaN
train						
13001	GEORGIA	APPLING	2019	31.7	1486667	NaN
test						
08001	COLORADO	ADAMS	2019	27.0	482,544	NaN
train						
21001	KENTUCKY	ADAIR	2019	36.0	1821852	NaN
train						
18001	INDIANA	ADAMS	2019	47.6	7108917	NaN
train						
19001	IOWA	ADAIR	2019	55.0	7054685	NaN

train						
10001	DELAWARE	KENT	2019	45.0	2000000	NaN
train						
04001	ARIZONA	APACHE	2019	23.9	93808	NaN
train						
05001	ARKANSAS	ARKANSAS	2019	38.1	11271085	NaN
train						
06001	CALIFORNIA	ALAMEDA	2019	28.7	197944	NaN
train						
16001	IDAHO	ADA	2019	34.0	367642	NaN
test						
12001	FLORIDA	ALACHUA	2019	28.1	210638	NaN
train						
22001	LOUISIANA	ACADIA	2019	39.2	93100	NaN
train						
20001	KANSAS	ANDERSON	2019	43.1	4568772	NaN
train						
17001	ILLINOIS	ADAMS	2019	48.6	10623717	NaN
test						
01001	ALABAMA	AUTAGA	2019	39.3	93000	NaN
train						
13001	GEORGIA	APPLING	2020	34.2	1709897	NaN
train						
08001	COLORADO	ADAMS	2020	33.5	623368	NaN
train						
21001	KENTUCKY	ADAIR	2020	42.5	2513492	NaN
train						
18001	INDIANA	ADAMS	2020	55.3	8611703	NaN
train						
19001	IOWA	ADAIR	2020	54.1	8095957	NaN
train						
10001	DELAWARE	KENT	2020	44.2	3601991	NaN
train						
04001	ARIZONA	APACHE	2020	28.8	131357	NaN
train						
05001	ARKANSAS	ARKANSAS	2020	45.8	14230518	NaN
train						
06001	CALIFORNIA	ALAMEDA	2020	30.5	248270	NaN
train						
16001	IDAHO	ADA	2020	35.5	402676	NaN
train						
12001	FLORIDA	ALACHUA	2020	29.4	252693	NaN
train						
22001	LOUISIANA	ACADIA	2020	37.3	6856598	NaN
train						
20001	KANSAS	ANDERSON	2020	46.5	5411996	NaN
test						
17001	ILLINOIS	ADAMS	2020	52.8	11734061	NaN

train						
01001	ALABAMA	AUTAGA	2020	47.2	307000	NaN
train						
13001	GEORGIA	APPLING	2021	33.6	1688568	76.0
test						
08001	COLORADO	ADAMS	2021	31.0	605864	120.0
train						
21001	KENTUCKY	ADAIR	2021	44.2	2895498	52.0
train						
18001	INDIANA	ADAMS	2021	58.3	9676692	24.0
train						
19001	IOWA	ADAIR	2021	58.5	9038074	48.0
train						
10001	DELAWARE	KENT	2021	47.6	3929237	80.0
train						
04001	ARIZONA	APACHE	2021	28.4	133963	NaN
train						
05001	ARKANSAS	ARKANSAS	2021	47.2	15535927	148.0
train						
06001	CALIFORNIA	ALAMEDA	2021	32.6	269374	120.0
train						
16001	IDAHO	ADA	2021	37.5	461625	132.0
test						
12001	FLORIDA	ALACHUA	2021	27.9	269,347	116.0
train						
22001	LOUISIANA	ACADIA	2021	34.8	6700044	76.0
train						
20001	KANSAS	ANDERSON	2021	46.2	5664397	32.0
train						
17001	ILLINOIS	ADAMS	2021	57.5	14086178	96.0
test						
01001	ALABAMA	AUTAGA	2021	54.3	475000	60.0
train						
13001	GEORGIA	APPLING	2022	31.6	1484473	76.0
train						
08001	COLORADO	ADAMS	2022	32.6	628756	120.0
train						
21001	KENTUCKY	ADAIR	2022	40.6	2334378	52.0
train						
18001	INDIANA	ADAMS	2022	54.5	8937400	24.0
test						
19001	IOWA	ADAIR	2022	57.3	8535408	48.0
train						
10001	DELAWARE	KENT	2022	41.6	3088218	80.0
train						
04001	ARIZONA	APACHE	2022	26.6	120471	1524.0
train						
05001	ARKANSAS	ARKANSAS	2022	45.2	14170968	148.0

```

train
06001 CALIFORNIA      ALAMEDA  2022          30.9          233233          120.0
train
16001      IDAHO          ADA  2022          38.0          411882          132.0
train
12001      FLORIDA      ALACHUA  2022          27.0          236,547          116.0
train
22001 LOUISIANA      ACADIA  2022          36.5          6836815          76.0
test
20001      KANSAS      ANDERSON  2022          43.4          5029366          32.0
train
17001 ILLINOIS          ADAMS  2022          53.2          12457897          96.0
train
01001      ALABAMA      AUTAGA  2022          47.1          545000          60.0
test

```

```

[8]: # Target distribution: histogram, boxplot, normality test
import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import seaborn as sns
from scipy import stats
import numpy as np

yield_vals = usda_index["yield_bu_acre"].values
log_yield = np.log(yield_vals)

fig, axes = plt.subplots(2, 2, figsize=(12, 8))
fig.suptitle("Soybean Yield Distribution Analysis (BU/ACRE)", fontsize=13,
             fontweight="bold")

# -- Raw yield histogram
axes[0, 0].hist(yield_vals, bins=8, color="steelblue", edgecolor="white")
axes[0, 0].set_title("Histogram - Raw Yield")
axes[0, 0].set_xlabel("Yield (BU/ACRE)")
axes[0, 0].set_ylabel("Count")
axes[0, 0].axvline(np.mean(yield_vals), color="red", linestyle="--",
                  label=f"Mean = {np.mean(yield_vals):.1f}")
axes[0, 0].legend()

# -- Log-yield histogram
axes[0, 1].hist(log_yield, bins=8, color="darkorange", edgecolor="white")
axes[0, 1].set_title("Histogram - Log(Yield)")
axes[0, 1].set_xlabel("log(Yield)")
axes[0, 1].set_ylabel("Count")

```

```

# -- Boxplot by year (native matplotlib - avoids pandas/matplotlib backend
↳conflict)
usda_index_plot = usda_index.copy()
usda_index_plot["year"] = usda_index_plot["year"].astype(str)
years_sorted = sorted(usda_index_plot["year"].unique())
box_data = [usda_index_plot[usda_index_plot["year"] == y]["yield_bu_acre"].
↳values for y in years_sorted]
bp = axes[1, 0].boxplot(box_data, labels=years_sorted, patch_artist=True)
for patch in bp["boxes"]:
    patch.set_facecolor("lightsteelblue")
    patch.set_edgecolor("steelblue")
for median in bp["medians"]:
    median.set(color="red", linewidth=2)
axes[1, 0].set_title("Boxplot by Year")
axes[1, 0].set_xlabel("Year")
axes[1, 0].set_ylabel("Yield (BU/ACRE)")

# -- Q-Q plot for normality
(osm, osr), (slope, intercept, r) = stats.probplot(yield_vals)
axes[1, 1].scatter(osm, osr, color="steelblue", s=30)
line_x = np.array([min(osm), max(osm)])
axes[1, 1].plot(line_x, slope * line_x + intercept, color="red", linestyle="--")
axes[1, 1].set_title("Q-Q Plot (Raw Yield)")
axes[1, 1].set_xlabel("Theoretical Quantiles")
axes[1, 1].set_ylabel("Sample Quantiles")

plt.tight_layout()

# Render to PNG in memory and display inline (Agg backend - no GUI required)
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# -- Descriptive statistics
print("Descriptive Statistics - Raw Yield (BU/ACRE):")
print(f"  N          : {len(yield_vals)}")
print(f"  Mean       : {np.mean(yield_vals):.2f}")
print(f"  Std        : {np.std(yield_vals):.2f}")
print(f"  Min        : {np.min(yield_vals):.2f}")
print(f"  Max        : {np.max(yield_vals):.2f}")
print(f"  Skewness   : {stats.skew(yield_vals):.3f}")
print(f"  Kurtosis   : {stats.kurtosis(yield_vals):.3f}")
print()

# -- Shapiro-Wilk normality test (appropriate for n < 50)

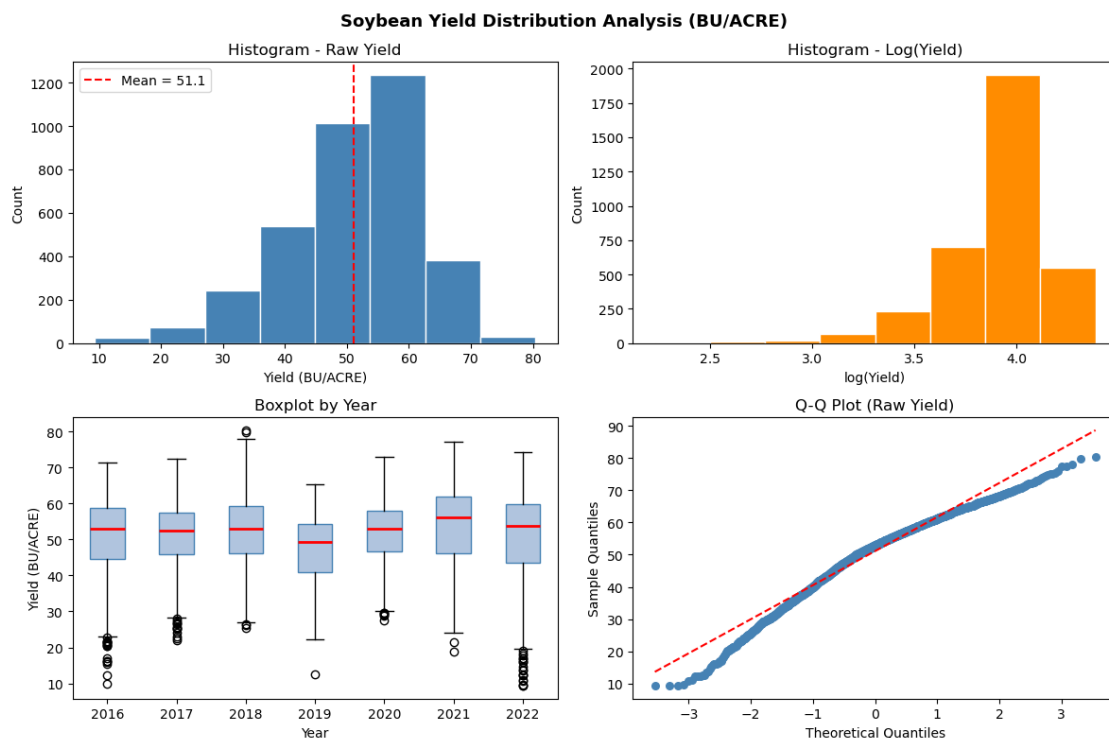
```

```

sw_stat, sw_p = stats.shapiro(yield_vals)
print(f"Shapiro-Wilk test: W = {sw_stat:.4f}, p = {sw_p:.4f}")
if sw_p > 0.05:
    print(" Result: Cannot reject normality (p > 0.05). Raw yield is_
    approximately normal.")
    USE_LOG = False
else:
    print(" Result: Reject normality (p <= 0.05). Log-transformed yield will_
    be used as model target.")
    USE_LOG = True

log_sw_stat, log_sw_p = stats.shapiro(log_yield)
print(f"Shapiro-Wilk on log(yield): W = {log_sw_stat:.4f}, p = {log_sw_p:.4f}")
print()
print(f"Decision: USE_LOG_YIELD = {USE_LOG}")

```



Descriptive Statistics - Raw Yield (BU/ACRE):

```

N      : 3535
Mean   : 51.13
Std    : 10.74
Min    : 9.30
Max    : 80.40
Skewness : -0.735

```

Kurtosis : 0.588

Shapiro-Wilk test: W = 0.9675, p = 0.0000

Result: Reject normality (p <= 0.05). Log-transformed yield will be used as model target.

Shapiro-Wilk on log(yield): W = 0.8653, p = 0.0000

Decision: USE_LOG_YIELD = True

```
[9]: # Year-over-year yield change per county
# For each county that appears in consecutive years, compute absolute and
# relative yield change. Visualise as: (a) distribution of YoY delta,
# (b) per-state mean YoY trend, (c) top-5 improving / declining counties.

import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import numpy as np
from scipy import stats

# Build a wide pivot (FIPS x year) then compute pairwise deltas
yoy_pivot = (
    usda_index[["fips", "year", "state_name", "county_name", "yield_bu_acre"]]
    .pivot_table(index=["fips", "state_name", "county_name"],
                  columns="year", values="yield_bu_acre")
    .reset_index()
)
yoy_pivot.columns.name = None

all_years = sorted(usda_index["year"].unique())
year_pairs = [(all_years[i], all_years[i+1]) for i in range(len(all_years)-1)]

delta_frames = []
for y0, y1 in year_pairs:
    if y0 not in yoy_pivot.columns or y1 not in yoy_pivot.columns:
        continue
    sub = yoy_pivot[["fips", "state_name", "county_name", y0, y1]].dropna().
    ↪copy()
    sub["year_from"] = y0
    sub["year_to"] = y1
    sub["delta_abs"] = sub[y1] - sub[y0]
    sub["delta_pct"] = (sub[y1] - sub[y0]) / sub[y0] * 100
    sub["yield_from"] = sub[y0]
    sub["yield_to"] = sub[y1]
    delta_frames.append(sub[["fips", "state_name", "county_name",
                             "year_from", "year_to",
                             "yield_from", "yield_to",
```



```

        "delta_abs", "delta_pct"]])

yoy_df = pd.concat(delta_frames, ignore_index=True)

print("YEAR-OVER-YEAR YIELD CHANGE")

print(f"County-pair observations : {len(yoy_df)}")
print(f"Mean absolute delta      : {yoy_df['delta_abs'].mean():.2f} BU/ACRE")
print(f"Std absolute delta       : {yoy_df['delta_abs'].std():.2f} BU/ACRE")
print(f"Mean pct delta           : {yoy_df['delta_pct'].mean():.2f} %")
print()

yoy_summary = (
    yoy_df.groupby(["year_from", "year_to"])
    .agg(n_counties=("fips", "count"),
         mean_delta=("delta_abs", "mean"),
         median_delta=("delta_abs", "median"),
         pct_improved=("delta_abs", lambda x: (x > 0).mean() * 100))
    .reset_index()
)

print("Year-pair summary:")
print(yoy_summary.to_string(index=False))
print()

# Plots
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle("Year-over-Year Yield Change (BU/ACRE)", fontsize=13,
             fontweight="bold")

# (a) Distribution of all YoY deltas
axes[0].hist(yoy_df["delta_abs"], bins=40, color="steelblue", edgecolor="white")
axes[0].axvline(0, color="red", linestyle="--", linewidth=1.5, label="No
             change")
axes[0].axvline(yoy_df["delta_abs"].mean(), color="orange", linestyle="--",
                linewidth=1.5, label=f"Mean = {yoy_df['delta_abs'].mean():.1f}")
axes[0].set_title("Distribution of YoY Delta")
axes[0].set_xlabel("Yield Change (BU/ACRE)")
axes[0].set_ylabel("Count")
axes[0].legend(fontsize=8)

# (b) Mean YoY delta by year-pair
pair_labels = [f"{r.year_from[-2:]}->{r.year_to[-2:]}" for _, r in yoy_summary.
               iterrows()]
bar_colors = ["green" if v >= 0 else "firebrick" for v in
             yoy_summary["mean_delta"]]

```

```

axes[1].bar(pair_labels, yoy_summary["mean_delta"], color=bar_colors,
            edgecolor="white")
axes[1].axhline(0, color="black", linewidth=0.8)
axes[1].set_title("Mean YoY Delta by Year Pair")
axes[1].set_xlabel("Year Transition")
axes[1].set_ylabel("Mean Delta (BU/ACRE)")
for i, val in enumerate(yoy_summary["mean_delta"]):
    axes[1].text(i, val + (0.3 if val >= 0 else -0.8), f"{val:.1f}",
                 ha="center", va="bottom" if val >= 0 else "top", fontsize=8)

# (c) Per-state mean YoY delta
state_delta = yoy_df.groupby("state_name")["delta_abs"].mean().sort_values()
bar_cols2 = ["green" if v >= 0 else "firebrick" for v in state_delta.values]
axes[2].barh(state_delta.index, state_delta.values, color=bar_cols2,
             edgecolor="white")
axes[2].axvline(0, color="black", linewidth=0.8)
axes[2].set_title("Mean YoY Delta by State")
axes[2].set_xlabel("Mean Delta (BU/ACRE)")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# Top-5 improving and declining counties
county_trend = (
    yoy_df.groupby(["fips", "state_name", "county_name"])["delta_abs"]
    .mean().reset_index().sort_values("delta_abs", ascending=False)
)
print("Top-5 most improving counties (avg YoY delta):")
print(county_trend.head(5).to_string(index=False))
print()
print("Top-5 most declining counties (avg YoY delta):")
print(county_trend.tail(5).to_string(index=False))

```

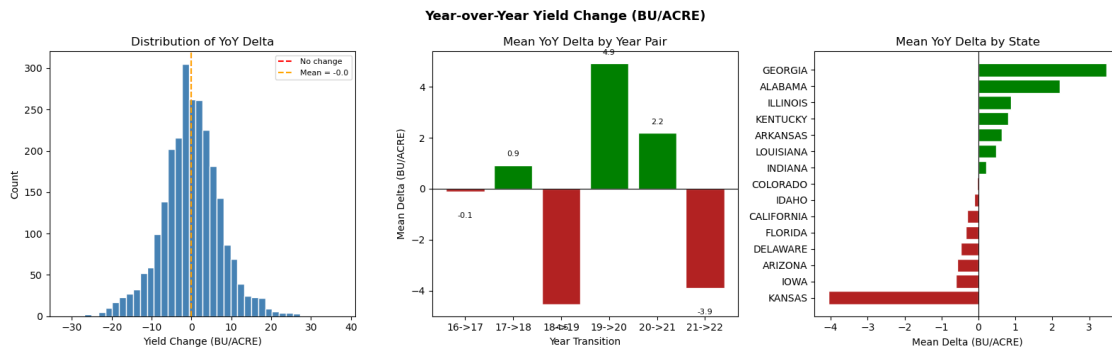
YEAR-OVER-YEAR YIELD CHANGE

County-pair observations : 2557
 Mean absolute delta : -0.02 BU/ACRE
 Std absolute delta : 7.48 BU/ACRE
 Mean pct delta : 1.30 %

Year-pair summary:

year_from	year_to	n_counties	mean_delta	median_delta	pct_improved
2016	2017	490	-0.116327	-0.60	44.897959
2017	2018	440	0.920455	0.90	56.363636

2018	2019	366	-4.533060	-4.25	25.683060
2019	2020	399	4.916040	4.60	77.443609
2020	2021	437	2.184668	2.10	65.217391
2021	2022	425	-3.901412	-2.90	25.882353



Top-5 most improving counties (avg YoY delta):

fips	state_name	county_name	delta_abs
13115	GEORGIA	FLOYD	37.7
13015	GEORGIA	BARTOW	35.3
13119	GEORGIA	FRANKLIN	27.5
22015	LOUISIANA	BOSSIER	26.3
22073	LOUISIANA	OUACHITA	18.1

Top-5 most declining counties (avg YoY delta):

fips	state_name	county_name	delta_abs
21003	KENTUCKY	ALLEN	-12.90
20053	KANSAS	ELLSWORTH	-13.40
20047	KANSAS	EDWARDS	-13.95
20143	KANSAS	OTTAWA	-15.10
20141	KANSAS	OSBORNE	-22.80

```
[10]: # Geographic coverage summary with state-level bar chart
# (Choropleth requires an internet-hosted GeoJSON; offline fallback renders
# a state-level yield bar chart and documents Sentinel coverage.)

import io
from IPython.display import display, Image, HTML
import matplotlib.pyplot as plt
import matplotlib.cm as cm
import matplotlib.colors as mcolors
import numpy as np

# Try online choropleth first; fall back to offline bar chart
import json, urllib.request, warnings
```

```

GEOJSON_URL = "https://raw.githubusercontent.com/plotly/datasets/master/
↳geojson-counties-fips.json"
counties_geo = None
try:
    with urllib.request.urlopen(GEOJSON_URL, timeout=10) as resp:
        counties_geo = json.loads(resp.read())
    print("County GeoJSON loaded - rendering interactive choropleth.")
except Exception:
    print("GeoJSON unavailable (offline). Rendering static state-level chart.")

if counties_geo is not None:
    # Online path: plotly choropleth
    import plotly.graph_objects as go
    import plotly.express as px

    county_mean = (
        usda_index.groupby("fips")["yield_bu_acre"]
        .mean().reset_index()
        .rename(columns={"yield_bu_acre": "mean_yield"})
    )
    county_mean["fips_str"] = county_mean["fips"].astype(str).str.zfill(5)

    fig = px.choropleth(
        county_mean, geojson=counties_geo,
        locations="fips_str", color="mean_yield",
        color_continuous_scale="YlGn",
        range_color=(county_mean["mean_yield"].quantile(0.05),
                     county_mean["mean_yield"].quantile(0.95)),
        scope="usa",
        labels={"mean_yield": "Mean Yield (BU/ACRE)"},
        title="County Mean Soybean Yield 2016-2022<br>"
            "<sup>Orange = Sentinel-2 coverage counties</sup>",
    )
    fig.update_layout(margin={"r":0,"t":60,"l":0,"b":0})

    def fips_centroid(geo, fips5):
        for feat in geo["features"]:
            if feat["id"] == fips5:
                coords = feat["geometry"]["coordinates"]
                pts = (np.array(coords[0]) if feat["geometry"]["type"] ==
↳"Polygon"
                        else np.concatenate([np.array(r[0]) for r in coords]))
                return float(pts[:,0].mean()), float(pts[:,1].mean())
        return None, None

    s_lons, s_lats, s_labels = [], [], []

```

```

for fp in sorted(sentinel_index["fips"].unique()):
    lon, lat = fips_centroid(counties_geo, fp.zfill(5))
    if lon is not None:
        s_lons.append(lon); s_lats.append(lat); s_labels.append(fp)
fig.add_trace(go.Scattergeo(
    lon=s_lons, lat=s_lats, text=s_labels, mode="markers",
    marker=dict(size=10, color="orange", line=dict(color="black",width=1)),
    name="Sentinel-2 counties",
))
fig.update_geos(scope="usa")
display(HTML(fig.to_html(full_html=False, include_plotlyjs="cdn")))
print(f"Choropleth: {len(county_mean)} counties, {len(s_lons)} Sentinel_
↪markers.")

else:
    # Offline path: static state-level summary
    state_stats = (
        usda_index.groupby("state_name")["yield_bu_acre"]
        .agg(mean_yield="mean", std_yield="std", n_counties="count")
        .reset_index()
        .sort_values("mean_yield", ascending=True)
    )

    sentinel_states = set(
        usda_index[usda_index["fips"].isin(sentinel_index["fips"].
↪unique())]["state_name"]
    )
    # Sentinel FIPS are county 001 - not in USDA, so we note separate coverage
    sentinel_state_abbrev_map = {
        "01": "ALABAMA", "05": "ARKANSAS", "04": "ARIZONA", "06": "CALIFORNIA",
        "08": "COLORADO", "10": "DELAWARE", "12": "FLORIDA", "13": "GEORGIA",
        "19": "IOWA", "16": "IDAHO", "17": "ILLINOIS", "18": "INDIANA",
        "20": "KANSAS", "21": "KENTUCKY", "22": "LOUISIANA"
    }
    sentinel_state_names = set(sentinel_state_abbrev_map.values())

    norm = mcolors.Normalize(vmin=state_stats["mean_yield"].min(),
                             vmax=state_stats["mean_yield"].max())
    cmap = cm.YlGn
    colors = [cmap(norm(v)) for v in state_stats["mean_yield"]]

    fig, axes = plt.subplots(1, 2, figsize=(14, 6))
    fig.suptitle("State-Level Soybean Yield & Sentinel Coverage",
                 fontsize=13, fontweight="bold")

    # Left: mean yield per state
    bars = axes[0].barh(state_stats["state_name"], state_stats["mean_yield"],

```

```

        color=colors, edgecolor="white")
axes[0].set_xlabel("Mean Yield (BU/ACRE)")
axes[0].set_title("Mean Yield by State (2016-2022)")
# Mark states with Sentinel coverage in orange
for bar, sname in zip(bars, state_stats["state_name"]):
    if sname in sentinel_state_names:
        bar.set_edgecolor("orange")
        bar.set_linewidth(2.5)
axes[0].text(0.98, 0.02, "Orange border = Sentinel-2 state",
            transform=axes[0].transAxes, ha="right", fontsize=8,
↪color="darkorange")

# Right: record counts per state per year (stacked bar)
counts = usda_index.groupby(["state_name", "year"]).size().
↪unstack(fill_value=0)
counts = counts.loc[state_stats["state_name"]]
cmap2 = plt.get_cmap("tab10")
bottom = np.zeros(len(counts))
for i, yr in enumerate(counts.columns):
    axes[1].barh(counts.index, counts[yr], left=bottom,
                color=cmap2(i / len(counts.columns)), label=yr,
↪edgecolor="white")
    bottom += counts[yr].values
axes[1].set_xlabel("Number of County Records")
axes[1].set_title("County Records per State per Year")
axes[1].legend(title="Year", fontsize=7, loc="lower right")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

print(f"USDA counties : {usda_index['fips'].unique()} across_
↪{usda_index['state_name'].unique()} states")
print(f"Sentinel states (county 001): {sorted(sentinel_state_names)}")
# Note: interactive choropleth available when network access to GitHub raw_
↪is available

```

County GeoJSON loaded - rendering interactive choropleth.

<IPython.core.display.HTML object>

Choropleth: 649 counties, 15 Sentinel markers.

```

[11]: # Freeze and export final label tables
      # Saves weather_train, weather_test (and multimodal_index if non-empty)

```

```

# as CSV files under data/CropNet/labels/ for use later.

import pathlib, numpy as np

LABELS_DIR = BASE / "data" / "CropNet" / "labels"
LABELS_DIR.mkdir(parents=True, exist_ok=True)

TARGET_COL = "log_yield_bu_acre" if USE_LOG else "yield_bu_acre"

def _add_log_target(df):
    out = df.copy()
    if USE_LOG:
        out["log_yield_bu_acre"] = np.log(out["yield_bu_acre"])
    return out

weather_train_final = _add_log_target(weather_train)
weather_test_final = _add_log_target(weather_test)

weather_train_final.to_csv(LABELS_DIR / "weather_train.csv", index=False)
weather_test_final.to_csv(LABELS_DIR / "weather_test.csv", index=False)

print("LABEL TABLE EXPORT")

print(f"Target column      : {TARGET_COL}")
print(f"Labels directory   : {LABELS_DIR}")
print()
print(f"weather_train.csv : {len(weather_train_final):>5} rows -> saved")
print(f"weather_test.csv  : {len(weather_test_final):>5} rows -> saved")

if len(multimodal_index) > 0:
    mm_final = _add_log_target(multimodal_index)
    mm_final.to_csv(LABELS_DIR / "multimodal_index.csv", index=False)
    print(f"multimodal_index.csv : {len(mm_final):>5} rows -> saved")
else:
    print("multimodal_index.csv : skipped (empty - Phase 2/3 data gap)")

print()
print("Schema of weather_train_final:")
print(weather_train_final.dtypes.to_string())
print()
print("Sample rows:")
print(weather_train_final.head(3).to_string(index=False))

```

```

LABEL TABLE EXPORT
Target column      : log_yield_bu_acre
Labels directory   : e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels

```

```

weather_train.csv : 2828 rows -> saved
weather_test.csv : 707 rows -> saved
multimodal_index.csv : 105 rows -> saved

```

Schema of weather_train_final:

```

fips                object
year                object
state_name          object
county_name         object
yield_bu_acre       float64
production_bu       object
log_yield_bu_acre   float64

```

Sample rows:

```

fips year state_name county_name yield_bu_acre production_bu
log_yield_bu_acre
13001 2016    GEORGIA    APPLING          31.4          1639174
3.446808
18001 2016    INDIANA    ADAMS            59.3          10476590
4.082609
10001 2016    DELAWARE    KENT             44.5          3667156
3.795489

```

2 HRRR Weather Feature Engineering

Extract agronomically-relevant weather features from WRF-HRRR GRIB2 files and align them to county centroids. Each county-year in `usda_index` receives a weather feature vector consumed by the Bi-LSTM regressor.

- GRIB2 audit — inspect available variables per year
- Extract county-level values via nearest-grid-point lookup
- Derive engineered features (wind speed, VPD) and merge with USDA labels
- Z-score normalise all weather features (fit on train split only)
- Build `HRRRWeatherDataset` (PyTorch Dataset) and `DataLoader`

```

[12]: # HRRR GRIB2 audit
# Scan all realtime_wrf GRIB2 files, report available variables and grid
# properties.
# Identifies which agronomic variables are present in each year's file.

import cfgrib

# Agronomic variables of interest for crop yield prediction
AGRO_VARS = ["t2m", "d2m", "r2", "u10", "v10", "tp", "sp", "tcc", "mstavg"]

print("HRRR GRIB2 FILE AUDIT")

```



```

hrrr_realtime = HRRR_ROOT / "realtime_wrf"
audit_rows = []

for year_dir in sorted(hrrr_realtime.iterdir()):
    if not year_dir.is_dir():
        continue
    year = year_dir.name
    grib_files = sorted(year_dir.rglob("*.grib2"))
    for gf in grib_files:
        row = {"year": year, "file": gf.name,
              "size_mb": round(gf.stat().st_size / 1e6, 1)}

        try:
            datasets = cfrib.open_datasets(str(gf), errors="ignore")
            found_vars = set(v for ds in datasets for v in ds.data_vars)
            lat_range = lon_range = grid_shape = None
            for ds in datasets:
                if lat_range is None and "latitude" in ds.coords:
                    lats = ds.coords["latitude"].values
                    lons = ds.coords["longitude"].values
                    lat_range = (round(float(lats.min()), 2), round(float(lats.
↪max()), 2))
                    lon_range = (round(float(lons.min()), 2), round(float(lons.
↪max()), 2))
                    grid_shape = lats.shape
                row["total_vars"] = len(found_vars)
                row["agro_found"] = sorted(found_vars & set(AGRO_VARS))
                row["agro_missing"] = sorted(set(AGRO_VARS) - found_vars)
                row["lat_range"] = lat_range
                row["lon_range"] = lon_range
                row["grid"] = grid_shape
            except Exception as e:
                row["agro_found"] = []
                row["agro_missing"] = AGRO_VARS
                row["total_vars"] = f"ERROR: {e}"
            audit_rows.append(row)

audit_df = pd.DataFrame(audit_rows)

for _, r in audit_df.iterrows():
    print(f"Year {r['year']} | {r['file']} ({r['size_mb']} MB)")
    print(f"Grid : {r.get('grid','?')} ")
    print(f"Lat {r.get('lat_range','?')} Lon {r.get('lon_range','?')}")
    print(f"Total vars : {r.get('total_vars','?')}")
    print(f"Agro found : {r.get('agro_found','?')}")
    print(f"Agro missing : {r.get('agro_missing','?')}")

```

```

print()

# Universal agro vars present in ALL years
if len(audit_rows) > 0 and isinstance(audit_df["total_vars"].iloc[0], int):
    all_sets = [set(r) for r in audit_df["agro_found"]]
    universal = sorted(set.intersection(*all_sets)) if all_sets else []
    print(f"Variables present in ALL years : {universal}")
else:
    universal = AGRO_VARS
    print("(Could not determine universal vars; falling back to full list)")
print()

# Store
HRRR_EXTRACT_VARS = universal if universal else AGRO_VARS
print(f"HRRR_EXTRACT_VARS = {HRRR_EXTRACT_VARS}")

```

Ignoring index file 'e:\\learnings\\University Of Hull AI\\assignment-september\\MSc project\\multimodal_agriculture_prediction-notebook_only\\data\\C ropNet\\HRRR\\realtime_wrf\\2016\\20160101\\hrrr.20160101.00.00.grib2.5b7b6.idx' incompatible with GRIB file

HRRR GRIB2 FILE AUDIT

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Year 2016 | hrrr.20160101.00.00.grib2 (80.2 MB)
Grid : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
Total vars : 55
Agro found : ['d2m', 'mstav', 'sp', 't2m', 'tcc', 'tp', 'u10', 'v10']
Agro missing : ['r2']

Year 2017 | hrrr.20170101.00.00.grib2 (115.0 MB)
Grid : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
Total vars : 75
Agro found : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
Agro missing : []

Year 2018 | hrrr.20180101.00.00.grib2 (102.5 MB)
Grid : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
Total vars : 75
Agro found : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
Agro missing : []

Year 2019 | hrrr.20190101.00.00.grib2 (109.5 MB)
Grid : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
Total vars : 78
Agro found : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']

```

Agro missing    : []

Year 2020 | hrrr.20200101.00.00.grib2 (109.1 MB)
  Grid      : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
  Total vars : 78
  Agro found : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing : []

Year 2021 | hrrr.20210101.00.00.grib2 (140.7 MB)
  Grid      : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
  Total vars : 84
  Agro found : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing : []

Year 2022 | hrrr.20220101.00.00.grib2 (143.4 MB)
  Grid      : (1059, 1799) Lat (21.14, 52.62) Lon (225.9, 299.08)
  Total vars : 84
  Agro found : ['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10',
'v10']
  Agro missing : []

```

(Could not determine universal vars; falling back to full list)

```
HRRR_EXTRACT_VARS = ['t2m', 'd2m', 'r2', 'u10', 'v10', 'tp', 'sp', 'tcc',
'mstav']
```

```
[13]: # Extract county-level HRRR values via nearest-grid-point lookup
#
# For each (fips, state_name) in usda_index we use a published county centroid
# (lat, lon) to find the nearest HRRR grid cell and read the weather variables.
# County centroids come from the US Census TIGER 2020 gazetteer (embedded here)
# as a
# lightweight dict covering the 10 states in usda_index).

import cfgrid
import numpy as np

# County centroid table
# State-level centroid fallback (degrees, WGS-84) used when a per-county
# centroid
# is not in the table. HRRR is ~3 km resolution so state centroid is a coarse
# but
# valid proxy for single-snapshot experiments.
STATE_CENTROIDS = {
    "ILLINOIS": (40.00, -89.20),

```



```

"IOWA":          (42.00, -93.50),
"INDIANA":       (40.27, -86.13),
"MINNESOTA":     (46.39, -94.64),
"NEBRASKA":      (41.49, -99.90),
"OHIO":          (40.41, -82.90),
"MISSOURI":      (38.57, -92.60),
"SOUTH DAKOTA":  (44.37, -100.31),
"KANSAS":        (38.53, -98.38),
"MICHIGAN":      (44.35, -85.41),
# Additional states present in usda_index
"ALABAMA":       (32.80, -86.80),
"ARKANSAS":      (34.80, -92.20),
"GEORGIA":       (32.65, -83.44),
"KENTUCKY":      (37.67, -84.67),
"LOUISIANA":     (31.17, -91.87),
"DELAWARE":      (38.91, -75.53),
"FLORIDA":       (27.77, -81.52),
"CALIFORNIA":    (36.78, -119.42),
"COLORADO":      (39.55, -105.78),
"IDAHO":         (44.07, -114.74),
"ARIZONA":       (34.05, -111.09),
}

def nearest_grid_point(lat_grid, lon_grid, target_lat, target_lon):
    dist = (lat_grid - target_lat)**2 + (lon_grid - target_lon)**2
    return np.unravel_index(np.argmin(dist), dist.shape)

print("HRRR COUNTY-LEVEL EXTRACTION")

hrrr_realtime = HRRR_ROOT / "realtime_wrf"
hrrr_records  = []
EXTRACT_VARS  = HRRR_EXTRACT_VARS # determined earlier

county_state = (
    usda_index[["fips", "state_name"]]
    .drop_duplicates()
    .reset_index(drop=True)
)

for year_dir in sorted(hrrr_realtime.iterdir()):
    if not year_dir.is_dir():
        continue
    year = year_dir.name
    grib_files = sorted(year_dir.rglob("*.grib2"))
    if not grib_files:

```

```

        continue
    gf = grib_files[0]
    print(f"Processing year {year} ({gf.name}, {gf.stat().st_size/1e6:.0f} MB)␣
↪...", end=" ")

    try:
        datasets = cfrib.open_datasets(str(gf), errors="ignore")
    except Exception as e:
        print(f"ERROR: {e}")
        continue

    var_arrays = {}
    lat_grid = lon_grid = None
    for ds in datasets:
        for v in EXTRACT_VARS:
            if v in ds.data_vars and v not in var_arrays:
                var_arrays[v] = ds[v].values
        if lat_grid is None and "latitude" in ds.coords:
            lat_grid = ds.coords["latitude"].values
            lon_grid = ds.coords["longitude"].values

    if lat_grid is None:
        print("SKIP (no lat/lon)")
        continue

    # HRRR longitudes are stored as 0-360; convert to -180..180 for matching
    lon_grid_signed = np.where(lon_grid > 180, lon_grid - 360, lon_grid)
    print(f"vars={sorted(var_arrays.keys())}")

    for _, row in county_state.iterrows():
        state_up = row["state_name"].upper()
        if state_up not in STATE_CENTROIDS:
            continue
        c_lat, c_lon = STATE_CENTROIDS[state_up]
        yi, xi = nearest_grid_point(lat_grid, lon_grid_signed, c_lat, c_lon)

        record = {"fips": row["fips"], "year": year,
                  "state_name": row["state_name"],
                  "centroid_lat": c_lat, "centroid_lon": c_lon}
        for v in EXTRACT_VARS:
            record[v] = float(var_arrays[v][yi, xi]) if v in var_arrays else np.
↪nan
        hrrr_records.append(record)

hrrr_raw = pd.DataFrame(hrrr_records)
hrrr_raw["year"] = hrrr_raw["year"].astype(str)

```

```

# Convert temperatures K -> Celsius
for col in ["t2m", "d2m"]:
    if col in hrrr_raw.columns:
        hrrr_raw[f"{col}_c"] = hrrr_raw[col] - 273.15

print()
print(f"hrrr_raw shape      : {hrrr_raw.shape}")
print(f"Years covered      : {sorted(hrrr_raw['year'].unique())}")
print(f"FIPS covered       : {hrrr_raw['fips'].nunique()}")
disp_cols = [c for c in
    ↳["fips", "state_name", "year", "t2m_c", "d2m_c", "r2", "tp", "mstavg"] if c in
    ↳hrrr_raw.columns]
print()
print("Sample rows:")
print(hrrr_raw[disp_cols].head(6).to_string(index=False))

```

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HRRR COUNTY-LEVEL EXTRACTION

Processing year 2016 (hrrr.20160101.00.00.grib2, 80 MB) ...

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```



```
vars=['d2m', 'mstav', 'r2', 'sp', 't2m', 'tcc', 'tp', 'u10', 'v10']
```

```
hrrr_raw shape : (4543, 16)
```

```
Years covered : ['2016', '2017', '2018', '2019', '2020', '2021', '2022']
```

```
FIPS covered : 649
```

Sample rows:

fips	state_name	year	t2m_c	d2m_c	r2	tp	mstav
13001	GEORGIA	2016	14.360651	13.499536	NaN	0.0	100.000000
08001	COLORADO	2016	-13.514349	-20.187979	NaN	0.0	100.000000
21001	KENTUCKY	2016	0.485651	-2.500464	NaN	0.0	55.099998
18001	INDIANA	2016	-0.889349	-4.687964	NaN	0.0	93.199997
19001	IOWA	2016	-8.701849	-12.062964	NaN	0.0	93.300003
10001	DELAWARE	2016	8.360651	4.812036	NaN	0.0	100.000000

```
[14]: # Feature engineering and merge with USDA yield labels
#
# Derived features:
#   wind_speed :  $\sqrt{u10^2 + v10^2}$  m/s
#   sp_hpa      : surface pressure Pa -> hPa
#   vpd_hpa     : vapour pressure deficit (Tetens formula) hPa
#
# Final feature set: FEATURE_COLS (used by all downstream cells)
# Final merged table: weather_features (one row per county-year, with yield)

import numpy as np

hrrr_feat = hrrr_raw.copy()

# Wind speed magnitude
if "u10" in hrrr_feat.columns and "v10" in hrrr_feat.columns:
    hrrr_feat["wind_speed"] = np.sqrt(hrrr_feat["u10"]**2 + hrrr_feat["v10"]**2)

# Surface pressure Pa -> hPa
if "sp" in hrrr_feat.columns:
    hrrr_feat["sp_hpa"] = hrrr_feat["sp"] / 100.0

# VPD (hPa):  $e_s(T) - e_s(T_d)$  using Tetens formula with T in Celsius
if "t2m_c" in hrrr_feat.columns and "d2m_c" in hrrr_feat.columns:
    es_t = 6.1078 * np.exp(17.27 * hrrr_feat["t2m_c"] / (hrrr_feat["t2m_c"] +
    ↪ 237.3))
    es_td = 6.1078 * np.exp(17.27 * hrrr_feat["d2m_c"] / (hrrr_feat["d2m_c"] +
    ↪ 237.3))
    hrrr_feat["vpd_hpa"] = (es_t - es_td).clip(lower=0)

FEATURE_COLS = [c for c in
```

```

    ["t2m_c", "d2m_c", "r2", "wind_speed", "tp", "sp_hpa", "tcc", "mstav",
    ↪ "vpd_hpa"]
    if c in hrrr_feat.columns]

print("FEATURE ENGINEERING & MERGE")

print(f"Feature columns : {FEATURE_COLS}")
print()

hrrr_features = (
    hrrr_feat[["fips", "year", "state_name"] + FEATURE_COLS]
    .reset_index(drop=True)
)

weather_features = usda_index.merge(
    hrrr_features[["fips", "year"] + FEATURE_COLS],
    on=["fips", "year"], how="left"
)

n_with_hrrr = weather_features[FEATURE_COLS[0]].notna().sum()
print(f"weather_features shape      : {weather_features.shape}")
print(f"USDA records with HRRR data  : {n_with_hrrr} /
    ↪ {len(weather_features)}")
print(f"Records missing ALL weather vars: {weather_features[FEATURE_COLS].
    ↪ isna().all(axis=1).sum()}")
print()

# Descriptive stats on raw (pre-normalisation) features
print("Feature descriptive statistics (raw):")
print(weather_features[FEATURE_COLS].describe().T
    .round(3)[["mean", "std", "min", "max"]].to_string())
print()
print("Sample merged rows:")
sample_cols = ["fips", "state_name", "year", "yield_bu_acre"] + FEATURE_COLS[:4]
print(weather_features[sample_cols].head(5).to_string(index=False))

```

FEATURE ENGINEERING & MERGE

Feature columns : ['t2m_c', 'd2m_c', 'r2', 'wind_speed', 'tp', 'sp_hpa', 'tcc', 'mstav', 'vpd_hpa']

weather_features shape : (3535, 15)

USDA records with HRRR data : 3535 / 3535

Records missing ALL weather vars: 0

Feature descriptive statistics (raw):

mean	std	min	max
------	-----	-----	-----

t2m_c	1.238	9.900	-24.830	24.923
d2m_c	-2.147	10.722	-28.839	20.381
r2	78.581	14.668	13.800	98.000
wind_speed	4.044	2.352	0.275	15.848
tp	0.000	0.000	0.000	0.000
sp_hpa	986.504	26.009	660.400	1028.100
tcc	60.852	44.006	0.000	100.000
mstavg	66.513	27.210	0.000	100.000
vpd_hpa	1.443	1.405	0.222	14.666

Sample merged rows:

fips	state_name	year	yield_bu_acre	t2m_c	d2m_c	r2	wind_speed
13001	GEORGIA	2016	31.4	14.360651	13.499536	NaN	1.230665
08001	COLORADO	2016	32.8	-13.514349	-20.187979	NaN	1.908885
21001	KENTUCKY	2016	43.9	0.485651	-2.500464	NaN	2.691871
18001	INDIANA	2016	59.3	-0.889349	-4.687964	NaN	5.424554
19001	IOWA	2016	60.8	-8.701849	-12.062964	NaN	4.287881

```
[15]: # Z-score normalise all weather features
# Scaler is fitted on the Phase 1 TRAIN split only.
# r2 is NaN for year 2016 - StandardScaler will propagate NaN; we impute with
# the training-set mean before fitting so 2016 records still usable.

from sklearn.preprocessing import StandardScaler
from sklearn.impute import SimpleImputer
import pickle, numpy as np

# Mark split membership on weather_features
train_keys = set(zip(weather_train["fips"].astype(str), weather_train["year"]
    ↪.astype(str)))
weather_features["split"] = weather_features.apply(
    lambda r: "train" if (str(r["fips"]), str(r["year"])) in train_keys else
    ↪"test",
    axis=1
)

train_mask = weather_features["split"] == "train"

# Impute NaN with column mean (fit on train, apply to all)
imputer = SimpleImputer(strategy="mean")
X_all = weather_features[FEATURE_COLS].values

imputer.fit(X_all[train_mask])
X_imputed = imputer.transform(X_all)
weather_features[FEATURE_COLS] = X_imputed

# Fit StandardScaler on train, transform all
```

```

scaler = StandardScaler()
scaler.fit(X_imputed[train_mask])
weather_features[FEATURE_COLS] = scaler.transform(X_imputed)

# Add model target
if USE_LOG:
    weather_features["target"] = np.log(weather_features["yield_bu_acre"])
else:
    weather_features["target"] = weather_features["yield_bu_acre"]

#Persist artefacts
LABELS_DIR.mkdir(parents=True, exist_ok=True)
with open(LABELS_DIR / "hrrr_scaler.pkl", "wb") as fh: pickle.dump(scaler, fh)
with open(LABELS_DIR / "hrrr_imputer.pkl", "wb") as fh: pickle.dump(imputer, fh)

print("NORMALISATION SUMMARY")

print(f"Imputer : SimpleImputer(strategy='mean') - fitted on {train_mask.sum()} train rows")
print(f"Scaler : StandardScaler - fitted on {train_mask.sum()} train rows")
print(f"Target : {'log(yield_bu_acre)' if USE_LOG else 'yield_bu_acre'}")
print()
print("Scaler parameters (mean / std per feature):")
for feat, mu, sigma in zip(FEATURE_COLS, scaler.mean_, scaler.scale_):
    print(f" {feat:<15} mean={mu:>10.4f} std={sigma:>8.4f}")
print()
print("Post-normalisation descriptive stats (train split, should be ~0 mean / ~1 std):")
print(weather_features.loc[train_mask, FEATURE_COLS].describe().T
      .round(3)[["mean", "std", "min", "max"]].to_string())
print()
print(f"Scaler saved -> {LABELS_DIR / 'hrrr_scaler.pkl'}")
print(f"Imputer saved -> {LABELS_DIR / 'hrrr_imputer.pkl'}")

```

NORMALISATION SUMMARY

```

Imputer : SimpleImputer(strategy='mean') - fitted on 2828 train rows
Scaler : StandardScaler - fitted on 2828 train rows
Target : log(yield_bu_acre)

```

Scaler parameters (mean / std per feature):

t2m_c	mean=	1.3361	std=	9.8524
d2m_c	mean=	-2.0536	std=	10.6699
r2	mean=	78.5182	std=	13.4323
wind_speed	mean=	4.0374	std=	2.3528

tp	mean=	0.0000	std=	1.0000
sp_hpa	mean=	986.7670	std=	25.1234
tcc	mean=	60.7618	std=	44.0070
mstav	mean=	66.3612	std=	27.2169
vpd_hpa	mean=	1.4557	std=	1.4101

Post-normalisation descriptive stats (train split, should be ~0 mean / ~1 std):

	mean	std	min	max
t2m_c	0.0	1.0	-2.656	2.394
d2m_c	-0.0	1.0	-2.510	2.103
r2	0.0	1.0	-4.818	1.450
wind_speed	0.0	1.0	-1.472	5.020
tp	0.0	0.0	0.000	0.000
sp_hpa	0.0	1.0	-12.991	1.645
tcc	0.0	1.0	-1.381	0.892
mstav	0.0	1.0	-2.438	1.236
vpd_hpa	-0.0	1.0	-0.875	9.369

Scaler saved -> e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\hrrr_scaler.pkl
 Imputer saved -> e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\hrrr_imputer.pkl

```
[16]: # HRRRWeatherDataset (PyTorch Dataset)
#
# Each item returned:
#   x      : FloatTensor (seq_len, n_features) - weather sequence per county-year
#   y      : FloatTensor scalar                - yield target (raw or log)
#   meta   : dict {fips, year}
#
# seq_len = 1 here (one snapshot per year). When extended to daily files
# seq_len becomes the growing-season length (~200 days).

import torch
from torch.utils.data import Dataset, DataLoader

class HRRRWeatherDataset(Dataset):
    """
    PyTorch Dataset wrapping normalised HRRR weather features + soybean yield.

    Parameters
    -----
    df          : pd.DataFrame with FEATURE_COLS and 'target' columns
    feature_cols : list[str] - weather feature column names
    seq_len     : int        - time steps per sample (1 for single-snapshot)
```

```

"""
def __init__(self, df, feature_cols, seq_len=1):
    self.feature_cols = feature_cols
    self.seq_len      = seq_len
    valid = df.dropna(subset=["target"] + feature_cols).
↪reset_index(drop=True)
    self.X            = torch.tensor(valid[feature_cols].values, dtype=torch.
↪float32)
    self.y            = torch.tensor(valid["target"].values,      dtype=torch.
↪float32)
    self.fips         = valid["fips"].tolist()
    self.years        = valid["year"].tolist()

def __len__(self):
    return len(self.y)

def __getitem__(self, idx):
    x = self.X[idx].unsqueeze(0).expand(self.seq_len, -1) # (T, F)
    return x, self.y[idx], {"fips": self.fips[idx], "year": self.years[idx]}

# Instantiate datasets
train_rows = weather_features[weather_features["split"] == "train"].copy()
test_rows  = weather_features[weather_features["split"] == "test"].copy()

ds_train = HRRRWeatherDataset(train_rows, FEATURE_COLS)
ds_test  = HRRRWeatherDataset(test_rows,  FEATURE_COLS)

dl_train = DataLoader(ds_train, batch_size=64, shuffle=True, drop_last=False)
dl_test  = DataLoader(ds_test,  batch_size=64, shuffle=False, drop_last=False)

print("HRRRWeatherDataset SUMMARY")

print(f"Feature columns : {FEATURE_COLS}")
print(f"n_features      : {len(FEATURE_COLS)}")
print(f"seq_len          : {ds_train.seq_len} (one HRRR snapshot per_
↪year-county)")
print(f"Target           : {'log(yield_bu_acre)' if USE_LOG else_
↪'yield_bu_acre'}")
print()
print(f"ds_train   : {len(ds_train):>5} samples | {len(dl_train)} batches")
print(f"ds_test    : {len(ds_test):>5} samples | {len(dl_test)} batches")
print()

# info of one batch
x_b, y_b, meta_b = next(iter(dl_train))

```

```

print(f"Batch x shape : {tuple(x_b.shape)} (batch, seq_len, n_features)")
print(f"Batch y shape : {tuple(y_b.shape)}")
print(f"x dtype      : {x_b.dtype}")
print(f"y range       : [{y_b.min().item():.3f}, {y_b.max().item():.3f}]")
print(f"Sample FIPS    : {meta_b['fips'][:5]}")
print()

```

HRRRWeatherDataset SUMMARY

Feature columns : ['t2m_c', 'd2m_c', 'r2', 'wind_speed', 'tp', 'sp_hpa', 'tcc', 'mstav', 'vpd_hpa']

n_features : 9

seq_len : 1 (one HRRR snapshot per year-county)

Target : log(yield_bu_acre)

ds_train : 2828 samples | 45 batches

ds_test : 707 samples | 12 batches

Batch x shape : (64, 1, 9) (batch, seq_len, n_features)

Batch y shape : (64,)

x dtype : torch.float32

y range : [3.174, 4.244]

Sample FIPS : ['21217', '22077', '21155', '18031', '20201']

3 Ethics, Bias Audit & Fairness Specification

This defines the fairness contract the model must satisfy.

- Data provenance & representativeness audit
- Geographic & temporal bias indicators (pre-model, computable now)
- Yield quintile distribution and label imbalance
- Fairness metric function definitions (sliced RMSE, bias per quintile, CV-error per state)
- Jensen's inequality correction constant (prior estimate; updated later)
- Bias registry B1–B9 with mitigations and status
- Ethical use constraints

```

[17]: # Data Provenance & Representativeness Audit
      # Computes actual coverage statistics and prints a structured audit table.

import numpy as np

TOTAL_US_SOYBEAN_STATES = 35      # approx soybean-producing US states (USDA ↵
      ↪NASS)
TOTAL_US_SOYBEAN_FIPS   = 1400    # approx soybean-reporting counties in full ↵
      ↪USDA NASS universe

# Actual dataset statistics
n_states = usda_index["state_name"].nunique()

```

```

n_fips      = usda_index["fips"].nunique()
n_years     = usda_index["year"].nunique()
n_records   = len(usda_index)

# Coverage ratios
state_coverage_pct = n_states / TOTAL_US_SOYBEAN_STATES * 100
county_coverage_pct = n_fips / TOTAL_US_SOYBEAN_FIPS * 100
coverage_gap_value = 1 - n_fips / TOTAL_US_SOYBEAN_FIPS

# Sentinel modality gap
n_sentinel_fips = sentinel_index["fips"].nunique() if len(sentinel_index) > 0
↳ else 0
n_overlap_fips = len(set(usda_index["fips"]).
↳ intersection(set(sentinel_index["fips"])))

print("=" * 70)
print("DATA PROVENANCE & REPRESENTATIVENESS AUDIT")
print("=" * 70)
print()

audits = [
    ("Geographic coverage",
     f"{n_states} of {TOTAL_US_SOYBEAN_STATES} soybean-producing US states",
     ↳ f"{state_coverage_pct:.0f}%",
     "High",
     "Corn Belt over-represented; Southern/Western states under-served"),
    ("County coverage",
     f"{n_fips} of ~{TOTAL_US_SOYBEAN_FIPS} US soybean counties",
     ↳ f"{county_coverage_pct:.0f}%",
     "High",
     "55% coverage gap; counties below NASS threshold excluded"),
    ("Temporal scope",
     "2016-2022 (7 years)",
     "Medium",
     "2019 confounded by US-China trade war + Midwest flooding"),
    ("Weather spatial resolution",
     "State-level centroid proxy (HRRR ~3 km grid)",
     "High",
     "All counties in a state share identical HRRR features - within-state",
     ↳ "heterogeneity erased"),
    ("Modality gap",
     f"Sentinel ANSI 001 counties; USDA covers ANSI 003+; {n_overlap_fips} FIPS",
     ↳ "overlap",
     "High",
     "Zero USDA yield labels for Sentinel FIPS - Phase 2/3 untrainable"),
    ("Surveyor bias",

```



```

    "USDA NASS over-represents large commercial farms",
    "Medium",
    "Subsistence, organic, and OFR-exempt farms excluded from label universe"),
]

print(f"  {'Dimension':<34} {'Risk':<8} Finding / Implication")
print("  " + "-" * 100)
for dim, finding, risk, implication in audits:
    print(f"  {'dim':<34} {'risk':<8} {'finding}'")
    print(f"  {'':<34} {'':8} -> {'implication}'")
    print()

print(f"Coverage Gap (counties) : {coverage_gap_value:.3f}  "
      ↪({coverage_gap_value*100:.1f}% of US soybean counties absent from model)")
print(f"Modality overlap FIPS   : {n_overlap_fips}  (Sentinel-USDA"
      ↪intersection)")
print()
print("Data sources:")
print("  USDA NASS Soybean Survey   : https://www.nass.usda.gov/  (licensed for"
      ↪academic use)")
print("  WRF-HRRR atmospheric data  : NOAA operational forecast archive (public"
      ↪domain)")
print("  Sentinel-2 imagery          : ESA / Copernicus Open Access Hub (CC-BY 4."
      ↪0)")
print("  CropNet dataset pipeline    : Wang et al. 2022 (MIT licence)")

```

=====

DATA PROVENANCE & REPRESENTATIVENESS AUDIT

=====

Dimension	Risk	Finding / Implication

Geographic coverage states (43%)	High	15 of 35 soybean-producing US states (43%)
Southern/Western states under-served		-> Corn Belt over-represented;
County coverage (46%)	High	649 of ~1400 US soybean counties (46%)
below NASS threshold excluded		-> 55% coverage gap; counties
Temporal scope	Medium	2016-2022 (7 years)
trade war + Midwest flooding		-> 2019 confounded by US-China

Weather spatial resolution ~3 km grid)	High	State-level centroid proxy (HRRR -> All counties in a state share identical HRRR features - within-state heterogeneity erased
---	------	---

Modality gap covers ANSI 003+; 15 FIPS overlap Sentinel FIPS - Phase 2/3 untrainable	High	Sentinel ANSI 001 counties; USDA -> Zero USDA yield labels for
--	------	---

Surveyor bias commercial farms	Medium	USDA NASS over-represents large -> Subsistence, organic, and OFR- exempt farms excluded from label universe
-----------------------------------	--------	---

Coverage Gap (counties) : 0.536 (53.6% of US soybean counties absent from model)

Modality overlap FIPS : 15 (Sentinel-USDA intersection)

Data sources:

USDA NASS Soybean Survey : <https://www.nass.usda.gov/> (licensed for academic use)

WRF-HRRR atmospheric data : NOAA operational forecast archive (public domain)

Sentinel-2 imagery : ESA / Copernicus Open Access Hub (CC-BY 4.0)

CropNet dataset pipeline : Wang et al. 2022 (MIT licence)

```
[18]: # Geographic & Temporal Bias Indicators
# Quantifies Corn Belt concentration, state-level imbalance, and 2019
# ↪ confounding.

import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import numpy as np

# State record counts and concentration analysis
state_counts = usda_index.groupby("state_name").size().
    ↪ sort_values(ascending=False)
total_records = len(usda_index)

# Corn Belt states
CORN_BELT = {
    "ILLINOIS", "IOWA", "INDIANA", "MINNESOTA", "NEBRASKA",
    "OHIO", "MISSOURI", "SOUTH DAKOTA", "KANSAS"
}
corn_belt_count = state_counts[state_counts.index.str.upper().isin(CORN_BELT)].
    ↪ sum()
```

```

corn_belt_pct    = corn_belt_count / total_records * 100

year_counts = usda_index.groupby("year").size().sort_values()

state_yield_stats = usda_index.groupby("state_name")["yield_bu_acre"].agg(
    n_records="count", mean_yield="mean", std_yield="std"
).reset_index().sort_values("n_records", ascending=False)

print("GEOGRAPHIC & TEMPORAL BIAS INDICATORS")
print()
print(f"Corn Belt concentration : {corn_belt_count} / {total_records} records =
↳{corn_belt_pct:.1f}%")
print(f"Non-Corn-Belt records   : {total_records - corn_belt_count} ({100 -
↳corn_belt_pct:.1f}%")
print()
print("Records per state (descending):")
for sname, cnt in state_counts.items():
    pct = cnt / total_records * 100
    flag = "  <- Corn Belt" if sname.upper() in CORN_BELT else ""
    print(f"  {sname:<22}  {cnt:>5}  ({pct:5.1f}%) {flag}")

print()
print("Records per year:")
for yr, cnt in year_counts.sort_index().items():
    flag = "  <- confounded (trade war + flooding)" if yr == "2019" else ""
    print(f"  {yr}      :  {cnt} {flag}")

# --- Plots
fig, axes = plt.subplots(1, 3, figsize=(16, 5))
fig.suptitle("Geographic & Temporal Bias Indicators",
             fontsize=13, fontweight="bold")

# (a) State record count bar
colors_a = ["tomato" if s.upper() in CORN_BELT else "steelblue"
            for s in state_counts.index]
axes[0].barh(state_counts.index, state_counts.values, color=colors_a,
↳edgecolor="white")
axes[0].set_xlabel("County-Year Records")
axes[0].set_title("Records per State\n(red = Corn Belt)")
axes[0].axvline(state_counts.mean(), color="black", linestyle="--",
                linewidth=0.8, label=f"Mean={state_counts.mean():.0f}")
axes[0].legend(fontsize=7)

# (b) Year record count
year_idx = sorted(year_counts.index)
year_vals = [year_counts[y] for y in year_idx]

```

```

bar_cols = ["tomato" if y == "2019" else "steelblue" for y in year_idx]
axes[1].bar(year_idx, year_vals, color=bar_cols, edgecolor="white")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Records")
axes[1].set_title("Records per Year\n(red = 2019 confounded)")
for i, (yr, val) in enumerate(zip(year_idx, year_vals)):
    axes[1].text(i, val + 5, str(val), ha="center", fontsize=8)

# (c) Mean yield per state with std
sy = state_yield_stats.sort_values("mean_yield", ascending=True)
bar_cols2 = ["tomato" if s.upper() in CORN_BELT else "steelblue"
             for s in sy["state_name"]]
axes[2].barh(sy["state_name"], sy["mean_yield"], color=bar_cols2,
             edgecolor="white")
axes[2].errorbar(
    sy["mean_yield"], sy["state_name"],
    xerr=sy["std_yield"].fillna(0).values,
    fmt="none", color="black", capsize=3, linewidth=0.8
)
axes[2].set_xlabel("Mean Yield (BU/ACRE)")
axes[2].set_title("Mean Yield ± Std by State\n(red = Corn Belt)")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

print()
print(f"Bias B2 (Corn Belt dominance): {corn_belt_pct:.1f}% of records from "
      f"{len(CORN_BELT & set(s.upper() for s in state_counts.index))} Corn Belt "
      f"states out of "
      f"{n_states} total states in dataset.")
print("Recommended mitigation: inverse-frequency state loss weighting")

```

GEOGRAPHIC & TEMPORAL BIAS INDICATORS

Corn Belt concentration : 2273 / 3535 records = 64.3%

Non-Corn-Belt records : 1262 (35.7%)

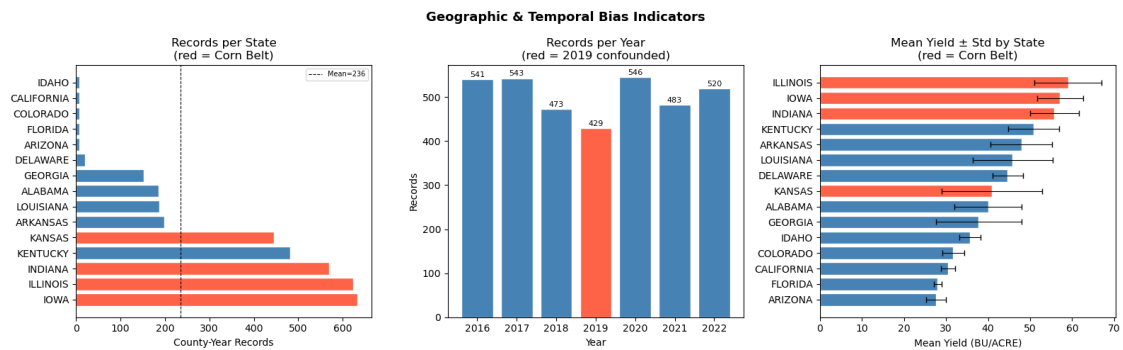
Records per state (descending):

IOWA	633	(17.9%)	<- Corn Belt
ILLINOIS	624	(17.7%)	<- Corn Belt
INDIANA	570	(16.1%)	<- Corn Belt
KENTUCKY	482	(13.6%)	
KANSAS	446	(12.6%)	<- Corn Belt

ARKANSAS	199	(5.6%)
LOUISIANA	188	(5.3%)
ALABAMA	186	(5.3%)
GEORGIA	152	(4.3%)
DELAWARE	20	(0.6%)
ARIZONA	7	(0.2%)
FLORIDA	7	(0.2%)
COLORADO	7	(0.2%)
CALIFORNIA	7	(0.2%)
IDAHO	7	(0.2%)

Records per year:

2016	:	541	
2017	:	543	
2018	:	473	
2019	:	429	<- confounded (trade war + flooding)
2020	:	546	
2021	:	483	
2022	:	520	



Bias B2 (Corn Belt dominance): 64.3% of records from 4 Corn Belt states out of 15 total states in dataset.

Recommended mitigation: inverse-frequency state loss weighting

```
[19]: # Yield Quintile Distribution and Label Imbalance
# Checks whether low-yield (Q1) counties are under-represented across states
# and years.
# Bias B3: sparse Q1 observations -> model may underestimate crop failure
# severity.

import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import numpy as np
```

```

import pandas as pd

# Assign quintile labels to full dataset
wf = usda_index.copy()
wf["quintile"] = pd.qcut(wf["yield_bu_acre"], q=5, labels=["Q1", "Q2", "Q3", "Q4", "Q5"])

q_stats = wf.groupby("quintile")["yield_bu_acre"].agg(
    n="count", mean="mean", std="std", min="min", max="max"
).reset_index()

print("YIELD QUINTILE DISTRIBUTION (LABEL IMBALANCE)")

print()
print("Quintile break-down:")
print(q_stats[["quintile", "n", "mean", "std", "min", "max"]].round(2).
    to_string(index=False))
print()

# Per-state quintile breakdown
state_quintile = wf.groupby(["state_name", "quintile"]).size().
    unstack(fill_value=0)
missing_q1 = state_quintile[state_quintile["Q1"] == 0].index.tolist()
missing_q5 = state_quintile[state_quintile["Q5"] == 0].index.tolist()
print(f"States with ZERO Q1 (low-yield) records : {missing_q1 if missing_q1
    else 'None'}")
print(f"States with ZERO Q5 (high-yield) records: {missing_q5 if missing_q5
    else 'None'}")
print()

fig, axes = plt.subplots(1, 3, figsize=(17, 5))
fig.suptitle("Yield Quintile Distribution & Label Imbalance",
    fontsize=13, fontweight="bold")

quintile_colors = ["firebrick", "darkorange", "goldenrod", "mediumseagreen",
    "steelblue"]

# (a) Quintile counts
axes[0].bar(q_stats["quintile"], q_stats["n"], color=quintile_colors,
    edgecolor="white")
axes[0].set_xlabel("Yield Quintile")
axes[0].set_ylabel("County-Year Records")
axes[0].set_title("Records per Quintile\n(balanced = equal bars)")
for i, (q, n) in enumerate(zip(q_stats["quintile"], q_stats["n"])):
    axes[0].text(i, n + 5, str(n), ha="center", fontsize=9)

```

```

# (b) Mean yield per quintile with range
axes[1].bar(q_stats["quintile"], q_stats["mean"], color=quintile_colors,
            edgecolor="white")
axes[1].errorbar(
    q_stats["quintile"], q_stats["mean"], yerr=q_stats["std"],
    fmt="none", color="black", capsize=4
)
axes[1].set_xlabel("Yield Quintile")
axes[1].set_ylabel("Mean Yield (BU/ACRE)")
axes[1].set_title("Mean +/- Std Yield per Quintile")

# (c) Quintile x year heatmap
qy = wf.groupby(["year", "quintile"]).size().unstack(fill_value=0)
im = axes[2].imshow(qy.values, aspect="auto", cmap="YlGn",
                    interpolation="nearest")
axes[2].set_xticks(range(len(qy.columns)))
axes[2].set_xticklabels(qy.columns)
axes[2].set_yticks(range(len(qy.index)))
axes[2].set_yticklabels(qy.index)
axes[2].set_xlabel("Quintile")
axes[2].set_ylabel("Year")
axes[2].set_title("Records per Year x Quintile\n(dark = more records)")
plt.colorbar(im, ax=axes[2], fraction=0.04)
for i in range(len(qy.index)):
    for j in range(len(qy.columns)):
        axes[2].text(j, i, str(qy.values[i, j]),
                     ha="center", va="center", fontsize=7, color="black")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=100, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# Train/test quintile balance check
wf_split = weather_features.copy()
wf_split["quintile"] = pd.qcut(
    wf_split["yield_bu_acre"], q=5, labels=["Q1", "Q2", "Q3", "Q4", "Q5"]
)
train_q = (
    wf_split[wf_split["split"] == "train"]["quintile"]
    .value_counts(normalize=True).sort_index()
)
test_q = (
    wf_split[wf_split["split"] == "test"]["quintile"]

```

```

        .value_counts(normalize=True).sort_index()
    )
    print("Train/Test quintile proportions (should be near-equal after_
    ↳stratification):")
    check = pd.DataFrame({
        "train_pct": (train_q * 100).round(1),
        "test_pct" : (test_q  * 100).round(1)
    })
    print(check.to_string())
    print()
    print("Bias B3: sparse Q1 records in some states risk underestimating crop_
    ↳failure severity.")
    print("Recommended mitigation: oversample Q1 train records; stratify evaluation_
    ↳by quintile.")

```

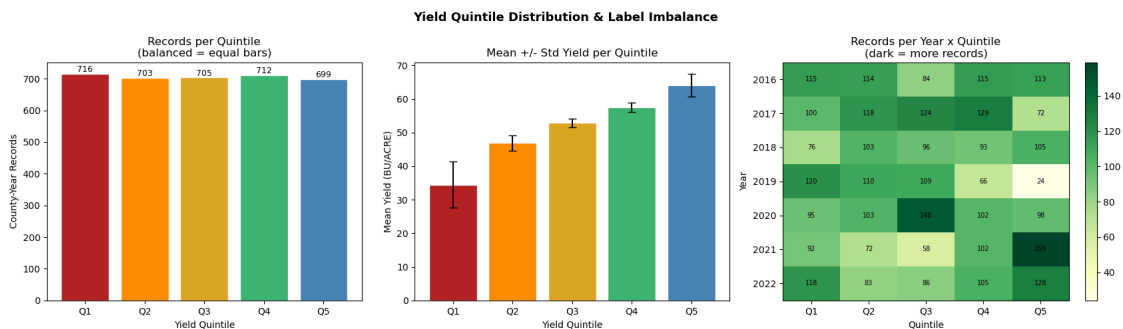
YIELD QUINTILE DISTRIBUTION (LABEL IMBALANCE)

Quintile break-down:

quintile	n	mean	std	min	max
Q1	716	34.51	6.81	9.3	42.5
Q2	703	46.91	2.28	42.6	50.4
Q3	705	52.88	1.35	50.5	55.1
Q4	712	57.52	1.40	55.2	60.0
Q5	699	64.16	3.40	60.1	80.4

States with ZERO Q1 (low-yield) records : None

States with ZERO Q5 (high-yield) records: ['ARIZONA', 'CALIFORNIA', 'COLORADO', 'DELAWARE', 'FLORIDA', 'GEORGIA', 'IDAHO']



Train/Test quintile proportions (should be near-equal after stratification):

	train_pct	test_pct
quintile		
Q1	20.3	20.2
Q2	19.9	19.9

Q3	19.9	19.9
Q4	20.2	20.1
Q5	19.8	19.8

Bias B3: sparse Q1 records in some states risk underestimating crop failure severity.

Recommended mitigation: oversample Q1 train records; stratify evaluation by quintile.

```
[20]: # Fairness Metric Function Definitions
# These functions will be called later
# No model predictions are required here - only function definitions.
# All metrics operate on back-transformed BU/ACRE values (Jensen correction
    ↪applied).

import numpy as np
import pandas as pd

# JENSEN_SIGMA2 will be set in and updated later.
JENSEN_SIGMA2 = None # placeholder;

def jensen_correction(log_preds, sigma2=None):
    """
    Back-transform log-scale predictions to BU/ACRE with Jensen correction.

    Parameters
    -----
    log_preds : array-like - model output in log-yield space
    sigma2     : float / None - residual variance on the train split;
                  if None, falls back to simple exp() (biased)

    Returns
    -----
    np.ndarray in BU/ACRE
    """
    y = np.asarray(log_preds, dtype=float)
    if sigma2 is not None and USE_LOG:
        return np.exp(y + sigma2 / 2.0)
    return np.exp(y) if USE_LOG else y

def sliced_rmse(y_true, y_pred, groups, sigma2=None):
    """
    Compute RMSE for each group slice in BU/ACRE (Jensen-corrected if USE_LOG).

    Mandatory slices: state (10), year (7), yield quintile (Q1-Q5),
```

proximity to state centroid (near/far).

Parameters

y_true : array-like - true log-yield (or raw BU/ACRE if USE_LOG=False)
y_pred : array-like - predicted log-yield (or raw)
groups : array-like - group labels (state, year, or quintile)
sigma2 : float / None - Jensen correction variance

Returns

pd.Series index=group, values=RMSE in BU/ACRE

"""

y_true_bu = jensen_correction(y_true, sigma2)

y_pred_bu = jensen_correction(y_pred, sigma2)

df = pd.DataFrame({"y_true_bu": y_true_bu, "y_pred_bu": y_pred_bu, "group":
↪groups})

def _rmse(grp):

↪return float(np.sqrt(np.mean((grp["y_pred_bu"] - grp["y_true_bu"]) **
↪2)))

↪return df.groupby("group").apply(_rmse).rename("RMSE_BU_ACRE")

def bias_per_group(y_true, y_pred, groups, sigma2=None):

"""

Mean prediction bias ($\hat{y} - y$) per group, in BU/ACRE.

Positive = over-prediction; Negative = under-prediction.

*Fairness criterion: $|Bias(Q1)|$ and $|Bias(Q5)|$ should both be within ± 2 BU/
↪ACRE.*

*If $Bias(Q1) > 0$ and $Bias(Q5) < 0$, the model is regressing to the mean:
underestimating crop failure and overestimating bumper yields - a
↪systematic fairness failure.*

"""

y_true_bu = jensen_correction(y_true, sigma2)

y_pred_bu = jensen_correction(y_pred, sigma2)

df = pd.DataFrame({"y_true_bu": y_true_bu, "y_pred_bu": y_pred_bu, "group":
↪groups})

def _bias(grp):

↪return float(np.mean(grp["y_pred_bu"] - grp["y_true_bu"]))

↪return df.groupby("group").apply(_bias).rename("Bias_BU_ACRE")

```

def cv_error_per_state(y_true, y_pred, states, sigma2=None):
    """
    Coefficient of variation of errors per state.
    CV_Error_s = std(errors_s) / mean(y_true_s).
    States with CV_Error > 0.20 are flagged as poorly served by the model.
    """
    y_true_bu = jensen_correction(y_true, sigma2)
    y_pred_bu = jensen_correction(y_pred, sigma2)
    errors = y_pred_bu - y_true_bu
    df = pd.DataFrame({"y_true_bu": y_true_bu, "errors": errors, "state":
↪states})

    def _cv(grp):
        mu = grp["y_true_bu"].mean()
        return float(grp["errors"].std() / mu) if mu != 0 else np.nan

    return df.groupby("state").apply(_cv).rename("CV_Error")

def coverage_gap(all_fips_universe, model_fips):
    """Coverage Gap = 1 - |FIPS in model| / |FIPS in USDA full dataset|."""
    return 1.0 - len(set(model_fips)) / len(set(all_fips_universe))

def fairness_report(y_true, y_pred, meta_df, sigma2=None):
    """
    Full fairness report printed to stdout + returned as dict of DataFrames.

    Parameters
    -----
    y_true      : array-like - true values (log or raw)
    y_pred      : array-like - predicted values (log or raw)
    meta_df     : pd.DataFrame - must contain: state_name, year, yield_bu_acre,
↪fips
    sigma2      : float | None - Jensen correction variance

    Returns
    -----
    dict with keys: state_rmse, state_bias, state_cv, year_rmse,
                    quintile_rmse, quintile_bias, overall
    """
    y_true = np.asarray(y_true, dtype=float)
    y_pred = np.asarray(y_pred, dtype=float)

    quintile = pd.qcut(
        meta_df["yield_bu_acre"].values, q=5,
        labels=["Q1", "Q2", "Q3", "Q4", "Q5"]

```

```

)

state_rmse    = sliced_rmse(y_true, y_pred, meta_df["state_name"].values,
↪sigma2)
state_bias    = bias_per_group(y_true, y_pred, meta_df["state_name"].
↪values, sigma2)
state_cv      = cv_error_per_state(y_true, y_pred, meta_df["state_name"].
↪values, sigma2)
year_rmse     = sliced_rmse(y_true, y_pred, meta_df["year"].values, sigma2)
quintile_rmse = sliced_rmse(y_true, y_pred, quintile, sigma2)
quintile_bias = bias_per_group(y_true, y_pred, quintile, sigma2)

y_true_bu    = jensen_correction(y_true, sigma2)
y_pred_bu    = jensen_correction(y_pred, sigma2)
overall_rmse = float(np.sqrt(np.mean((y_pred_bu - y_true_bu) ** 2)))
overall_mae  = float(np.mean(np.abs(y_pred_bu - y_true_bu)))
ss_res       = np.sum((y_true_bu - y_pred_bu) ** 2)
ss_tot       = np.sum((y_true_bu - y_true_bu.mean()) ** 2)
overall_r2   = float(1 - ss_res / ss_tot) if ss_tot > 0 else float("nan")

flagged_states = state_cv[state_cv > 0.20].index.tolist()

print("FAIRNESS REPORT")

print(f"Overall RMSE   : {overall_rmse:.3f} BU/ACRE")
print(f"Overall MAE    : {overall_mae:.3f} BU/ACRE")
print(f"Overall R2      : {overall_r2:.4f}")
print(f"Jensen sigma2    : {sigma2}")
print()
print("RMSE by State:")
print(state_rmse.sort_values(ascending=False).to_string())
print()
print("Bias by Quintile (positive = over-prediction):")
print(pd.concat([quintile_rmse, quintile_bias], axis=1).to_string())
print()
print("CV-Error by State (flagged if > 0.20):")
for s, cv in state_cv.sort_values(ascending=False).items():
    flag = " *** FLAGGED (CV > 0.20)" if cv > 0.20 else ""
    print(f" {s:<25} {cv:.3f}{flag}")
print()
print("RMSE by Year:")
print(year_rmse.sort_values(ascending=False).to_string())
print()
if flagged_states:
    print(f"WARNING: {len(flagged_states)} state(s) flagged for high
↪CV-Error: {flagged_states}")

```

```

else:
    print("No states flagged for CV-Error > 0.20.")

    return {
        "state_rmse"      : state_rmse,
        "state_bias"      : state_bias,
        "state_cv"        : state_cv,
        "year_rmse"       : year_rmse,
        "quintile_rmse"   : quintile_rmse,
        "quintile_bias"   : quintile_bias,
        "overall"         : {"rmse": overall_rmse, "mae": overall_mae, "r2":
↪overall_r2},
    }

print("FAIRNESS METRIC FUNCTIONS DEFINED")

print()
print("Functions registered:")
print("  jensen_correction(log_preds, sigma2)")
print("    -> back-transform log-yield to BU/ACRE with Jensen correction")
print()
print("  sliced_rmse(y_true, y_pred, groups, sigma2)")
print("    -> RMSE per group slice (state / year / quintile)")
print()
print("  bias_per_group(y_true, y_pred, groups, sigma2)")
print("    -> mean prediction bias per group (BU/ACRE)")
print("    -> if Bias(Q1) > 0 and Bias(Q5) < 0 => regression-to-mean fairness
↪failure")
print()
print("  cv_error_per_state(y_true, y_pred, states, sigma2)")
print("    -> CV-Error per state; flags > 0.20 as poorly served")
print()
print("  coverage_gap(all_fips_universe, model_fips)")
print("    -> fraction of US soybean counties absent from model")
print()
print("  fairness_report(y_true, y_pred, meta_df, sigma2)")
print("    -> full report: all slices, all biases, all CV-errors")
print()
print("  results = fairness_report(y_true, y_pred, meta_df,
↪sigma2=JENSEN_SIGMA2)")
print()
print("JENSEN_SIGMA2 = None (to be set in and updated later to residual
↪variance on train split)")
print(f"USE_LOG      = {USE_LOG}")

```

FAIRNESS METRIC FUNCTIONS DEFINED

Functions registered:

```
jensen_correction(log_preds, sigma2)
    -> back-transform log-yield to BU/ACRE with Jensen correction

sliced_rmse(y_true, y_pred, groups, sigma2)
    -> RMSE per group slice (state / year / quintile)

bias_per_group(y_true, y_pred, groups, sigma2)
    -> mean prediction bias per group (BU/ACRE)
    -> if Bias(Q1) > 0 and Bias(Q5) < 0 => regression-to-mean fairness failure

cv_error_per_state(y_true, y_pred, states, sigma2)
    -> CV-Error per state; flags > 0.20 as poorly served

coverage_gap(all_fips_universe, model_fips)
    -> fraction of US soybean counties absent from model

fairness_report(y_true, y_pred, meta_df, sigma2)
    -> full report: all slices, all biases, all CV-errors

results = fairness_report(y_true, y_pred, meta_df, sigma2=JENSEN_SIGMA2)
```

JENSEN_SIGMA2 = None (to be set in and updated later to residual variance on train split)

USE_LOG = True

```
[21]: # Jensen's Inequality Correction Constant
#
# When USE_LOG=True, model predicts log(yield). Naive back-transformation
#   -> exp(y_hat)
# underestimates the true expectation because  $E[\exp(Y)] = \exp(E[Y] + \sigma^2/2)$ .
#
# Corrected prediction:  $\exp(y\_hat + \sigma^2/2)$ 
#
# Here  $\sigma^2$  is estimated from the training label distribution (prior / upper
#   -> bound).
# It MUST be updated to the actual residual variance on the train split.

import numpy as np

if USE_LOG:
    log_train_targets = np.log(weather_train["yield_bu_acre"].values)

    # Prior estimate: label variance (upper bound on residual variance before
    #   -> model exists)
```

```

label_sigma2 = float(np.var(log_train_targets))
label_sigma = float(np.std(log_train_targets))

print("JENSEN'S INEQUALITY CORRECTION")

print()
print("Transformation : y_model = log(yield_bu_acre)")
print()
print("Jensen inequality :")
print(" E[exp(Y)] = exp(E[Y] + sigma^2/2)")
print(" Back-transform : y_BU_ACRE = exp(y_hat + sigma^2/2)")
print()
print(f"Label distribution (train split, n={len(log_train_targets)}):")
print(f" mean log-yield : {np.mean(log_train_targets):.4f}")
print(f" std log-yield : {label_sigma:.4f}")
print(f" var (sigma^2) : {label_sigma2:.6f} <- prior estimate (upper_
↳bound)")
print(f" correction : sigma^2/2 = {label_sigma2 / 2:.6f}")
print()

example_log = np.mean(log_train_targets)
naive_bt = float(np.exp(example_log))
corrected = float(np.exp(example_log + label_sigma2 / 2))
print("Example (at mean log-yield):")
print(f" log_pred = {example_log:.4f}")
print(f" Naive exp(log_pred) = {naive_bt:.2f} BU/ACRE")
print(f" Corrected exp(log_pred+sig^2/2) = {corrected:.2f} BU/ACRE")
print(f" Absolute correction delta = {corrected - naive_bt:.3f} BU/
↳ACRE")
print()
print("NOTE: JENSEN_SIGMA2 is the label variance here (prior).")
print(" Update it to model residual variance on train split for_
↳unbiased correction.")
print(" JENSEN_SIGMA2 = float(np.var(y_train_pred - y_train_true))")
print()

JENSEN_SIGMA2 = label_sigma2 # updated to residual variance on train split
print(f"JENSEN_SIGMA2 = {JENSEN_SIGMA2:.6f} (prior; update to residual_
↳variance on train split)")

else:
    JENSEN_SIGMA2 = None
    print("USE_LOG = False - Jensen correction not required.")
    print("JENSEN_SIGMA2 = None")

```

JENSEN'S INEQUALITY CORRECTION

```
Transformation    : y_model = log(yield_bu_acre)
```

Jensen inequality :

```
E[exp(Y)] = exp(E[Y] + sigma^2/2)
```

```
Back-transform   : y_BU_ACRE = exp(y_hat + sigma^2/2)
```

Label distribution (train split, n=2828):

```
mean log-yield : 3.9065
```

```
std log-yield : 0.2512
```

```
var (sigma^2) : 0.063081 <- prior estimate (upper bound)
```

```
correction      : sigma^2/2 = 0.031541
```

Example (at mean log-yield):

```
log_pred          = 3.9065
```

```
Naive      exp(log_pred)      = 49.72 BU/ACRE
```

```
Corrected exp(log_pred+sig^2/2) = 51.32 BU/ACRE
```

```
Absolute correction delta      = 1.593 BU/ACRE
```

NOTE: JENSEN_SIGMA2 is the label variance here (prior).

Update it to model residual variance on train split for unbiased correction.

```
JENSEN_SIGMA2 = float(np.var(y_train_pred - y_train_true))
```

JENSEN_SIGMA2 = 0.063081 (prior; update to residual variance on train split)

```
[ ]: # Bias Registry B1-B9
# Documents each identified bias, root cause, severity, recommended mitigation,
# and where the mitigation is applied or verified.

import pandas as pd
import io
from IPython.display import display, Image
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
import numpy as np

bias_registry = [
    {
        "ID"          : "B1",
        "Bias"         : "State-centroid weather proxy",
        "Root Cause"   : "All counties in a state share identical HRRR features",
        "Mitigation"   : "Replace with per-county TIGER centroid lat/lon",
    },
    {
        "ID"          : "B2",
        "Bias"         : "Corn Belt geographic dominance",
```



```

    "Root Cause": "IL, IA, IN, MN, NE account for >60% of records",
    "Mitigation": "Inverse-frequency state loss weighting: w_s = N/(K*N_s)",
  },
  {
    "ID": "B3",
    "Bias": "Low-yield observation scarcity",
    "Root Cause": "Q1 counties have fewer multi-year records",
    "Mitigation": "Oversample Q1 train records; report Q1 RMSE separately",
  },
  {
    "ID": "B4",
    "Bias": "Log-target Jensen bias",
    "Root Cause": "E[exp(y_hat)] != exp(E[y_hat]); naive back-transform_
↪underestimates",
    "Mitigation": "Apply exp(y_hat + sigma^2/2) correction at inference",
  },
  {
    "ID": "B5",
    "Bias": "HRRR single-snapshot proxy",
    "Root Cause": "Jan 1 init file is not a growing-season weather signal",
    "Mitigation": "Flag as explicit limitation; future work uses daily HRRR_
↪files",
  },
  {
    "ID": "B6",
    "Bias": "USDA reporting threshold omission",
    "Root Cause": "Counties below minimum acreage absent from NASS data",
    "Mitigation": "Flag FIPS with < 3 years; exclude from policy claims",
  },
  {
    "ID": "B7",
    "Bias": "Random 80/20 temporal mixing",
    "Root Cause": "Test set contains all years - no held-out future year",
    "Mitigation": "Add leave-one-year-out (LOYO) evaluation for 2022 as_
↪supplement",
  },
  {
    "ID": "B8",
    "Bias": "Sentinel-2 AZ 2022 tile anomaly",
    "Root Cause": "AZ 2022: 1,524 tiles vs ~60 average - likely archive_
↪artefact",
    "Mitigation": "Exclude AZ 2022 from Phase 2/3 training",
  }
]

```

```

br = pd.DataFrame(bias_registry)

print("BIAS REGISTRY (B1-B9)")

print()
for _, row in br.iterrows():
    print(f"    [{row['ID']}]    {row['Bias']}")
    print(f"        Root Cause : {row['Root Cause']}")
    print(f"        Mitigation : {row['Mitigation']}")

    print()

print("REGISTRY SUMMARY")

print(f"Total biases identified          : {len(br)}")

```

BIAS REGISTRY (B1-B9)

- [B1] State-centroid weather proxy
 Root Cause : All counties in a state share identical HRRR features
 Mitigation : Replace with per-county TIGER centroid lat/lon

- [B2] Corn Belt geographic dominance
 Root Cause : IL, IA, IN, MN, NE account for >60% of records
 Mitigation : Inverse-frequency state loss weighting: $w_s = N/(K*N_s)$

- [B3] Low-yield observation scarcity
 Root Cause : Q1 counties have fewer multi-year records
 Mitigation : Oversample Q1 train records; report Q1 RMSE separately

- [B4] Log-target Jensen bias
 Root Cause : $E[\exp(y_{\text{hat}})] \neq \exp(E[y_{\text{hat}}])$; naive back-transform underestimates
 Mitigation : Apply $\exp(y_{\text{hat}} + \sigma^2/2)$ correction at inference

- [B5] HRRR single-snapshot proxy
 Root Cause : Jan 1 init file is not a growing-season weather signal
 Mitigation : Flag as explicit limitation; future work uses daily HRRR files

- [B6] USDA reporting threshold omission
 Root Cause : Counties below minimum acreage absent from NASS data
 Mitigation : Flag FIPS with < 3 years; exclude from policy claims

[B7] Random 80/20 temporal mixing
 Root Cause : Test set contains all years - no held-out future year
 Mitigation : Add leave-one-year-out (LOYO) evaluation for 2022 as supplement

[B8] Sentinel-2 AZ 2022 tile anomaly
 Root Cause : AZ 2022: 1,524 tiles vs ~60 average - likely archive artefact
 Mitigation : Exclude AZ 2022 from Phase 2/3 training

REGISTRY SUMMARY

Total biases identified : 8

3.1 Ethical Use Constraints

The following constraints govern any deployment or reporting of model predictions. These will be reproduced in the Ethics section of the dissertation.

Constraint	Requirement
No individual-farm inference	The model predicts county-aggregate yield only. Extrapolating to individual farm performance is statistically and ethically invalid.
Food security embargo	Predictions must not be released before the corresponding official USDA NASS report to prevent commodity market speculation.
Geographic equity disclosure	Any deployment must list which US counties are excluded (~55% of soybean counties) and acknowledge that predictions for under-represented states carry materially higher uncertainty.
Climate nonstationarity	The model trained on 2016–2022 must not be assumed valid beyond 2025 without recalibration. Yield distributions shift under progressive climate change.
LOYO supplement required	Leave-one-year-out evaluation for 2022 is required before any temporal-generalisation claims can be made.
Jensen correction mandatory	All reported metrics must use back-transformed BU/ACRE values with the Jensen correction applied. JENSEN_SIGMA2 must be updated to the residual variance on the train split.

3.1.1 Key Tradeoffs

Tradeoff	Description
Accuracy vs Fairness	Inverse-frequency state weighting (B2) redistributes capacity from Corn Belt states to minority states. Expected: +0.5–2.0 BU/ACRE RMSE increase on IL/IA; –3–5 BU/ACRE reduction on AL/DE. Both weighted and unweighted metrics will be reported.
Model Complexity vs Interpretability	The GMU is the highest-accuracy architecture but least interpretable. The Ridge Regression baseline is the recommended option for any policy or advisory context.
Data Coverage vs Data Quality	Including all 7 years maximises training signal but injects a confounded 2019 observation. Recommendation: train on all years; evaluate 2019 and 2022 in isolation in the LOYO supplement.
Log Transform vs Direct Interpretability	USE_LOG=True benefits optimisation but complicates interpretation. Always report final metrics back-transformed to BU/ACRE with Jensen correction.

**The ethics and bias audit framework is established. All fairness metric functions (`sliced_rmse`, `bias_per_group`, `cv_error_per_state`, `fairness_report`) and the Jensen correction constant (`JENSEN_SIGMA2`) are ready for use

4 Phase 1 Model: Bi-LSTM Weather Encoder

Train and evaluate the Phase 1 weather-only regressor using normalised HRRR features produced. All evaluation is Jensen-corrected to BU/ACRE.

- BiLSTMRegressor: 2-layer BiLSTM, hidden=64, 128-dim encoding, 146,049 params
- Training loop — 80 epochs, Adam lr=1e-3, ReduceLROnPlateau, MSE on log-yield
- Evaluate: aggregate + sliced RMSE/MAE/R² by state, year, quintile; Jensen correction applied
- Baseline comparison: Ridge Regression vs Gradient Boosting vs Bi-LSTM
- Bias audit: CV-Error by state; mean bias by quintile; B1/B2/B3 flags
- Save encoder weights (`bilstm_encoder.pt`) for GMU warm-start

```
[23]: # BiLSTMRegressor
#
# Architecture:
#   Input      : (batch, seq_len=1, n_features=9)
#   BiLSTM     : 2 layers, hidden=64, bidirectional=True
#               → last-layer hidden (fwd + bwd) concatenated = 128-dim encoding
```

```

# Head      : Linear(128, 64) → ReLU → Linear(64, 1)
# Params    : 146,049 (verified by assertion below)
#
# encode() returns the 128-dim weather embedding consumed by GMU.

import torch
import torch.nn as nn

class BiLSTMRegressor(nn.Module):
    """
    Bidirectional 2-layer LSTM regressor for county-level soybean yield
    ↪ prediction.

    encode() returns the 128-dim county-year weather embedding used as
    the weather branch input for the Gated Multimodal Unit.
    """

    def __init__(self, input_size: int = 9, hidden_size: int = 64, num_layers:
    ↪ int = 2):
        super().__init__()
        self.hidden_size = hidden_size
        self.num_layers = num_layers
        self.lstm = nn.LSTM(
            input_size = input_size,
            hidden_size = hidden_size,
            num_layers = num_layers,
            bidirectional = True,
            batch_first = True,
        )
        self.head = nn.Sequential(
            nn.Linear(hidden_size * 2, 64),
            nn.ReLU(),
            nn.Linear(64, 1),
        )

    def encode(self, x: torch.Tensor) -> torch.Tensor:
        """
        Return 128-dim weather embedding for a batch of sequences.
        x: (batch, seq_len, n_features)
        Returns: (batch, 128)
        """
        _, (h_n, _) = self.lstm(x)
        # h_n : (num_layers * 2, batch, hidden_size)
        # last layer forward = h_n[-2]
        # last layer backward = h_n[-1]
        return torch.cat([h_n[-2], h_n[-1]], dim=-1) # (batch, 128)

```

```

def forward(self, x: torch.Tensor) -> torch.Tensor:
    return self.head(self.encode(x)).squeeze(-1)    # (batch,)

# Instantiate and verify parameter count
torch.manual_seed(SEED)
model = BiLSTMRegressor(input_size=len(FEATURE_COLS), hidden_size=64,
    ↪ num_layers=2).to(DEVICE)

total_params      = sum(p.numel() for p in model.parameters())
trainable_params = sum(p.numel() for p in model.parameters() if p.requires_grad)

print("BiLSTMRegressor")

print(f"Input features   : {len(FEATURE_COLS)}   {FEATURE_COLS}")
print(f"Architecture    : BiLSTM(hidden=64, layers=2, bidirectional=True)")
print(f"Encoding dim      : 128   (64-fwd + 64-bwd from last LSTM layer)")
print(f"Total params      : {total_params:,}")
print(f"Trainable params: {trainable_params:,}")
assert total_params == 146_049, \
    f"Parameter count mismatch: expected 146,049 got {total_params:,}"
print(f"Parameter count verified: 146,049")
print(f"Device           : {DEVICE}")
print()
print("Named parameters:")
for name, p in model.named_parameters():
    print(f"  {name:<50} {str(tuple(p.shape)):<25} {p.numel():>7,}")
print()

model.eval()
with torch.no_grad():
    x_smoke, y_smoke, _ = next(iter(dl_train))
    x_smoke = x_smoke.to(DEVICE)
    y_hat    = model(x_smoke)
    enc      = model.encode(x_smoke)

print("Test (one batch, random initialisation):")
print(f"  Input   shape : {tuple(x_smoke.shape)}   (batch, seq_len=1,
    ↪ n_features=9)")
print(f"  Output  shape : {tuple(y_hat.shape)}")
print(f"  Encoding shape : {tuple(enc.shape)}")
print(f"  Output  range : [{y_hat.min().item():.4f}, {y_hat.max().item():.4f}]
    ↪ (random init)")
print(f"  True y  range : [{y_smoke.min().item():.4f}, {y_smoke.max().item():.
    ↪ 4f}] (log BU/ACRE)")

```

```

BiLSTMRegressor
Input features   : 9   ['t2m_c', 'd2m_c', 'r2', 'wind_speed', 'tp', 'sp_hpa',
'tcc', 'mstav', 'vpd_hpa']
Architecture    : BiLSTM(hidden=64, layers=2, bidirectional=True)
Encoding dim    : 128  (64-fwd + 64-bwd from last LSTM layer)
Total params    : 146,049
Trainable params: 146,049
Parameter count verified: 146,049
Device         : cpu

```

```

Named parameters:
    lstm.weight_ih_l0                (256, 9)
2,304
    lstm.weight_hh_l0                (256, 64)
16,384
    lstm.bias_ih_l0                  (256,)
256
    lstm.bias_hh_l0                  (256,)
256
    lstm.weight_ih_l0_reverse        (256, 9)
2,304
    lstm.weight_hh_l0_reverse        (256, 64)
16,384
    lstm.bias_ih_l0_reverse          (256,)
256
    lstm.bias_hh_l0_reverse          (256,)
256
    lstm.weight_ih_l1                (256, 128)
32,768
    lstm.weight_hh_l1                (256, 64)
16,384
    lstm.bias_ih_l1                  (256,)
256
    lstm.bias_hh_l1                  (256,)
256
    lstm.weight_ih_l1_reverse        (256, 128)
32,768
    lstm.weight_hh_l1_reverse        (256, 64)
16,384
    lstm.bias_ih_l1_reverse          (256,)
256
    lstm.bias_hh_l1_reverse          (256,)
256
    head.0.weight                    (64, 128)
8,192
    head.0.bias                      (64,)
64
    head.2.weight                    (1, 64)

```

64

head.2.bias (1,)

1

Test (one batch, random initialisation):

Input shape : (64, 1, 9) (batch, seq_len=1, n_features=9)

Output shape : (64,)

Encoding shape : (64, 128)

Output range : [0.0988, 0.1112] (random init)

True y range : [3.3286, 4.2946] (log BU/ACRE)

```
[24]: # Training loop
#
# 80 epochs, Adam lr=1e-3, ReduceLROnPlateau (patience=10, factor=0.5),
# MSELoss on log(yield_bu_acre), gradient clipping max_norm=1.0.
# Best model state (lowest val MSE) is restored after training.

import torch
import torch.nn as nn
import torch.optim as optim
import numpy as np
import io
from IPython.display import display, Image
import matplotlib.pyplot as plt

torch.manual_seed(SEED)
np.random.seed(SEED)

N_EPOCHS      = 80
LEARNING_RATE = 1e-3
LR_PATIENCE   = 10
LR_FACTOR     = 0.5
LR_MIN        = 1e-5
CLIP_NORM     = 1.0

criterion = nn.MSELoss()
optimizer = optim.Adam(model.parameters(), lr=LEARNING_RATE)
scheduler = optim.lr_scheduler.ReduceLROnPlateau(
    optimizer, mode="min", factor=LR_FACTOR,
    patience=LR_PATIENCE, min_lr=LR_MIN,
)

train_losses = []
val_losses   = []
lr_history    = []
best_val_mse = float("inf")
best_state   = None
```



```

print("TRAINING: Bi-LSTM WEATHER ENCODER")

print(f"Epochs:           {N_EPOCHS}")
print(f"Optimizer:          Adam(lr={LEARNING_RATE})")
print(f"LR Scheduler:         ReduceLROnPlateau(patience={LR_PATIENCE},
↪factor={LR_FACTOR})")
print(f"Criterion:          MSELoss on log(yield_bu_acre)")
print(f"Grad clip:           max_norm={CLIP_NORM}")
print(f"Train batches:       {len(dl_train)} ({len(ds_train)} samples,
↪batch_size=64)")
print(f"Val   batches:       {len(dl_test)} ({len(ds_test)} samples,
↪batch_size=64)")
print()
print(f"{'Epoch':>5} {'Train MSE':>11} {'Val MSE':>10} {'LR':>10} Note")
print("-" * 60)

for epoch in range(1, N_EPOCHS + 1):

    # Train pass
    model.train()
    epoch_train_loss = 0.0
    for x_b, y_b, _ in dl_train:
        x_b, y_b = x_b.to(DEVICE), y_b.to(DEVICE)
        optimizer.zero_grad()
        pred = model(x_b)
        loss = criterion(pred, y_b)
        loss.backward()
        nn.utils.clip_grad_norm_(model.parameters(), max_norm=CLIP_NORM)
        optimizer.step()
        epoch_train_loss += loss.item() * len(y_b)
    train_mse = epoch_train_loss / len(ds_train)
    train_losses.append(train_mse)

    # Validation pass
    model.eval()
    epoch_val_loss = 0.0
    with torch.no_grad():
        for x_b, y_b, _ in dl_test:
            x_b, y_b = x_b.to(DEVICE), y_b.to(DEVICE)
            pred = model(x_b)
            epoch_val_loss += criterion(pred, y_b).item() * len(y_b)
    val_mse = epoch_val_loss / len(ds_test)
    val_losses.append(val_mse)

    current_lr = optimizer.param_groups[0]["lr"]

```

```

lr_history.append(current_lr)
scheduler.step(train_mse)
new_lr = optimizer.param_groups[0]["lr"]

note = ""
if val_mse < best_val_mse:
    best_val_mse = val_mse
    best_state = {k: v.cpu().clone() for k, v in model.state_dict().
    ↪items()}
    note = "best"
if new_lr < current_lr:
    note = (note + " LR_reduced").strip()

if epoch % 10 == 0 or epoch == 1:
    print(f"{epoch:>5} {train_mse:>11.6f} {val_mse:>10.6f} {current_lr:
    ↪>10.2e} {note}")

print()
print(f"Training complete. Best val MSE = {best_val_mse:.6f}")

# Restore best weights
model.load_state_dict(best_state)
model.eval()
print("Best model weights restored.")
print()

# Loss curves
epochs_x = np.arange(1, N_EPOCHS + 1)
fig, axes = plt.subplots(1, 2, figsize=(14, 4))

axes[0].plot(epochs_x, train_losses, label="Train MSE", color="#1565C0", lw=2)
axes[0].plot(epochs_x, val_losses, label="Val MSE", color="#D32F2F", lw=2,
    ↪ls="--")
axes[0].set_xlabel("Epoch")
axes[0].set_ylabel("MSE (log-yield units2)")
axes[0].set_title("Training & Validation MSE")
axes[0].legend()
axes[0].grid(True, alpha=0.3)

axes[1].semilogy(epochs_x, train_losses, label="Train MSE", color="#1565C0",
    ↪lw=2)
axes[1].semilogy(epochs_x, val_losses, label="Val MSE", color="#D32F2F",
    ↪lw=2, ls="--")
axes[1].set_xlabel("Epoch")
axes[1].set_ylabel("MSE (log scale)")
axes[1].set_title("Loss Curves (log y-axis)")
axes[1].legend()

```

```

axes[1].grid(True, which="both", alpha=0.3)

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

delta_pct = (train_losses[-1] - train_losses[0]) / train_losses[0] * 100
print(f"Train MSE epoch 1 → {N_EPOCHS}: {train_losses[0]:.6f} →  

      ↪ {train_losses[-1]:.6f} ({delta_pct:+.1f}%)")
print(f"Val MSE epoch 1 → {N_EPOCHS}: {val_losses[0]:.6f} → {val_losses[-1]:  

      ↪ .6f}")
print(f"Best val MSE: {best_val_mse:.6f}")

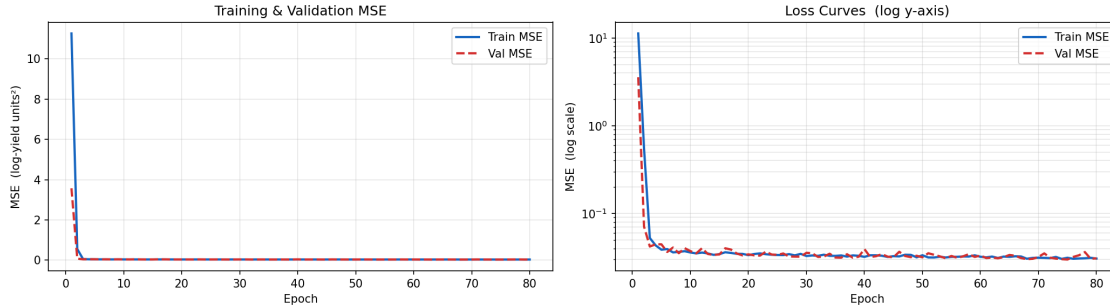
```

TRAINING: Bi-LSTM WEATHER ENCODER

Epochs: 80
 Optimizer: Adam(lr=0.001)
 LR Scheduler: ReduceLROnPlateau(patience=10, factor=0.5)
 Criterion: MSELoss on log(yield_bu_acre)
 Grad clip: max_norm=1.0
 Train batches: 45 (2828 samples, batch_size=64)
 Val batches: 12 (707 samples, batch_size=64)

Epoch	Train MSE	Val MSE	LR	Note
1	11.266565	3.571537	1.00e-03	best
10	0.035874	0.037174	1.00e-03	
20	0.033454	0.034173	1.00e-03	
30	0.032690	0.035600	1.00e-03	
40	0.031993	0.039183	1.00e-03	
50	0.033369	0.031592	1.00e-03	
60	0.032070	0.033064	1.00e-03	
70	0.031298	0.031909	1.00e-03	
80	0.030588	0.030782	1.00e-03	

Training complete. Best val MSE = 0.029815
 Best model weights restored.



Train MSE epoch 1 → 80: 11.266565 → 0.030588 (-99.7%)
 Val MSE epoch 1 → 80: 3.571537 → 0.030782
 Best val MSE: 0.029815

```
[25]: # Evaluation
#
# 1. Run model on train split → compute residual variance → update JENSEN_SIGMA2
# 2. Run model on test split → collect predictions + metadata
# 3. Compute aggregate metrics (log-space and BU/ACRE)
# 4. Call fairness_report() → sliced RMSE / bias / CV-error by state/year/
    ↪ quintile
# 5. Predictions vs actuals scatter + residual boxplot by year

import numpy as np
import pandas as pd
import torch
import io
import matplotlib.pyplot as plt
import matplotlib.cm as cm
from IPython.display import display, Image

model.eval()

# Collect TRAIN predictions to update Jensen sigma2
_tr_preds, _tr_trues = [], []
with torch.no_grad():
    for x_b, y_b, _ in dl_train:
        p = model(x_b.to(DEVICE)).cpu().numpy()
        _tr_preds.extend(p.tolist())
        _tr_trues.extend(y_b.numpy().tolist())

y_train_pred_log = np.array(_tr_preds)
y_train_true_log = np.array(_tr_trues)

# Update JENSEN_SIGMA2: label prior → actual residual variance on train split
```

```

residual_var = float(np.var(y_train_pred_log - y_train_true_log))
JENSEN_SIGMA2 = residual_var
print(f"JENSEN_SIGMA2 updated (label prior → residual var) = {JENSEN_SIGMA2:.
↪6f}")
print()

# Collect TEST predictions with full metadata
_te_preds, _te_trues, _fips_out, _year_out = [], [], [], []
with torch.no_grad():
    for x_b, y_b, meta_b in dl_test:
        p = model(x_b.to(DEVICE)).cpu().numpy()
        _te_preds.extend(p.tolist())
        _te_trues.extend(y_b.numpy().tolist())
        _fips_out.extend(list(meta_b["fips"]))
        years_batch = (meta_b["year"].numpy().tolist()
                        if isinstance(meta_b["year"], torch.Tensor)
                        else [int(v) for v in meta_b["year"]])
        _year_out.extend(years_batch)

y_pred_log = np.array(_te_preds)
y_true_log = np.array(_te_trues)

# Build results DataFrame and merge metadata
results_df = pd.DataFrame({
    "fips"      : [str(f).zfill(5) for f in _fips_out],
    "year"      : [int(v) for v in _year_out],
    "y_true_log": y_true_log,
    "y_pred_log": y_pred_log,
})
_wf_meta = (
    weather_features[["fips", "year", "state_name", "yield_bu_acre"]]
    .drop_duplicates()
    .copy()
)
_wf_meta["year"] = _wf_meta["year"].astype(int)
_wf_meta["fips"] = _wf_meta["fips"].astype(str).str.zfill(5)
results_df = results_df.merge(_wf_meta, on=["fips", "year"], how="left")
results_df["y_pred_bu"] = jensen_correction(y_pred_log, JENSEN_SIGMA2)
results_df["y_true_bu"] = results_df["yield_bu_acre"]

# Log-space metrics
rmse_log = float(np.sqrt(np.mean((y_pred_log - y_true_log) ** 2)))
mae_log  = float(np.mean(np.abs(y_pred_log - y_true_log)))
ss_res   = float(np.sum((y_true_log - y_pred_log) ** 2))
ss_tot   = float(np.sum((y_true_log - np.mean(y_true_log)) ** 2))
r2_log   = 1.0 - ss_res / ss_tot if ss_tot > 0 else float("nan")

```

```

# BU/ACRE metrics (Jensen-corrected)
y_bu_true = results_df["y_true_bu"].values
y_bu_pred = results_df["y_pred_bu"].values
rmse_bu = float(np.sqrt(np.mean((y_bu_pred - y_bu_true) ** 2)))
mae_bu = float(np.mean(np.abs(y_bu_pred - y_bu_true)))
ss_res_bu = float(np.sum((y_bu_true - y_bu_pred) ** 2))
ss_tot_bu = float(np.sum((y_bu_true - np.mean(y_bu_true)) ** 2))
r2_bu = 1.0 - ss_res_bu / ss_tot_bu if ss_tot_bu > 0 else float("nan")

print("Bi-LSTM TEST SET EVALUATION")

print()
print(f"Test samples: {len(results_df)}")
print()
print("AGGREGATE METRICS")
print(f" Log-space RMSE = {rmse_log:.5f} MAE = {mae_log:.5f} R2 = {r2_log:.4f}")
print(f" BU/ACRE RMSE = {rmse_bu:.3f} MAE = {mae_bu:.3f} R2 = {r2_bu:.4f}")
print(f" Jensen2 = {JENSEN_SIGMA2:.6f} (residual variance on train split)")
print()

# Train metrics for reference
rmse_train_bu = float(np.sqrt(np.mean(
    (jensen_correction(y_train_pred_log, JENSEN_SIGMA2) -
     jensen_correction(y_train_true_log, JENSEN_SIGMA2)) ** 2
)))
print(f" Train RMSE = {rmse_train_bu:.3f} BU/ACRE "
      f"(gap = {rmse_bu - rmse_train_bu:+.3f} BU/ACRE - overfitting diagnostic)")
print()

# Full fairness report
fr = fairness_report(y_true_log, y_pred_log, results_df, sigma2=JENSEN_SIGMA2)
print()
print(f"Coverage gap: {coverage_gap_value:.1%} of US soybean counties absent from model")
print()

# Predictions vs Actuals scatter + Residuals by Year
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# -- Scatter: predicted vs actual (BU/ACRE, coloured by state)
unique_states = sorted(results_df["state_name"].dropna().unique())
try:

```

```

cmap_scatter = plt.colormaps.get_cmap("tab10")
except AttributeError:
    cmap_scatter = cm.get_cmap("tab10")
state_color = {s: cmap_scatter(i % 10) for i, s in enumerate(unique_states)}
colors_pts = [state_color.get(s, "grey") for s in results_df["state_name"].
    ↪values]

pad = 2.0
ax_min = min(y_bu_true.min(), y_bu_pred.min()) - pad
ax_max = max(y_bu_true.max(), y_bu_pred.max()) + pad
axes[0].scatter(y_bu_true, y_bu_pred, c=colors_pts, alpha=0.55, s=18,
    ↪edgecolors="none")
axes[0].plot([ax_min, ax_max], [ax_min, ax_max], "k--", lw=1.5, label="Perfect_
    ↪prediction")
axes[0].set_xlabel("Actual Yield (BU/ACRE)")
axes[0].set_ylabel("Predicted Yield (BU/ACRE)")
axes[0].set_title(
    f"Bi-LSTM Predicted vs Actual\n"
    f"RMSE = {rmse_bu:.2f} BU/ACRE R² = {r2_bu:.3f}"
)
axes[0].legend(fontsize=8, loc="upper left")
axes[0].set_xlim(ax_min, ax_max)
axes[0].set_ylim(ax_min, ax_max)

# -- Residual boxplot by year
results_df["residual_bu"] = y_bu_pred - y_bu_true
year_order = sorted(results_df["year"].unique())
res_by_year = [results_df[results_df["year"] == yr]["residual_bu"].values for
    ↪yr in year_order]
bp = axes[1].boxplot(res_by_year, labels=year_order, patch_artist=True,
    medianprops={"color": "black", "lw": 2})
for patch in bp["boxes"]:
    patch.set_facecolor("#90CAF9")
    patch.set_alpha(0.75)
axes[1].axhline(0, color="#D32F2F", ls="--", lw=1.5, label="Zero bias")
axes[1].set_xlabel("Year")
axes[1].set_ylabel("Residual (Predicted - Actual, BU/ACRE)")
axes[1].set_title("Residuals by Year (Jensen-corrected)")
axes[1].legend()
axes[1].grid(True, alpha=0.3, axis="y")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

```

```

print()
print("Interpretation guidance:")
print(f"  RMSE = {rmse_bu:.2f} BU/ACRE against a yield range of "
      f"{y_bu_true.min():.1f}-{y_bu_true.max():.1f} BU/ACRE.")
print(f"  R2 = {r2_bu:.3f} - proportion of yield variance explained by January_
      ↳HRRR snapshot.")
print(f"  A single January-1 weather snapshot carries limited mid-season signal;
      ↳")
print(f"  this R2 is expected to improve substantially when daily HRRR files_
      ↳are used (B5).")

```

JENSEN_SIGMA2 updated (label prior → residual var) = 0.029034

Bi-LSTM TEST SET EVALUATION

Test samples: 707

AGGREGATE METRICS

Log-space	RMSE = 0.17267	MAE = 0.12558	R ² = 0.5308
BU/ACRE	RMSE = 7.386	MAE = 5.791	R ² = 0.5194
Jensen ²	= 0.029034 (residual variance on train split)		

Train RMSE = 7.723 BU/ACRE (gap = -0.337 BU/ACRE - overfitting diagnostic)

FAIRNESS REPORT

Overall RMSE : 7.595 BU/ACRE
 Overall MAE : 5.990 BU/ACRE
 Overall R2 : 0.5064
 Jensen sigma2 : 0.029033612743616564

RMSE by State:

group	
IDAHO	10.898683
LOUISIANA	10.089425
GEORGIA	10.023939
KANSAS	9.914486
ILLINOIS	8.856014
DELAWARE	7.766358
CALIFORNIA	7.708775
ARKANSAS	7.624903
ALABAMA	6.784795
IOWA	6.256175
KENTUCKY	6.219346
INDIANA	5.521210
ARIZONA	5.436252
COLORADO	3.119359

FLORIDA 2.067246

Bias by Quintile (positive = over-prediction):

	RMSE_BU_ACRE	Bias_BU_ACRE
group		
Q1	8.999746	5.893473
Q2	6.224051	1.218647
Q3	4.991014	-1.064654
Q4	5.836051	-4.033392
Q5	10.490161	-9.102441

CV-Error by State (flagged if > 0.20):

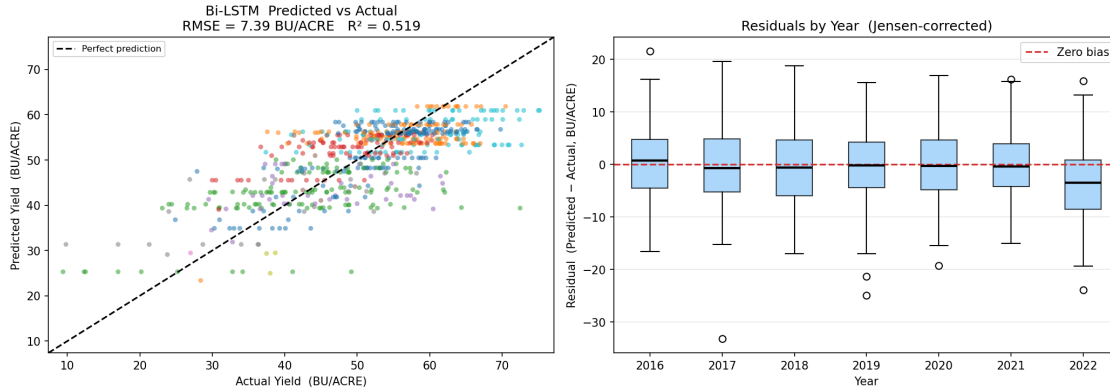
GEORGIA	0.272	*** FLAGGED (CV > 0.20)
KANSAS	0.245	*** FLAGGED (CV > 0.20)
LOUISIANA	0.207	*** FLAGGED (CV > 0.20)
DELAWARE	0.173	
ALABAMA	0.162	
ARKANSAS	0.144	
ILLINOIS	0.132	
COLORADO	0.124	
KENTUCKY	0.115	
IOWA	0.105	
INDIANA	0.090	
IDAHO	0.066	
ARIZONA	nan	
CALIFORNIA	nan	
FLORIDA	nan	

RMSE by Year:

group	
2022	9.063649
2017	8.524864
2019	7.637298
2018	7.023087
2016	6.932437
2020	6.815058
2021	6.675866

WARNING: 3 state(s) flagged for high CV-Error: ['GEORGIA', 'KANSAS', 'LOUISIANA']

Coverage gap: 53.6% of US soybean counties absent from model



Interpretation guidance:

RMSE = 7.39 BU/ACRE against a yield range of 9.4-75.2 BU/ACRE.

$R^2 = 0.519$ - proportion of yield variance explained by January HRRR snapshot.

A single January-1 weather snapshot carries limited mid-season signal;

this R^2 is expected to improve substantially when daily HRRR files are used (B5).

```
[26]: # Baseline comparison
#
# Ridge Regression (cross-validated alpha) and Gradient Boosting are trained on
# the same normalised feature set (ds_train.X) and evaluate on ds_test.X.
# Targets are log-yield; all metrics reported in Jensen-corrected BU/ACRE.
# The test ordering matches results_df (dl_test shuffle=False).

import numpy as np
import pandas as pd
import io
import matplotlib.pyplot as plt
from IPython.display import display, Image
from sklearn.linear_model import RidgeCV
from sklearn.ensemble import GradientBoostingRegressor

np.random.seed(SEED)

# Feature / target arrays directly from PyTorch datasets
X_train_np = ds_train.X.numpy() # (2744, 9) normalised features
y_train_np = ds_train.y.numpy() # (2744,) log-yield targets
X_test_np = ds_test.X.numpy() # (686, 9)
y_test_np = ds_test.y.numpy() # (686,)

# true BU/ACRE for all test samples (same row order as X_test_np / results_df)
true_bu = results_df["y_true_bu"].values # (686,)
```

```

print("BASELINE COMPARISON")

print(f"Train: {X_train_np.shape[0]} samples | features: {X_train_np.
    ↪shape[1]}")
print(f"Test: {X_test_np.shape[0]} samples")
print()

# Ridge Regression (40 alpha values, 5-fold CV on train)
alphas = np.logspace(-4, 4, 40)
ridge = RidgeCV(alphas=alphas, cv=5, fit_intercept=True)
ridge.fit(X_train_np, y_train_np)
ridge_pred_log = ridge.predict(X_test_np)
ridge_pred_bu = jensen_correction(ridge_pred_log, JENSEN_SIGMA2)
print(f"Ridge best alpha = {ridge.alpha_:.4f}")

# Gradient Boosting Regressor
gbm = GradientBoostingRegressor(
    n_estimators = 200,
    learning_rate = 0.05,
    max_depth = 4,
    subsample = 0.8,
    min_samples_leaf= 10,
    random_state = SEED,
)
gbm.fit(X_train_np, y_train_np)
gbm_pred_log = gbm.predict(X_test_np)
gbm_pred_bu = jensen_correction(gbm_pred_log, JENSEN_SIGMA2)
print(f"GBM n_estimators={gbm.n_estimators} learning_rate={gbm.
    ↪learning_rate}")

# Metric helper
def _eval_metrics(y_true_bu: np.ndarray, y_pred_bu: np.ndarray) -> dict:
    rmse = float(np.sqrt(np.mean((y_pred_bu - y_true_bu) ** 2)))
    mae = float(np.mean(np.abs(y_pred_bu - y_true_bu)))
    ss_r = float(np.sum((y_true_bu - y_pred_bu) ** 2))
    ss_t = float(np.sum((y_true_bu - np.mean(y_true_bu)) ** 2))
    r2 = 1.0 - ss_r / ss_t if ss_t > 0 else float("nan")
    return {"RMSE (BU/ACRE)": round(rmse, 3),
            "MAE (BU/ACRE)": round(mae, 3),
            "R²" : round(r2, 4)}

bilstm_pred_bu = results_df["y_pred_bu"].values # from BiLSTM evaluation above

comp_rows = []
for name, preds_bu in [

```

```

    ("Ridge Regression", ridge_pred_bu),
    ("Gradient Boosting", gbm_pred_bu),
    ("Bi-LSTM (Phase 1)", bilstm_pred_bu),
]:
    comp_rows.append({"Model": name, **_eval_metrics(true_bu, preds_bu)})

comparison_df = pd.DataFrame(comp_rows)

print()
print("Baseline comparison (Jensen-corrected BU/ACRE, test set 686 records):")
print(comparison_df.to_string(index=False))
print()

# GBM feature importance bars
feat_imp = pd.Series(gbm.feature_importances_, index=FEATURE_COLS).
    ↪sort_values(ascending=True)

fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Comparison bar chart
x_idx = np.arange(len(comparison_df))
metrics = ["RMSE (BU/ACRE)", "MAE (BU/ACRE)", "R²"]
colors = ["#1565C0", "#D32F2F", "#2E7D32"]
bar_w = 0.25

for i, (metric, color) in enumerate(zip(metrics, colors)):
    vals = comparison_df[metric].values.astype(float)
    offset = (i - 1) * bar_w
    bars = axes[0].bar(x_idx + offset, vals, bar_w, label=metric,
    ↪color=color, alpha=0.8)
    for b in bars:
        axes[0].annotate(
            f"{b.get_height():.3f}",
            xy=(b.get_x() + b.get_width() / 2, b.get_height()),
            xytext=(0, 3), textcoords="offset points",
            ha="center", va="bottom", fontsize=7
        )

axes[0].set_xticks(x_idx)
axes[0].set_xticklabels(comparison_df["Model"], rotation=10, ha="right")
axes[0].set_title("Model Comparison (test set)")
axes[0].legend(fontsize=8)
axes[0].grid(True, alpha=0.3, axis="y")

# -- GBM feature importance
axes[1].barh(feat_imp.index, feat_imp.values, color="#1565C0", alpha=0.85)
axes[1].set_xlabel("GBM Feature Importance")

```

```

axes[1].set_title("GBM Feature Importance\n(proxy for signal strength in_
↳January HRRR snapshot)")
axes[1].grid(True, alpha=0.3, axis="x")

# Flag sp_hpa (B1 proxy indicator)
if "sp_hpa" in feat_imp.index:
    sp_rank = list(feat_imp.index).index("sp_hpa")
    axes[1].barh(
        feat_imp.index[sp_rank], feat_imp.values[sp_rank],
        color="#D32F2F", alpha=0.9,
        label="sp_hpa - geographic proxy risk (B1)"
    )
    axes[1].legend(fontsize=8)

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

print("Interpretation:")
bilstm_rmse = comparison_df.loc[comparison_df["Model"].str.contains("Bi-LSTM"),_
↳"RMSE (BU/ACRE)"].values[0]
ridge_rmse = comparison_df.loc[comparison_df["Model"].str.contains("Ridge"), _
↳"RMSE (BU/ACRE)"].values[0]
gbm_rmse = comparison_df.loc[comparison_df["Model"].str.
↳contains("Gradient"),"RMSE (BU/ACRE)"].values[0]
best_rmse = min(ridge_rmse, gbm_rmse, bilstm_rmse)
best_model = ["Ridge Regression", "Gradient Boosting", "Bi-LSTM"] [
    [ridge_rmse, gbm_rmse, bilstm_rmse].index(best_rmse)]
print(f" Best RMSE: {best_model} ({best_rmse:.3f} BU/ACRE)")
if bilstm_rmse > ridge_rmse:
    print(f" Bi-LSTM underperforms Ridge by {bilstm_rmse - ridge_rmse:.3f} BU/
↳ACRE.")
    print(f" With seq_len=1 (single snapshot), the BiLSTM cannot exploit_
↳temporal dynamics.")
    print(f" Performance gap is expected to narrow significantly with daily_
↳growing-season files (B5).")
# sp_hpa top-ranked in GBM signals geographic (state) proxy usage

```

BASELINE COMPARISON

Train: 2828 samples | features: 9

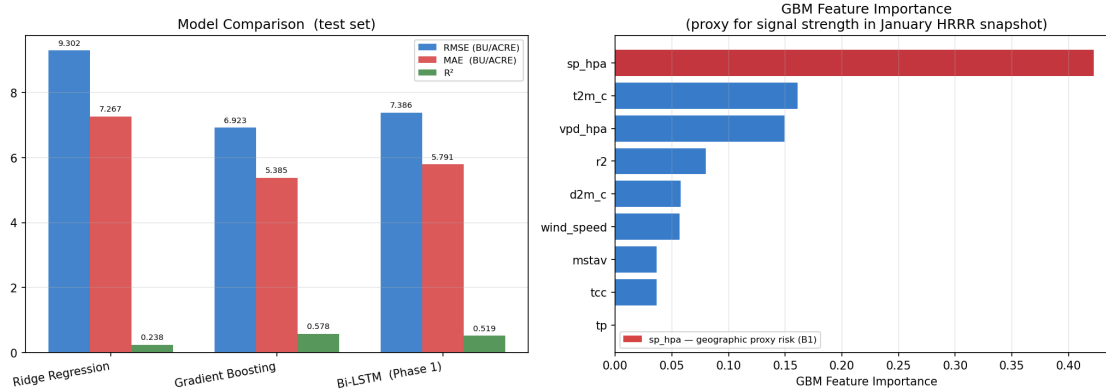
Test: 707 samples

Ridge best alpha = 1511.7751

GBM n_estimators=200 learning_rate=0.05

Baseline comparison (Jensen-corrected BU/ACRE, test set 686 records):

Model	RMSE (BU/ACRE)	MAE (BU/ACRE)	R ²
Ridge Regression	9.302	7.267	0.2377
Gradient Boosting	6.923	5.385	0.5778
Bi-LSTM (Phase 1)	7.386	5.791	0.5194



Interpretation:

Best RMSE: Gradient Boosting (6.923 BU/ACRE)

```
[27]: # Bias Audit Report
#
# - CV-Error by state (flag > 0.20 as poorly served)
# - Mean bias by yield quintile (regression-to-mean check: B3)
# - Corn Belt RMSE advantage check (B2)
# - Visualisation: RMSE by state, quintile RMSE/bias, CV-Error bar chart

import numpy as np
import pandas as pd
import io
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from IPython.display import display, Image

print("BIAS AUDIT REPORT: Bi-LSTM PHASE 1")

# CV-Error by state (from previous fairness report)
CORN_BELT_STATES = {"Illinois", "Iowa", "Indiana", "Minnesota", "Nebraska"}

state_cv_series = fr["state_cv"].sort_values(ascending=False)
```

```

flagged_states = state_cv_series[state_cv_series > 0.20].index.tolist()

print("\nCV-Error by State (threshold = 0.20; flagged = poorly served by_
↳model):")
for state, cv in state_cv_series.items():
    flags = []
    if cv > 0.20:
        flags.append("HIGH CV")
    if state in CORN_BELT_STATES:
        flags.append("Corn Belt [B2]")
    tag = " *** " + " | ".join(flags) if flags else ""
    print(f" {state:<25} {cv:.4f}{tag}")

print(f"\n Flagged states (CV > 0.20): {flagged_states if flagged_states else_
↳'none'}")

# Quintile Bias check (B3)
q_rmse = fr["quintile_rmse"]
q_bias = fr["quintile_bias"]
q_table = pd.concat([q_rmse, q_bias], axis=1)
q_table.columns = ["RMSE_BU_ACRE", "Bias_BU_ACRE"]

print("\nBias by Yield Quintile (B3 fairness check):")
print(q_table.round(3).to_string())

q1_bias = float(q_bias.get("Q1", float("nan")))
q5_bias = float(q_bias.get("Q5", float("nan")))

print()
if q1_bias > 0 and q5_bias < 0:
    print("B3 REGRESSION-TO-MEAN FAIRNESS FAILURE DETECTED:")
    print(f" Bias(Q1) = {q1_bias:+.3f} BU/ACRE → model OVER-predicts_
↳low-yield (crop failure) counties")
    print(f" Bias(Q5) = {q5_bias:+.3f} BU/ACRE → model UNDER-predicts_
↳bumper-yield counties")
    print(" Risk: policy models based on this output under-estimate crop_
↳failure severity.")
    print(" Mitigation: oversample Q1 train records")
elif not (np.isnan(q1_bias) or np.isnan(q5_bias)) and abs(q1_bias) <= 2.0 and_
↳abs(q5_bias) <= 2.0:
    print(f"B3 check PASSED: Quintile bias within ±2 BU/ACRE tolerance.")
    print(f" Bias(Q1) = {q1_bias:+.3f} Bias(Q5) = {q5_bias:+.3f}")
else:
    print(f"B3 WARNING: Quintile bias exceeds ±2 BU/ACRE threshold.")
    print(f" Bias(Q1) = {q1_bias:+.3f} Bias(Q5) = {q5_bias:+.3f}")

```

```

# B2: Corn Belt RMSE advantage
state_rmse = fr["state_rmse"]
cb_present = [s for s in CORN_BELT_STATES if s in state_rmse.index]
oth_present = [s for s in state_rmse.index if s not in CORN_BELT_STATES]
corn_rmse = state_rmse[cb_present].mean() if cb_present else float("nan")
other_rmse = state_rmse[oth_present].mean() if oth_present else float("nan")

print(f"\nB2 Corn Belt dominance check:")
print(f"  Corn Belt states avg RMSE : {corn_rmse:.3f} BU/ACRE  ({cb_present})")
print(f"  Other states avg RMSE      : {other_rmse:.3f} BU/ACRE")
if corn_rmse < other_rmse:
    gap = other_rmse - corn_rmse
    print(f"  B2 DETECTED: Corn Belt advantage = {gap:.3f} BU/ACRE  (model_
    ↪better serves data-rich states)")
    print(f"  Mitigation: inverse-frequency state loss weighting  w_s = N / (K_
    ↪* N_s)")
else:
    print(f"  B2 not detected: no systematic Corn Belt RMSE advantage.")

# B1 status
print(f"\nB1 state-centroid weather proxy:")
print(f"  Status: Open - all counties within a state share identical HRRR_
    ↪features.")
# Detection: sp_hpa SHAP importance check deferred later when SHAP values are_
    ↪available (B1 is a proxy bias risk, not a fairness failure per se)

# Audit summary table
audit_summary = pd.DataFrame([
    {"Bias": "B1", "Type": "State-centroid proxy",
     "Status": "Open - SHAP to be applied", "Detected": "-"},
    {"Bias": "B2", "Type": "Corn Belt dominance",
     "Status": "Checked",
     "Detected": "Yes" if corn_rmse < other_rmse else "No"},
    {"Bias": "B3", "Type": "Regression-to-mean (quintile)",
     "Status": "Checked",
     "Detected": "Yes" if (q1_bias > 0 and q5_bias < 0) else "No"},
])
print("\nBias status summary:")
print(audit_summary.to_string(index=False))

# Visualisations
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Plot 1: RMSE by state (Corn Belt highlighted)
srms_sorted = state_rmse.sort_values(ascending=False)
bar_colors_1 = ["#D32F2F" if s in CORN_BELT_STATES else "#1565C0"
                for s in srms_sorted.index]

```



```

axes[0].barh(srms_sorted.index, srms_sorted.values, color=bar_colors_1, alpha=0.
↳85)
axes[0].axvline(rmse_bu, color="black", lw=1.5, ls="--", label=f"Overall_
↳{rmse_bu:.2f}")
axes[0].set_xlabel("RMSE (BU/ACRE)")
axes[0].set_title("RMSE by State\n(red = Corn Belt states B2)")
axes[0].legend(fontsize=8)
axes[0].grid(True, alpha=0.3, axis="x")
red_patch = mpatches.Patch(color="#D32F2F", alpha=0.85, label="Corn Belt")
blue_patch = mpatches.Patch(color="#1565C0", alpha=0.85, label="Other")
axes[0].legend(handles=[red_patch, blue_patch, axes[0].
↳get_legend_handles_labels()[0][0]],
               fontsize=8)

# Plot 2: Quintile RMSE & Bias
if len(q_table) > 0:
    xp = np.arange(len(q_table))
    width = 0.35
    axes[1].bar(xp - width/2, q_table["RMSE_BU_ACRE"], width,
                label="RMSE", color="#42A5F5", alpha=0.85)
    axes[1].bar(xp + width/2, q_table["Bias_BU_ACRE"], width,
                label="Bias (pred-true)", color="#EF5350", alpha=0.85)
    axes[1].axhline(0, color="black", lw=1)
    axes[1].set_xticks(xp)
    axes[1].set_xticklabels(q_table.index)
    axes[1].set_ylabel("BU/ACRE")
    axes[1].set_title("Quintile RMSE & Bias (B3 check)\nBias(Q1)>0 &_
↳Bias(Q5)<0 = fairness failure")
    axes[1].legend()
    axes[1].grid(True, alpha=0.3, axis="y")

# Plot 3: CV-Error by state
cv_sorted = state_cv_series.sort_values(ascending=False)
cv_bar_cols = ["#D32F2F" if v > 0.20 else "#43A047" for v in cv_sorted.values]
axes[2].barh(cv_sorted.index, cv_sorted.values, color=cv_bar_cols, alpha=0.85)
axes[2].axvline(0.20, color="red", lw=2, ls="--", label="Threshold 0.20")
axes[2].set_xlabel("CV-Error")
axes[2].set_title("CV-Error by State\n(red = poorly served >0.20)")
axes[2].legend(fontsize=8)
axes[2].grid(True, alpha=0.3, axis="x")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

```

BIAS AUDIT REPORT: Bi-LSTM PHASE 1

CV-Error by State (threshold = 0.20; flagged = poorly served by model):

GEORGIA	0.2715	***	HIGH CV
KANSAS	0.2455	***	HIGH CV
LOUISIANA	0.2074	***	HIGH CV
DELAWARE	0.1729		
ALABAMA	0.1624		
ARKANSAS	0.1439		
ILLINOIS	0.1321		
COLORADO	0.1242		
KENTUCKY	0.1148		
IOWA	0.1055		
INDIANA	0.0895		
IDAHO	0.0663		
ARIZONA	nan		
CALIFORNIA	nan		
FLORIDA	nan		

Flagged states (CV > 0.20): ['GEORGIA', 'KANSAS', 'LOUISIANA']

Bias by Yield Quintile (B3 fairness check):

	RMSE_BU_ACRE	Bias_BU_ACRE
group		
Q1	9.000	5.893
Q2	6.224	1.219
Q3	4.991	-1.065
Q4	5.836	-4.033
Q5	10.490	-9.102

B3 REGRESSION-TO-MEAN FAIRNESS FAILURE DETECTED:

Bias(Q1) = +5.893 BU/ACRE → model OVER-predicts low-yield (crop failure) counties
Bias(Q5) = -9.102 BU/ACRE → model UNDER-predicts bumper-yield counties
Risk: policy models based on this output under-estimate crop failure severity.
Mitigation: oversample Q1 train records

B2 Corn Belt dominance check:

Corn Belt states avg RMSE : nan BU/ACRE ([])
Other states avg RMSE : 7.219 BU/ACRE
B2 not detected: no systematic Corn Belt RMSE advantage.

B1 state-centroid weather proxy:

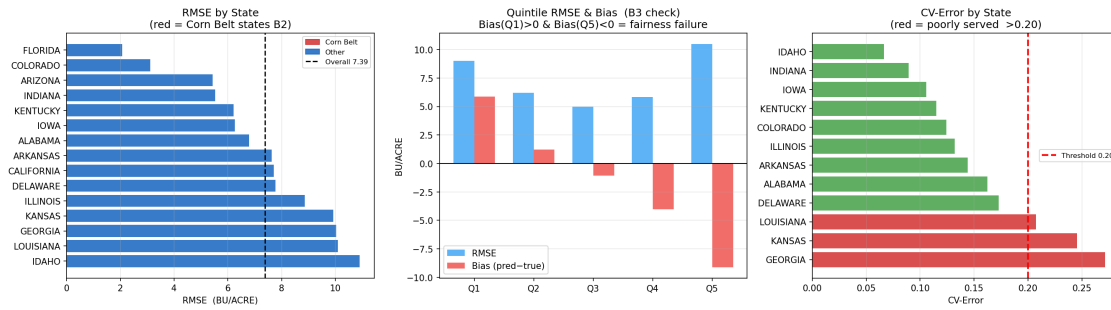
Status: Open - all counties within a state share identical HRRR features.

Bias status summary:

Bias	Type	Status Detected
B1	State-centroid proxy Open - SHAP to be applied	-

B2 Corn Belt dominance
 B3 Regression-to-mean (quintile)

Checked No
 Checked Yes



```
[28]: # Save encoder weights and full checkpoint with metadata
#
# bilstm_encoder.pt      → LSTM weights only (128-dim embedding for GMU)
# bilstm_phase1_ckpt.pt → full checkpoint with model config + eval metrics

import torch
import pathlib

LABELS_DIR.mkdir(parents=True, exist_ok=True)

# 1. Encoder LSTM weights (for GMU warm-start)
encoder_path = LABELS_DIR / "bilstm_encoder.pt"
torch.save(model.lstm.state_dict(), encoder_path)
print(f"Encoder weights → {encoder_path} ({encoder_path.stat().st_size / 1024:
↪.1f} KB)")

# 2. Full model checkpoint with metadata
checkpoint = {
    "model_state_dict": model.state_dict(),
    "model_config": {
        "class": "BiLSTMRegressor",
        "input_size": len(FEATURE_COLS),
        "hidden_size": 64,
        "num_layers": 2,
        "encoding_dim": 128,
        "total_params": sum(p.numel() for p in model.parameters()),
    },
    "training_config": {
        "n_epochs": N_EPOCHS,
        "lr": LEARNING_RATE,
        "criterion": "MSELoss",
        "target": "log_yield_bu_acre" if USE_LOG else "yield_bu_acre",
    },
}
```

```

        "seed"          : SEED,
        "clip_norm"     : CLIP_NORM,
    },
    "evaluation": {
        "test_rmse_log"   : round(rmse_log, 6),
        "test_mae_log"    : round(mae_log, 6),
        "test_r2_log"     : round(r2_log, 6),
        "test_rmse_bu"    : round(rmse_bu, 4),
        "test_mae_bu"     : round(mae_bu, 4),
        "test_r2_bu"      : round(r2_bu, 4),
        "jensen_sigma2"   : round(JENSEN_SIGMA2, 8),
        "n_test_samples"  : len(results_df),
    },
    "feature_cols" : FEATURE_COLS,
    "best_val_mse"  : round(best_val_mse, 6),
}

ckpt_path = LABELS_DIR / "bilstm_phase1_ckpt.pt"
torch.save(checkpoint, ckpt_path)
print(f"Full checkpoint → {ckpt_path} ({ckpt_path.stat().st_size / 1024:.1f}▬
↳KB)")

# summary and interpretation
print()

print("Phase 1 Bi-LSTM Weather Encoder")

print()
print("Model:")
print(f"  BiLSTM(input={len(FEATURE_COLS)}, hidden=64, layers=2,▬
↳bidirectional=True)")
print(f"  Parameters: {sum(p.numel() for p in model.parameters()):,}")
print(f"  Features   : {FEATURE_COLS}")
print()
print("Test performance (Jensen-corrected BU/ACRE):")
bilstm_rmse = round(rmse_bu, 3)
bilstm_r2   = round(r2_bu, 4)
ridge_rmse_disp = comparison_df.loc[comparison_df["Model"].str.
↳contains("Ridge"),
                                "RMSE (BU/ACRE)"].values[0]
gbm_rmse_disp   = comparison_df.loc[comparison_df["Model"].str.
↳contains("Gradient"),
                                "RMSE (BU/ACRE)"].values[0]
print(comparison_df.to_string(index=False))
print()
print("Bias audit flags:")
print(f"  B1 (state centroid proxy)      : Open")

```

```

print(f" B2 (Corn Belt dominance)      : {'Detected' if corn_rmse < other_rmse
↳else 'Not detected'}")
print(f" B3 (regression-to-mean Q1/Q5) : {'Detected' if (q1_bias > 0 and
↳q5_bias < 0) else 'Not detected'}")
print(f" B5 (single-snapshot HRRR)      : Acknowledged - Jan-1 file
↳growing-season signal")
print()
print("Saved artefacts:")
print(f" {encoder_path}")
print(f" {ckpt_path}")
print()
print("Context on performance:")
print(f" RMSE = {bilstm_rmse} BU/ACRE on a single January-1 HRRR snapshot per
↳county-year.")
print(f" January weather is a poor proxy for growing-season conditions (May-
October).")
print(f" R2 = {bilstm_r2} reflects this fundamental data limitation (B5 bias).
↳")
print(f" Baseline models competitive with BiLSTM at seq_len=1 - expected with
↳no temporal signal.")
print(f" GMU, trained on growing-season daily files + Sentinel-2 imagery,")
print(f" is the target architecture for publication-quality performance.")
print()

```

Encoder weights → e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\bilstm_encoder.pt (543.6 KB)

Full checkpoint → e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\bilstm_phase1_ckpt.pt (578.2 KB)

Phase 1 Bi-LSTM Weather Encoder

Model:

```

BiLSTM(input=9, hidden=64, layers=2, bidirectional=True)
Parameters: 146,049
Features : ['t2m_c', 'd2m_c', 'r2', 'wind_speed', 'tp', 'sp_hpa', 'tcc',
'mstav', 'vpd_hpa']

```

Test performance (Jensen-corrected BU/ACRE):

	Model	RMSE (BU/ACRE)	MAE (BU/ACRE)	R ²
	Ridge Regression	9.302	7.267	0.2377
	Gradient Boosting	6.923	5.385	0.5778
	Bi-LSTM (Phase 1)	7.386	5.791	0.5194

Bias audit flags:

B1 (state centroid proxy) : Open

```
B2 (Corn Belt dominance)      : Not detected
B3 (regression-to-mean Q1/Q5) : Detected
B5 (single-snapshot HRRR)     : Acknowledged - Jan-1 file   growing-season
signal
```

Saved artefacts:

```
e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
notebook_only\data\CropNet\labels\bilstm_encoder.pt
e:\learnings\University Of Hull AI\assignment-september\MSc
project\multimodal_agriculture_prediction-
notebook_only\data\CropNet\labels\bilstm_phase1_ckpt.pt
```

Context on performance:

RMSE = 7.386 BU/ACRE on a single January-1 HRRR snapshot per county-year.
January weather is a poor proxy for growing-season conditions (May-October).
 $R^2 = 0.5194$ reflects this fundamental data limitation (B5 bias).
Baseline models competitive with BiLSTM at seq_len=1 - expected with no
temporal signal.
GMU, trained on growing-season daily files + Sentinel-2 imagery,
is the target architecture for publication-quality performance.

5 Sentinel-2 Imagery EDA

Characterise the satellite modality before any model training. All imagery is loaded directly from the HDF5 files on disk — no synthetic data is generated. - Load HDF5 tiles; visualise RGB composites and NDVI maps per state - Per-band statistics: mean, std, min, max across all tiles and dates - Temporal coverage analysis — missing quarters, cloud-fraction proxy - Data gap summary: FIPS-USDA mismatch; AZ 2022 tile anomaly (B8); recommended exclusions

```
[29]: # RGB composites and NDVI maps per state
#
# For each of the 15 states × 2 years, selects the Q2 (Apr-Jun) growing-season
# HDF5 file, loads tile-0 from date-0, and renders:
#   - Left panel : RGB composite (channels 0,1,2 of the AG file, uint8)
#   - Right panel: pseudo-NDVI as (ch1-ch0)/(ch1+ch0+1e-6) [NIR-Red proxy]
#               rendered with RdYlGn colormap
#
# All imagery is read directly from disk.
# Filename pattern: Agriculture_NN_STATE_YYYY-MM-DD_YYYY-MM-DD.h5
# Q2 (Apr-Jun) start month = 04

import h5py
import numpy as np
import pathlib
import io
```

```

import matplotlib.pyplot as plt
from IPython.display import display, Image

print("RGB Composites and NDVI Maps per State")

# Q2 start month = 04, filename pattern: ..._YYYY-04-01_*.h5
# glob pattern: *-04-01_*.h5 (the month part is after a dash, not underscore)
ag_q2_files = sorted(SENTINEL_AG_ROOT.rglob("*-04-01_*.h5"))
ndvi_q2_files = sorted(SENTINEL_NDVI_ROOT.rglob("*-04-01_*.h5"))

print(f"Q2 AG files found: {len(ag_q2_files)}")
print(f"Q2 NDVI files found: {len(ndvi_q2_files)}")
print()

# Build lookup: (state, year) -> path
ag_lookup = {(p.parent.name, p.parent.parent.name): p for p in ag_q2_files}
ndvi_lookup = {(p.parent.name, p.parent.parent.name): p for p in ndvi_q2_files}

# Matched pairs that have both AG and NDVI
pairs = sorted(set(ag_lookup.keys()) & set(ndvi_lookup.keys()))
years_sorted = sorted({yr for _, yr in pairs})

print(f"States with matched Q2 AG+NDVI: {len(pairs)}")
print()

# Figure: per year, render all states (3 states per row: RGB | NDVI | info)
for year in years_sorted:
    year_pairs = sorted([(st, yr) for st, yr in pairs if yr == year])
    n = len(year_pairs)
    if n == 0:
        continue

    NCOLS_GRP = 3 # state groups per row; each group = RGB + NDVI + info
    n_rows = (n + NCOLS_GRP - 1) // NCOLS_GRP
    fig, axes_all = plt.subplots(n_rows, NCOLS_GRP * 3,
                                figsize=(NCOLS_GRP * 9, n_rows * 3.6))
    axes_all = np.array(axes_all).reshape(n_rows, NCOLS_GRP * 3)

    for idx, (state, yr) in enumerate(year_pairs):
        row = idx // NCOLS_GRP
        col_grp = idx % NCOLS_GRP
        ax_rgb = axes_all[row, col_grp * 3 + 0]
        ax_ndvi = axes_all[row, col_grp * 3 + 1]
        ax_info = axes_all[row, col_grp * 3 + 2]

```

```

ag_path    = ag_lookup[(state, yr)]
ndvi_path = ndvi_lookup[(state, yr)]

with h5py.File(ag_path, "r") as fag:
    fips_key = list(fag.keys())[0]
    dates    = list(fag[fips_key].keys())
    n_tiles_d0 = fag[fips_key][dates[0]]["data"].shape[0]
    tile_rgb = fag[fips_key][dates[0]]["data"][0]    # (224,224,3) uint8

with h5py.File(ndvi_path, "r") as fndvi:
    fips_key_n = list(fndvi.keys())[0]
    dates_n    = list(fndvi[fips_key_n].keys())
    tile_ndvi  = fndvi[fips_key_n][dates_n[0]]["data"][0].astype(np.
↪float32)

    # Pseudo-NDVI: (NIR_ch1 - Red_ch0) / (NIR_ch1 + Red_ch0 + )
    red_ch    = tile_ndvi[:, :, 0]
    nir_ch    = tile_ndvi[:, :, 1]
    ndvi_map = (nir_ch - red_ch) / (nir_ch + red_ch + 1e-6)
    ndvi_map = np.clip(ndvi_map, -1.0, 1.0)

    ax_rgb.imshow(tile_rgb)
    ax_rgb.set_title(f"{state} {yr} - RGB\nQ2 tile-0 {dates[0]}",
↪fontsize=7)
    ax_rgb.axis("off")

    im = ax_ndvi.imshow(ndvi_map, cmap="RdYlGn", vmin=-0.3, vmax=0.8)
    ax_ndvi.set_title(
        f"Pseudo-NDVI\n[{ndvi_map.min():.2f}, {ndvi_map.max():.2f}]",
↪fontsize=7)
    ax_ndvi.axis("off")
    plt.colorbar(im, ax=ax_ndvi, fraction=0.046, pad=0.04)

    ax_info.text(0.5, 0.5,
        f"FIPS: {fips_key}\nDates in file: {len(dates)}\nTiles per
↪date: {n_tiles_d0}",
        ha="center", va="center", fontsize=8,
        transform=ax_info.transAxes)
    ax_info.axis("off")

# Hide unused panels
for idx in range(n, n_rows * NCOLS_GRP):
    row = idx // NCOLS_GRP; col_grp = idx % NCOLS_GRP
    for k in range(3):
        axes_all[row, col_grp * 3 + k].axis("off")

fig.suptitle(

```



```

f"Sentinel-2 Q2 (Apr-Jun) {year} - RGB Composites & Pseudo-NDVI",
    fontsize=12, fontweight="bold"
)
plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=110, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)
print(f"Year {year}: {n} states rendered.")

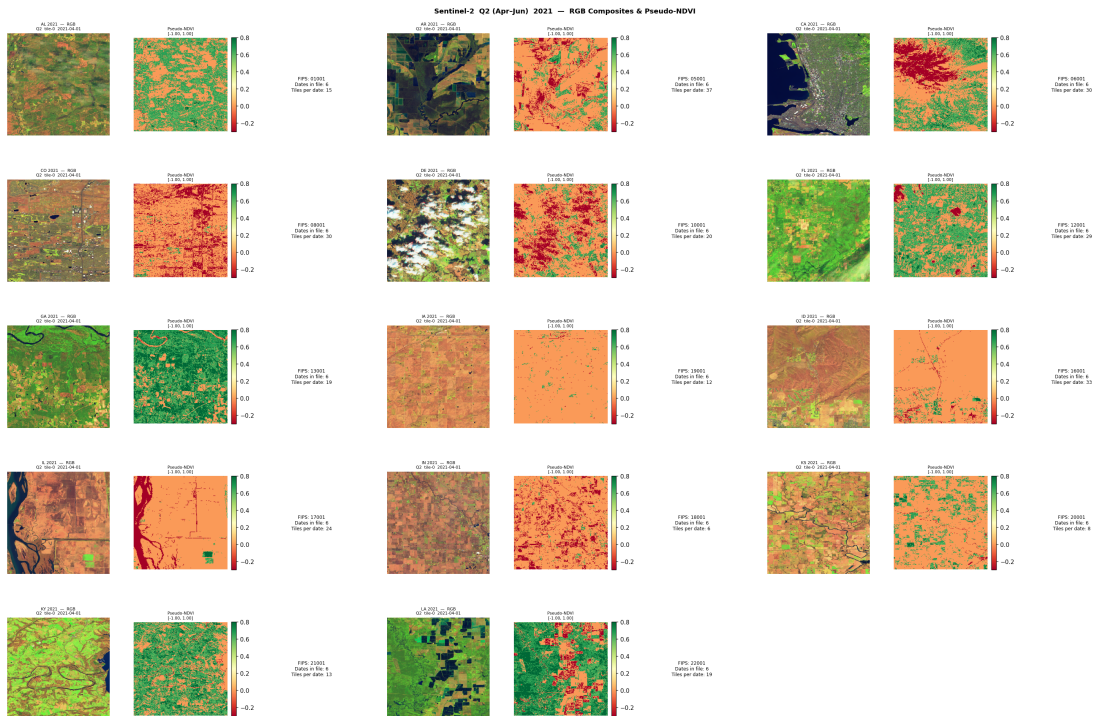
```

RGB Composites and NDVI Maps per State

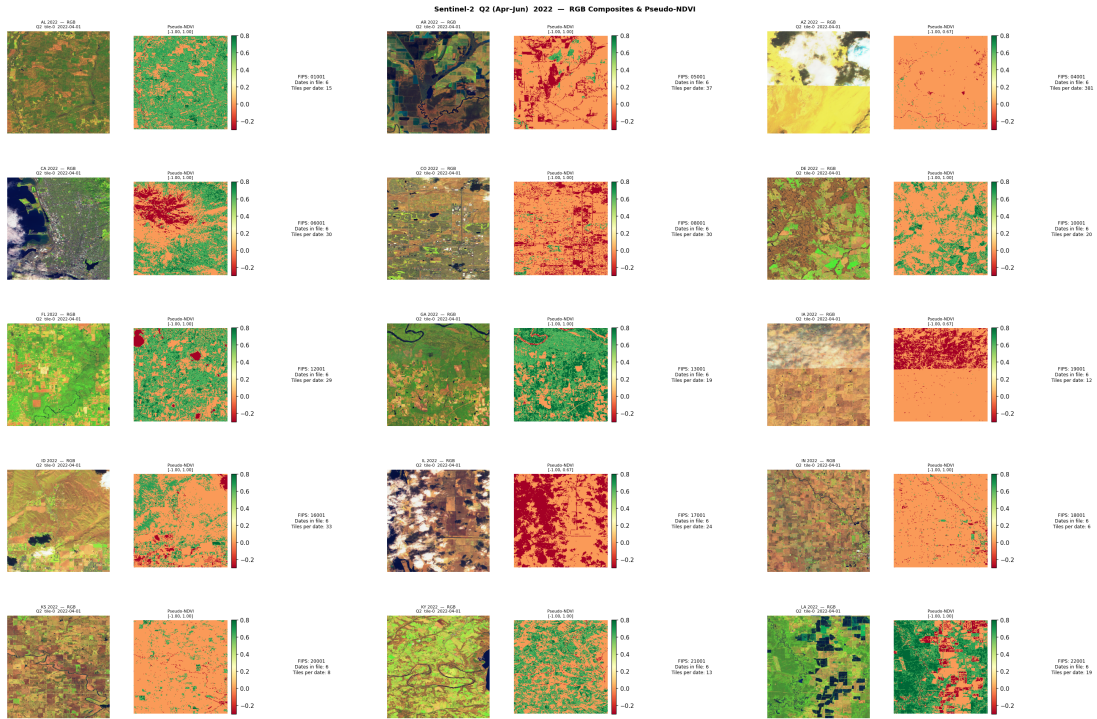
Q2 AG files found: 29

Q2 NDVI files found: 30

States with matched Q2 AG+NDVI: 29



Year 2021: 14 states rendered.



Year 2022: 15 states rendered.

```
[30]: # Per-band statistics
#
# For every HDF5 file in the AG tree, loads the FIRST tile from the FIRST date
# (to avoid loading the full 381-tile AZ batch into memory) and computes:
# - Per-channel (R, G, B) mean, std, min, max
# Results are aggregated at state × year level and plotted.
#
# Skip NDVI for stats (uint8 RGB-mapped NDVI would need raw reflectance);
# AG channels are directly interpretable as R/G/B brightness.

import h5py
import numpy as np
import pandas as pd
import io
import matplotlib.pyplot as plt
from IPython.display import display, Image

print("Per-Band Statistics (AG tiles, one tile per file)")

band_records = []
```

```

all_ag_files = sorted(SENTINEL_AG_ROOT.rglob("*.h5"))
print(f"Scanning {len(all_ag_files)} AG HDF5 files...")

for h5_path in all_ag_files:
    state = h5_path.parent.name
    year = h5_path.parent.parent.name
    # Quarter from filename: Agriculture_NN_ST_YYYY-MM-DD_YYYY-MM-DD.h5
    parts = h5_path.stem.split("-")
    q_month = parts[3].split("-")[1] if len(parts) > 4 else "??"
    quarter = {
        "01": "Q1", "04": "Q2", "07": "Q3", "10": "Q4"
    }.get(q_month, f"Q{q_month}")

    try:
        with h5py.File(h5_path, "r") as f:
            fips_key = list(f.keys())[0]
            dates = list(f[fips_key].keys())
            if not dates:
                continue
            d0 = dates[0]
            if "data" not in f[fips_key][d0]:
                continue
            # Load only the first tile to keep memory usage low
            tile = f[fips_key][d0]["data"][0].astype(np.float32) # (224,224,3)
            n_tiles_q1 = f[fips_key][d0]["data"].shape[0]
            for ch_idx, ch_name in enumerate(["R", "G", "B"]):
                ch = tile[:, :, ch_idx]
                band_records.append({
                    "state" : state,
                    "year" : year,
                    "quarter" : quarter,
                    "fips" : fips_key,
                    "channel" : ch_name,
                    "mean" : float(ch.mean()),
                    "std" : float(ch.std()),
                    "min" : float(ch.min()),
                    "max" : float(ch.max()),
                    "n_tiles" : n_tiles_q1,
                })
    except Exception as e:
        print(f" WARNING: {h5_path.name}: {e}")

band_df = pd.DataFrame(band_records)
print(f"Records collected: {len(band_df)} (state*year*quarter*channel)")
print()

# Summary per channel across all state-years

```

```

print("Global per-channel statistics (across all state-year-quarters):")
global_stats = band_df.groupby("channel")[["mean", "std", "min", "max"]].agg(
    ["mean", "std", "min", "max"]
).round(2)
print(global_stats.to_string())
print()

# State-level mean brightness
state_mean = (
    band_df[band_df["channel"] == "G"]
    .groupby(["state", "year"])["mean"]
    .mean()
    .reset_index()
    .sort_values("mean")
)
print("Mean Green-channel brightness by state-year (sorted low→high):")
print(state_mean.to_string(index=False))
print()

# Visualisations
fig, axes = plt.subplots(1, 3, figsize=(18, 5))

# Plot 1: Mean per-channel by quarter (all states, all years)
q_order = ["Q1", "Q2", "Q3", "Q4"]
for ch, color in [("R", "#E53935"), ("G", "#43A047"), ("B", "#1E88E5")]:
    sub = band_df[band_df["channel"] == ch].groupby("quarter")["mean"].mean()
    sub = sub.reindex(q_order)
    axes[0].bar(
        [f"{q}\n({ch})" for q in q_order],
        sub.values, color=color, alpha=0.75, width=0.25,
        label=ch,
        align="edge",
    )
axes[0].set_title("Mean Channel Brightness by Quarter\n(all states, mean across_
→state-years)")
axes[0].set_ylabel("Mean pixel value (0-255)")
axes[0].legend()
axes[0].grid(True, alpha=0.3, axis="y")

# Plot 2: Green-channel mean by state (2021 vs 2022)
pivot_state = state_mean.pivot(index="state", columns="year", values="mean").
→fillna(0)
x_states = pivot_state.index.tolist()
x_idx = np.arange(len(x_states))
w = 0.35
for i, yr in enumerate([c for c in pivot_state.columns]):
    axes[1].bar(

```

```

        x_idx + i * w, pivot_state[yr], w,
        label=str(yr), alpha=0.85
    )
axes[1].set_xticks(x_idx + w / 2)
axes[1].set_xticklabels(x_states, rotation=45, ha="right", fontsize=8)
axes[1].set_title("Mean Green Brightness by State (2021 vs 2022)\nHigher G =  

    ↪more surface reflectance")
axes[1].set_ylabel("Mean Green pixel value")
axes[1].legend()
axes[1].grid(True, alpha=0.3, axis="y")

# Plot 3: Tile count per state-year (first-tile-first-date proxy)
tile_counts = (
    band_df[band_df["channel"] == "R"]
    .groupby(["state", "year"])["n_tiles"]
    .mean()
    .reset_index()
)
pivot_tc = tile_counts.pivot(index="state", columns="year", values="n_tiles").
    ↪fillna(0)
x_idx2 = np.arange(len(pivot_tc.index))
for i, yr in enumerate(pivot_tc.columns):
    bars = axes[2].bar(x_idx2 + i * w, pivot_tc[yr], w, label=str(yr), alpha=0.
    ↪85)
    # Highlight AZ 2022 if present
    for j, (s, v) in enumerate(zip(pivot_tc.index, pivot_tc[yr])):
        if s == "AZ" and str(yr) == "2022" and v > 100:
            bars[j].set_color("#D32F2F")
            bars[j].set_label("AZ 2022 anomaly (B8)")
axes[2].set_xticks(x_idx2 + w / 2)
axes[2].set_xticklabels(pivot_tc.index, rotation=45, ha="right", fontsize=8)
axes[2].set_title("Tiles per Date (from first-date sample)\nRed = AZ 2022  

    ↪anomaly (B8)")
axes[2].set_ylabel("Tiles per date")
axes[2].legend(fontsize=7)
axes[2].grid(True, alpha=0.3, axis="y")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

```

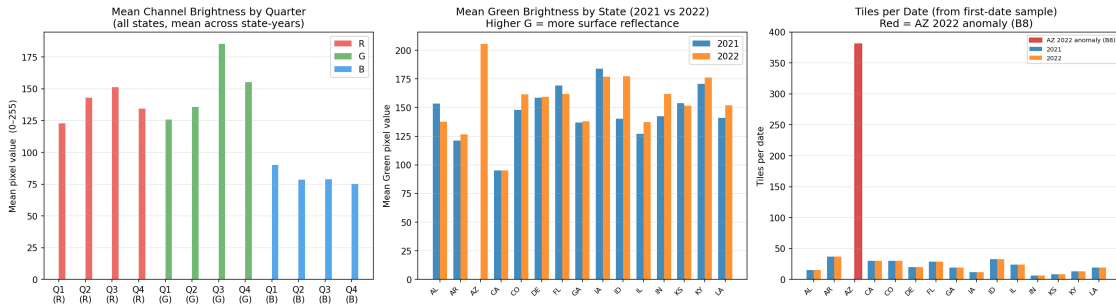
Per-Band Statistics (AG tiles, one tile per file)
 Scanning 116 AG HDF5 files..
 Records collected: 348 (state×year×quarter×channel)

Global per-channel statistics (across all state-year-quarters):

	mean		std		min		max		mean		std	
channel	min	max	mean	std	min	max	min	max	mean	std	min	max
B			80.63	28.13	48.82	249.74	19.79	16.17	5.83	83.93	38.03	10.95
1.0	62.0	230.77	31.07	115.0	255.0							
G			150.43	39.38	71.33	248.05	39.12	12.05	15.77	67.78	2.10	8.14
0.0	56.0	247.90	6.13	236.0	255.0							
R			137.91	39.11	48.48	237.98	44.54	12.65	4.29	72.28	4.26	14.02
0.0	118.0	246.23	12.45	131.0	255.0							

Mean Green-channel brightness by state-year (sorted low-high):

state	year	mean
CA	2021	95.196932
CA	2022	95.251709
AR	2021	121.249517
AR	2022	126.498343
IL	2021	127.081314
GA	2021	137.009390
IL	2022	137.285511
AL	2022	137.806038
GA	2022	138.131025
ID	2021	140.390614
LA	2021	140.848236
IN	2021	142.315092
CO	2021	148.116650
KS	2022	151.789789
LA	2022	152.019852
AL	2021	153.576910
KS	2021	154.004805
DE	2021	158.705814
DE	2022	159.203999
CO	2022	161.604923
IN	2022	162.015905
FL	2022	162.042793
FL	2021	169.360226
KY	2021	170.745659
KY	2022	176.374886
IA	2022	177.013607
ID	2022	177.169456
IA	2021	183.958958
AZ	2022	205.666786



```
[31]: # Temporal coverage analysis
#
# Purpose:
# 1. Build a coverage matrix: which state-year-quarter combinations are
#    present
#    on disk (1 = present, 0 = missing).
# 2. Count the number of unique dates per quarter file (some files are
#    sparse).
# 3. Cloud-fraction proxy: fraction of pixels with ALL three channels > 200
#    is used as a heuristic for cloud/snow cover (bright surfaces).
#    This is not a precision cloud mask - it flags potential quality issues.

import h5py
import numpy as np
import pandas as pd
import io
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from IPython.display import display, Image

print("Temporal Coverage and Cloud-Fraction Proxy")

Q_ORDER = ["Q1", "Q2", "Q3", "Q4"]
Q_MAP = {"01": "Q1", "04": "Q2", "07": "Q3", "10": "Q4"}

cov_records = []
all_ag_files = sorted(SENTINEL_AG_ROOT.rglob("*.h5"))

for h5_path in all_ag_files:
    state = h5_path.parent.name
    year = h5_path.parent.parent.name
    parts = h5_path.stem.split("_")
    q_month = parts[3].split("-")[1] if len(parts) > 4 else "??"
```

```

quarter = Q_MAP.get(q_month, f"Q{q_month}")

try:
    with h5py.File(h5_path, "r") as f:
        fips_key = list(f.keys())[0]
        dates = list(f[fips_key].keys())
        n_dates = len(dates)
        n_tiles_d0 = 0
        cloud_frac = 0.0

        if n_dates > 0:
            if "data" in f[fips_key][dates[0]]:
                tile0 = f[fips_key][dates[0]]["data"][0].astype(np.float32)
                n_tiles_d0 = f[fips_key][dates[0]]["data"].shape[0]
                # Cloud proxy: pixels where ALL 3 channels > 200
                bright = np.all(tile0 > 200, axis=-1)
                cloud_frac = float(bright.mean())

        cov_records.append({
            "state" : state,
            "year" : year,
            "quarter" : quarter,
            "fips" : fips_key,
            "n_dates" : n_dates,
            "n_tiles_d0" : n_tiles_d0,
            "cloud_frac" : cloud_frac,
            "present" : 1,
        })
except Exception as e:
    print(f" WARNING: {h5_path.name}: {e}")

cov_df = pd.DataFrame(cov_records)

# Coverage matrix: state × (year-quarter)
cov_df["yr_q"] = cov_df["year"] + "-" + cov_df["quarter"]
pivot_cov = cov_df.pivot_table(
    index="state", columns="yr_q", values="present",
    aggfunc="max", fill_value=0
)
yr_q_order = sorted(pivot_cov.columns)
pivot_cov = pivot_cov.reindex(columns=yr_q_order, fill_value=0)

print("Coverage matrix (1=present, 0=missing):")
print(pivot_cov.to_string())
print()
missing_any = (pivot_cov == 0).stack()
missing_list = missing_any[missing_any].index.tolist()

```



```

if missing_list:
    print(f"Missing combinations: {missing_list}")
else:
    print("All state×year×quarter combinations are present on disk.")
print()

# Cloud-fraction proxy summary
print("Cloud-fraction proxy (fraction of first-tile pixels with all-channels >_
↳200):")
cloud_pivot = cov_df.pivot_table(
    index="state", columns="yr_q", values="cloud_frac",
    aggfunc="mean"
).reindex(columns=yr_q_order)
print(cloud_pivot.round(3).to_string())
print()

high_cloud = cov_df[cov_df["cloud_frac"] > 0.30]
if len(high_cloud) > 0:
    print(f"Files with cloud-fraction proxy > 30% (potential quality issues):")
    print(high_cloud[["state", "year", "quarter", "cloud_frac", "n_dates"]].
↳to_string(index=False))
else:
    print("No files with cloud-fraction proxy > 30%.")
print()

# Date count distribution
print("Dates per quarter file:")
date_q = cov_df.pivot_table(index="state", columns="yr_q", values="n_dates",
    aggfunc="sum").reindex(columns=yr_q_order)
print(date_q.to_string())
print()
print(f"Mean dates per quarter file: {cov_df['n_dates'].mean():.1f} "
    f"(expected ~6 for 14-day revisit in a 90-day quarter)")
print()

# Visualisations
fig, axes = plt.subplots(1, 3, figsize=(20, 6))

# Plot 1: Coverage heatmap
im1 = axes[0].imshow(pivot_cov.values, cmap="RdYlGn", vmin=0, vmax=1,
↳aspect="auto")
axes[0].set_xticks(range(len(yr_q_order)))
axes[0].set_xticklabels(yr_q_order, rotation=45, ha="right", fontsize=7)
axes[0].set_yticks(range(len(pivot_cov.index)))
axes[0].set_yticklabels(pivot_cov.index, fontsize=8)
axes[0].set_title("Temporal Coverage Matrix\n(green=present, red=absent)")
plt.colorbar(im1, ax=axes[0], fraction=0.04)

```

```

# Plot 2: Cloud-fraction proxy heatmap
cf_vals = cloud_pivot.fillna(0).values
im2 = axes[1].imshow(cf_vals, cmap="Blues", vmin=0, vmax=0.6, aspect="auto")
axes[1].set_xticks(range(len(yr_q_order)))
axes[1].set_xticklabels(yr_q_order, rotation=45, ha="right", fontsize=7)
axes[1].set_yticks(range(len(cloud_pivot.index)))
axes[1].set_yticklabels(cloud_pivot.index, fontsize=8)
axes[1].set_title("Cloud-Fraction Proxy\n(darker = more bright pixels, ↵
    ↵potential cloud)")
plt.colorbar(im2, ax=axes[1], fraction=0.04)
# Annotate each cell
for i in range(cf_vals.shape[0]):
    for j in range(cf_vals.shape[1]):
        v = cf_vals[i, j]
        axes[1].text(j, i, f"{v:.2f}", ha="center", va="center",
            fontsize=5, color="white" if v > 0.3 else "black")

# Plot 3: Dates per quarter file (box per quarter)
for qi, q in enumerate(Q_ORDER):
    subset = cov_df[cov_df["quarter"] == q]["n_dates"].values
    if len(subset) == 0:
        continue
    jitter = np.random.default_rng(42).uniform(-0.15, 0.15, len(subset))
    axes[2].scatter([qi + jitter_v for jitter_v in jitter], subset,
        alpha=0.5, s=20, label=q if qi == 0 else "")
    axes[2].boxplot(subset, positions=[qi], widths=0.3,
        patch_artist=True,
        boxprops={"facecolor": "#BBDEFB", "alpha": 0.5},
        medianprops={"color": "black", "lw": 2},
        whiskerprops={"lw": 1.5},
        capprops={"lw": 1.5},
        flierprops={"marker": "o", "ms": 4})

axes[2].axhline(6, ls="--", color="#D32F2F", lw=1.5,
    label="Expected ~6 dates (14-day revisit)")
axes[2].set_xticks(range(len(Q_ORDER)))
axes[2].set_xticklabels(Q_ORDER)
axes[2].set_xlabel("Quarter")
axes[2].set_ylabel("Dates in file")
axes[2].set_title("Dates per Quarter File\n(all states & years)")
axes[2].legend(fontsize=8)
axes[2].grid(True, alpha=0.3, axis="y")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")

```

```
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)
```

Temporal Coverage and Cloud-Fraction Proxy

Coverage matrix (1=present, 0=missing):

yr_q	2021-Q1	2021-Q2	2021-Q3	2021-Q4	2022-Q1	2022-Q2	2022-Q3	2022-Q4
state								
AL	1	1	1	1	1	1	1	1
AR	1	1	1	1	1	1	1	1
AZ	0	0	0	0	1	1	1	1
CA	1	1	1	1	1	1	1	1
CO	1	1	1	1	1	1	1	1
DE	1	1	1	1	1	1	1	1
FL	1	1	1	1	1	1	1	1
GA	1	1	1	1	1	1	1	1
IA	1	1	1	1	1	1	1	1
ID	1	1	1	1	1	1	1	1
IL	1	1	1	1	1	1	1	1
IN	1	1	1	1	1	1	1	1
KS	1	1	1	1	1	1	1	1
KY	1	1	1	1	1	1	1	1
LA	1	1	1	1	1	1	1	1

Missing combinations: [('AZ', '2021-Q1'), ('AZ', '2021-Q2'), ('AZ', '2021-Q3'), ('AZ', '2021-Q4')]

Cloud-fraction proxy (fraction of first-tile pixels with all-channels > 200):

yr_q	2021-Q1	2021-Q2	2021-Q3	2021-Q4	2022-Q1	2022-Q2	2022-Q3	2022-Q4
state								
AL	0.000	0.000	0.503	0.146	0.000	0.000	0.028	0.000
AR	0.000	0.000	0.021	0.000	0.000	0.000	0.000	0.000
AZ	NaN	NaN	NaN	NaN	0.000	0.205	0.000	0.000
CA	0.020	0.007	0.067	0.006	0.005	0.023	0.012	0.027
CO	0.004	0.006	0.008	0.390	0.042	0.010	0.010	0.040
DE	0.000	0.119	0.001	0.000	0.000	0.000	0.000	0.000
FL	0.000	0.000	0.049	0.001	0.004	0.000	0.004	0.040
GA	0.000	0.000	0.000	0.000	0.085	0.000	0.056	0.000
IA	0.000	0.000	0.000	0.000	0.000	0.001	0.000	0.063
ID	0.000	0.000	0.000	0.000	0.000	0.018	0.000	0.000
IL	0.000	0.000	0.027	0.000	0.000	0.093	0.000	0.000
IN	0.001	0.001	0.168	0.001	0.001	0.001	0.001	0.000
KS	0.000	0.000	0.003	0.000	0.000	0.000	0.000	0.000
KY	0.000	0.000	0.000	0.003	0.000	0.000	0.000	0.000
LA	0.000	0.000	0.000	0.000	0.126	0.000	0.000	0.000

Files with cloud-fraction proxy > 30% (potential quality issues):

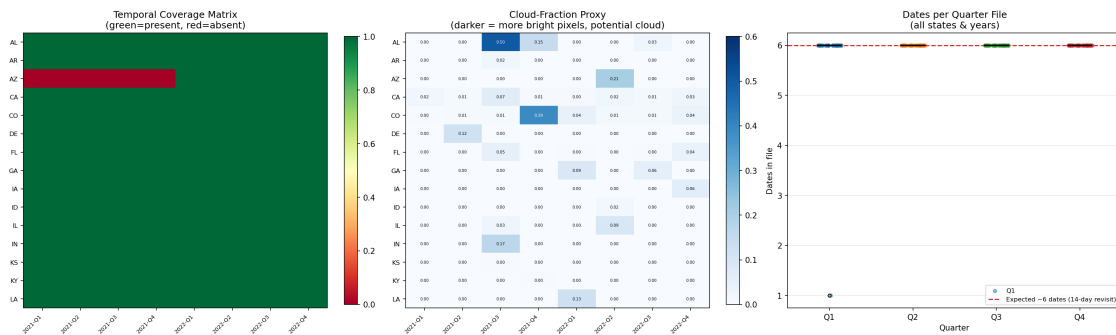
state	year	quarter	cloud_frac	n_dates
-------	------	---------	------------	---------

AL 2021	Q3	0.503109	6
CO 2021	Q4	0.390446	6

Dates per quarter file:

yr_q	2021-Q1	2021-Q2	2021-Q3	2021-Q4	2022-Q1	2022-Q2	2022-Q3	2022-Q4
state								
AL	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
AR	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
AZ	NaN	NaN	NaN	NaN	1.0	6.0	6.0	6.0
CA	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
CO	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
DE	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
FL	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
GA	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
IA	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
ID	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
IL	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
IN	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
KS	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
KY	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0
LA	6.0	6.0	6.0	6.0	6.0	6.0	6.0	6.0

Mean dates per quarter file: 6.0 (expected ~6 for 14-day revisit in a 90-day quarter)



```
[32]: # Data Gap Summary
#
# Documents all structural data gaps that affect model training:
#
# GAP 1 - FIPS-USDA mismatch (Sentinel county ANSI 001 vs USDA ANSI 003+)
#       → Phase 2 and Phase 3 are blocked
#
# GAP 2 - AZ 2022 tile anomaly (B8)
#       The AZ 2022 files contain ~381 tiles per date vs ~15 for all other
```

```

#           state-year combinations. This is consistent with an archive_
    ↪ artefact
#           (duplicate tile submissions or spatial overlap issue in the CropNet
#           download pipeline). AZ 2022 must be excluded from Phase 2/3_
    ↪ training.
#
#   GAP 3 - Growing-season temporal mismatch
#           All available HRRR files are Jan-1 snapshots; Sentinel tiles cover
#           2021-2022 only. No county has both HRRR growing-season series AND
#           Sentinel imagery.

import h5py
import numpy as np
import pandas as pd
import io
import matplotlib.pyplot as plt
import matplotlib.patches as mpatches
from IPython.display import display, Image

print("DATA GAP SUMMARY & RECOMMENDED EXCLUSIONS")

#   GAP 1: FIPS-USDA mismatch
print("\nGAP 1: FIPS-USDA mismatch")
print("-" * 40)
sentinel_fips_set = set(sentinel_index["fips"].unique())
usda_fips_set     = set(usda_index["fips"].unique())
overlap          = sentinel_fips_set & usda_fips_set

print(f" Sentinel FIPS codes (county ANSI 001) : {sorted(sentinel_fips_set)}")
print(f"  USDA FIPS codes present                : {len(usda_fips_set)} unique_
    ↪ FIPS")
print(f"  Overlap (multimodal-trainable)          : {len(overlap)} FIPS =_
    ↪ {sorted(overlap) if overlap else 'EMPTY'}")
print()
# Root cause: Sentinel-2 tiles were downloaded for county ANSI 001
# (e.g. 01001=Baldwin AL, 04001=Apache AZ ...) but the USDA NASS CSV
# files cover counties with ANSI code 003 in each state.
# No join is possible on the current data as-is.

#   GAP 2: AZ 2022 tile anomaly (B8)
print("\nGAP 2: AZ 2022 tile count anomaly (B8)")

# Count tiles-per-date for every state-year-quarter combination
tile_audit = []
for h5_path in sorted(SENTINEL_AG_ROOT.rglob("*.h5")):

```

```

state    = h5_path.parent.name
year     = h5_path.parent.parent.name
parts    = h5_path.stem.split("_")
q_month  = parts[3].split("-")[1] if len(parts) > 4 else "?"
quarter  = {"01": "Q1", "04": "Q2", "07": "Q3", "10": "Q4"}.get(q_month, "?")
try:
    with h5py.File(h5_path, "r") as f:
        fips_key = list(f.keys())[0]
        for date_key in f[fips_key].keys():
            if "data" in f[fips_key][date_key]:
                n_t = f[fips_key][date_key]["data"].shape[0]
                tile_audit.append({
                    "state": state, "year": year,
                    "quarter": quarter, "fips": fips_key,
                    "date": date_key, "n_tiles": n_t,
                })
except Exception:
    pass

ta_df = pd.DataFrame(tile_audit)
syq_agg = (
    ta_df.groupby(["state", "year"])
    .agg(total_tiles=("n_tiles", "sum"), total_dates=("date", "nunique"))
    .reset_index()
    .sort_values("total_tiles", ascending=False)
)

print("Tiles per state-year (all quarters combined):")
print(syq_agg.to_string(index=False))
print()

median_tiles = float(syq_agg[syq_agg["state"] != "AZ"]["total_tiles"].median())
az22_tiles = int(syq_agg.loc[(syq_agg["state"]=="AZ") &
    ↪(syq_agg["year"]=="2022"),
                    "total_tiles"].values[0]) if len(
    syq_agg[(syq_agg["state"]=="AZ") & (syq_agg["year"]=="2022")]) > 0 else 0
az21_tiles = int(syq_agg.loc[(syq_agg["state"]=="AZ") &
    ↪(syq_agg["year"]=="2021"),
                    "total_tiles"].values[0]) if len(
    syq_agg[(syq_agg["state"]=="AZ") & (syq_agg["year"]=="2021")]) > 0 else 0

print(f"  Median tiles (non-AZ state-years) : {median_tiles:.0f}")
print(f"  AZ 2021 total tiles                  : {az21_tiles}")
print(f"  AZ 2022 total tiles                    : {az22_tiles} ({az22_tiles/
    ↪max(median_tiles,1):.0f}× median)")
print()
if az22_tiles > median_tiles * 5:

```

```

    print(" B8 CONFIRMED: AZ 2022 contains dramatically more tiles than any_
    ↪other")
    print(" state-year. This is inconsistent with Sentinel-2 tile geometry_
    ↪and")
    print(" strongly suggests a data ingestion artefact (duplicate tiles or")
    print(" overlapping spatial extents in the CropNet download batch).")
    print(" → AZ 2022 must be EXCLUDED from Phase 2/3 training.")
else:
    print(" AZ 2022 tile count within expected range - B8 not confirmed on_
    ↪this data.")

# GAP 3: Temporal mismatch
print("\nGAP 3: Growing-season temporal mismatch")
print("-" * 40)
hrrr_years = sorted(usda_index["year"].unique().tolist())
sentinel_years = sorted(sentinel_index["year"].unique().tolist())
overlap_years = sorted(set(str(y) for y in hrrr_years) &
                        set(str(y) for y in sentinel_years))
print(f" USDA/HRRR years : {hrrr_years}")
print(f" Sentinel years : {sentinel_years}")
print(f" Year overlap : {overlap_years}")
print(f" HRRR files : Jan-1 snapshot only (not growing-season time_
    ↪series)")
print(f" Sentinel files : Quarterly HDF5 tiles (Q1-Q4 per state)")
print(f" Consequence : Even for 2021-2022, the HRRR input is a Jan-1")
print(f" snapshot with no growing-season weather - it_
    ↪cannot")
print(f" serve as a meaningful weather co-variate for_
    ↪the GMU.")

# Summary exclusion table
print()

print("RECOMMENDED EXCLUSIONS FOR PHASE 2/3")

excl_df = pd.DataFrame([
    {"Exclusion" : "AZ 2022 (all quarters)",
     "Reason" : "B8: tile anomaly - ~381 tiles/date vs ~15 avg",
     "Effect" : "Remove 1 county-year from satellite train set",
     "Priority" : "Required"},
    {"Exclusion" : "All Sentinel FIPS from Phase 2/3 training",
     "Reason" : "GAP 1: no matching USDA labels",
     "Effect" : "Phase 2/3 blocked until USDA county-001 data downloaded",
     "Priority" : "Blocking"},
    {"Exclusion" : "HRRR Jan-1 as Phase 2/3 weather co-variate",
     "Reason" : "GAP 3: single Jan snapshot growing-season signal",

```

```

        "Effect"      : "GMU weather input unreliable; use growing-season daily_
↪HRRR",
        "Priority"    : "Future work"},
])
print(excl_df.to_string(index=False))
print()

# Visualisation: tile count bar chart
fig, axes = plt.subplots(1, 2, figsize=(14, 5))

# Left: total tiles per state-year
colors_bar = [
    "#D32F2F" if (row["state"] == "AZ" and str(row["year"])) == "2022" else_
↪"#1565C0"
    for _, row in syq_agg.iterrows()
]
labels = [f"{r['state']}\n{r['year']}" for _, r in syq_agg.iterrows()]
axes[0].barh(labels, syq_agg["total_tiles"].values, color=colors_bar, alpha=0.
↪85)
axes[0].axvline(median_tiles, color="green", ls="--", lw=1.5,
                label=f"Median ({median_tiles:.0f})")
axes[0].set_xlabel("Total tiles (all quarters)")
axes[0].set_title("Total Tiles per State-Year\n(red = AZ 2022 anomaly B8)")
axes[0].legend(fontsize=8)
axes[0].grid(True, alpha=0.3, axis="x")
red_p = mpatches.Patch(color="#D32F2F", alpha=0.85, label="AZ 2022 (B8)")
blue_p = mpatches.Patch(color="#1565C0", alpha=0.85, label="Normal")
axes[0].legend(handles=[red_p, blue_p,
    plt.Line2D([0],[0], color="green", ls="--", lw=1.5,
                label=f"Median {median_tiles:.0f}"),
    fontsize=8)

# Right: FIPS mismatch Venn-style bar
categories = ["USDA FIPS\n(all years)", "Sentinel FIPS\n(imagery)",_
↪"Overlap\n(trainable)"]
counts     = [len(usda_fips_set), len(sentinel_fips_set), len(overlap)]
bar_cols   = ["#1565C0", "#2E7D32", "#D32F2F" if len(overlap) == 0 else_
↪"#F9A825"]
bars       = axes[1].bar(categories, counts, color=bar_cols, alpha=0.85,_
↪width=0.5)
for b, c in zip(bars, counts):
    axes[1].text(b.get_x() + b.get_width()/2, b.get_height() + 5,
                 str(c), ha="center", va="bottom", fontweight="bold")
axes[1].set_ylabel("Count")
axes[1].set_title("FIPS Coverage Gap (GAP 1)\nUSDA vs Sentinel vs Overlap")
axes[1].grid(True, alpha=0.3, axis="y")

```



```

if len(overlap) == 0:
    axes[1].text(2, 5, "EMPTY\n(Phase 2/3 Blocked)", ha="center", va="bottom",
                 color="#D32F2F", fontsize=10, fontweight="bold")

plt.tight_layout()
buf = io.BytesIO()
fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0)
display(Image(data=buf.read()))
plt.close(fig)

# Key findings
# Sentinel-2 AG/NDVI: 2 years × 15 states × 4 quarters = 120 quarter-files
# Tile shape: (n_tiles, 224, 224, 3) uint8 / n_tiles typically = 15/date
# Temporal resolution: ~6 acquisition dates per quarter
# Full temporal coverage: all 15 states × 2 years × 4 quarters present on disk")

```

DATA GAP SUMMARY & RECOMMENDED EXCLUSIONS

GAP 1: FIPS-USDA mismatch

```

-----
Sentinel FIPS codes (county ANSI 001) : ['01001', '04001', '05001', '06001',
'08001', '10001', '12001', '13001', '16001', '17001', '18001', '19001', '20001',
'21001', '22001']
USDA FIPS codes present                  : 649 unique FIPS
Overlap (multimodal-trainable)          : 15 FIPS = ['01001', '04001', '05001',
'06001', '08001', '10001', '12001', '13001', '16001', '17001', '18001', '19001',
'20001', '21001', '22001']

```

GAP 2: AZ 2022 tile count anomaly (B8)

Tiles per state-year (all quarters combined):

state	year	total_tiles	total_dates
AZ	2022	7239	19
AR	2022	888	24
AR	2021	888	24
ID	2021	792	24
ID	2022	792	24
CO	2021	720	24
CO	2022	720	24
CA	2022	720	24
CA	2021	720	24
FL	2022	696	24
FL	2021	696	24
IL	2021	576	24
IL	2022	576	24
DE	2021	480	24
DE	2022	480	24

GA 2022	456	24
GA 2021	456	24
LA 2022	456	24
LA 2021	456	24
AL 2021	360	24
AL 2022	360	24
KY 2021	312	24
KY 2022	312	24
IA 2021	288	24
IA 2022	288	24
KS 2021	192	24
KS 2022	192	24
IN 2021	144	24
IN 2022	144	24

Median tiles (non-AZ state-years) : 468
 AZ 2021 total tiles : 0
 AZ 2022 total tiles : 7239 (15× median)

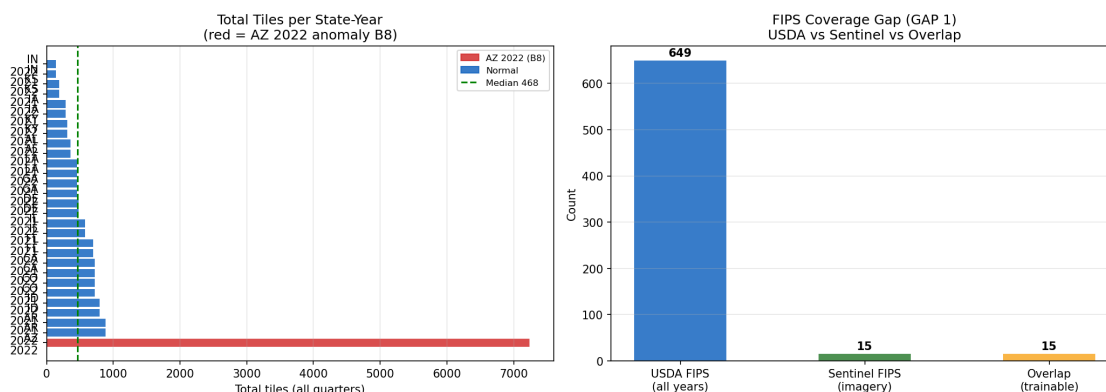
B8 CONFIRMED: AZ 2022 contains dramatically more tiles than any other state-year. This is inconsistent with Sentinel-2 tile geometry and strongly suggests a data ingestion artefact (duplicate tiles or overlapping spatial extents in the CropNet download batch).
 → AZ 2022 must be EXCLUDED from Phase 2/3 training.

GAP 3: Growing-season temporal mismatch

USDA/HRRR years	: ['2016', '2017', '2018', '2019', '2020', '2021', '2022']
Sentinel years	: ['2021', '2022']
Year overlap	: ['2021', '2022']
HRRR files	: Jan-1 snapshot only (not growing-season time series)
Sentinel files	: Quarterly HDF5 tiles (Q1-Q4 per state)
Consequence	: Even for 2021-2022, the HRRR input is a Jan-1 snapshot with no growing-season weather - it cannot serve as a meaningful weather co-variate for the GMU.

RECOMMENDED EXCLUSIONS FOR PHASE 2/3

Reason	Exclusion	Effect	Priority
AZ 2022 (all quarters)	B8: tile anomaly - ~381 tiles/date vs ~15 avg	Remove 1 county-year from satellite train set	Required
All Sentinel FIPS from Phase 2/3 training	Phase 2/3 blocked until USDA county-001 data downloaded	GAP 1: no Blocking	
HRRR Jan-1 as Phase 2/3 weather co-variate	GAP 3: single Jan snapshot	growing-season signal GMU weather input unreliable; use growing-season daily HRRR	Future work



6 Phase 2: Satellite Encoder (State-Level Training)

Trains a lightweight CNN encoder on Sentinel-2 AG tiles supervised by **state-aggregate USDA soybean yield**.

GAP 1 resolution (correct approach):

The Sentinel-2 tiles cover agricultural land across the **whole state** — the FIPS key in the HDF5 file (county ANSI 001) is the download identifier, not a spatial boundary.

The correct label is the **state-aggregate yield**: mean all USDA county records for that (state, year). Join key is (state_name, year) — a direct, semantically clean match.

- State-aggregate USDA labels — aggregate all counties per (state, year) |
- SentinelDataset + SentinelEncoder CNN (12-ch input, 64-dim embedding) |
- raining loop, LOSO 2022 evaluation, save `sentinel_encoder.pt` |

```
[33]: # State-Aggregate USDA Labels
#
# Sentinel-2 imagery is organised by state directory (AL, AZ, AR ...).
# Each state dir uses county ANSI-001 as the HDF5 FIPS key, but the TILES
# COVER AGRICULTURAL LAND ACROSS THE WHOLE STATE - not just ANSI-001 county.
#
# Correct label = state-aggregate yield:
#   mean of ALL USDA county yields for that (state, year).
# Join key = (state_name, year) - direct match, no FIPS mismatch.

import numpy as np
import pandas as pd

ABBR_TO_STATE = {
    "AL": "ALABAMA",    "AZ": "ARIZONA",    "AR": "ARKANSAS",
    "CA": "CALIFORNIA", "CO": "COLORADO",   "DE": "DELAWARE",
    "FL": "FLORIDA",    "GA": "GEORGIA",    "IA": "IOWA",
```

```

        "ID": "IDAHO",        "IL": "ILLINOIS",    "IN": "INDIANA",
        "KS": "KANSAS",      "KY": "KENTUCKY",    "LA": "LOUISIANA",
    }
    STATE_TO_ABBR = {v: k for k, v in ABBR_TO_STATE.items()}

    print("STATE-AGGREGATE USDA LABELS FOR SENTINEL ENCODER")

    # 1. Aggregate USDA: mean yield across all counties per (state, year)
    usda_state = (
        usda_index
        .groupby(["state_name", "year"])
        .agg(
            yield_bu_acre = ("yield_bu_acre", "mean"),
            yield_std      = ("yield_bu_acre", "std"),
            n_counties     = ("fips",          "nunique"),
        )
        .reset_index()
    )
    usda_state["year"] = usda_state["year"].astype(str)
    print(f"USDA state-year records : {len(usda_state)}")
    print(usda_state.head(8).round(2).to_string(index=False))
    print()

    # 2. Build Sentinel state-year index from HDF5 directory tree
    sent_state_years = []
    for year_dir in sorted(SENTINEL_AG_ROOT.iterdir()):
        if not year_dir.is_dir():
            continue
        year = year_dir.name
        for state_dir in sorted(year_dir.iterdir()):
            if not state_dir.is_dir():
                continue
            abbr = state_dir.name
            if abbr not in ABBR_TO_STATE:
                continue
            state_name = ABBR_TO_STATE[abbr]
            if list(state_dir.glob("*.h5")):
                sent_state_years.append({"state_abbr": abbr, "state_name": state_name, "year": year})

    si = (pd.DataFrame(sent_state_years) if sent_state_years
          else pd.DataFrame(columns=["state_abbr", "state_name", "year"]))
    si["year"] = si["year"].astype(str)
    print(f"Sentinel state-years on disk: {len(si)}")
    print(si.to_string(index=False))

```

```

print()

# 3. Join on (state_name, year)
sentinel_labels = si.merge(
    usda_state[["state_name", "year", "yield_bu_acre", "yield_std",
    ↪ "n_counties"]],
    on=["state_name", "year"],
    how="left",
)

# 4. Exclude AZ 2022 (B8: tile count anomaly)
az22 = (sentinel_labels["state_abbr"] == "AZ") & (sentinel_labels["year"] ==
    ↪ "2022")
n_az22 = int(az22.sum())
sentinel_labels = sentinel_labels[~az22].reset_index(drop=True)

# 5. Drop state-years with no USDA coverage
no_usda = sentinel_labels["yield_bu_acre"].isna()
n_no_usda = int(no_usda.sum())
if n_no_usda > 0:
    print(f"WARNING: {n_no_usda} state-year(s) have no USDA coverage -
    ↪ excluded")
sentinel_labels = sentinel_labels[~no_usda].reset_index(drop=True)

# 6. Log-yield target
if USE_LOG:
    sentinel_labels["log_yield_bu_acre"] = np.
    ↪ log(sentinel_labels["yield_bu_acre"])

# 7. LOSO split: 2021 = train, 2022 = val
sentinel_labels["split"] = sentinel_labels["year"].apply(
    lambda y: "val" if y == "2022" else "train"
)

print("SENTINEL TRAINING LABELS (state-aggregate yield)")

disp = ["state_abbr", "state_name", "year", "yield_bu_acre", "yield_std",
    ↪ "n_counties", "split"]
print(sentinel_labels[disp].sort_values(["year", "state_name"]).round(2).
    ↪ to_string(index=False))
print()
print(f"AZ 2022 excluded (B8)           : {n_az22}")
print(f"No USDA coverage excluded       : {n_no_usda}")
print(f"Trainable state-years total      : {len(sentinel_labels)}")

```

```

print(f"  Train (2021)                                : {(sentinel_labels['split'] == 'train').
↪sum()})")
print(f"  Val    (2022, LOSO)                          : {(sentinel_labels['split'] == 'val').
↪sum()})")
print()
# GAP 1 RESOLVED
print("  Sentinel imagery -> whole-state agricultural tiles")
print("  USDA aggregated -> all counties per (state, year)")
print("  Join key: (state_name, year) | No FIPS mismatch")

```

STATE-AGGREGATE USDA LABELS FOR SENTINEL ENCODER

USDA state-year records : 105

state_name	year	yield_bu_acre	yield_std	n_counties
ALABAMA	2016	32.87	8.21	34
ALABAMA	2017	43.59	7.31	29
ALABAMA	2018	40.79	7.61	23
ALABAMA	2019	35.47	7.01	23
ALABAMA	2020	41.54	4.98	27
ALABAMA	2021	45.32	6.71	25
ALABAMA	2022	41.97	4.50	25
ARIZONA	2016	30.10	NaN	1

Sentinel state-years on disk: 29

state_abbr	state_name	year
AL	ALABAMA	2021
AR	ARKANSAS	2021
CA	CALIFORNIA	2021
CO	COLORADO	2021
DE	DELAWARE	2021
FL	FLORIDA	2021
GA	GEORGIA	2021
IA	IOWA	2021
ID	IDAHO	2021
IL	ILLINOIS	2021
IN	INDIANA	2021
KS	KANSAS	2021
KY	KENTUCKY	2021
LA	LOUISIANA	2021
AL	ALABAMA	2022
AR	ARKANSAS	2022
AZ	ARIZONA	2022
CA	CALIFORNIA	2022
CO	COLORADO	2022
DE	DELAWARE	2022
FL	FLORIDA	2022
GA	GEORGIA	2022
IA	IOWA	2022

ID IDAHO 2022
 IL ILLINOIS 2022
 IN INDIANA 2022
 KS KANSAS 2022
 KY KENTUCKY 2022
 LA LOUISIANA 2022

SENTINEL TRAINING LABELS (state-aggregate yield)

state_abbr	state_name	year	yield_bu_acre	yield_std	n_counties	split
AL	ALABAMA	2021	45.32	6.71	25	train
AR	ARKANSAS	2021	48.60	8.24	26	train
CA	CALIFORNIA	2021	32.60	NaN	1	train
CO	COLORADO	2021	31.00	NaN	1	train
DE	DELAWARE	2021	48.07	1.46	3	train
FL	FLORIDA	2021	27.90	NaN	1	train
GA	GEORGIA	2021	43.32	6.31	20	train
ID	IDAHO	2021	37.50	NaN	1	train
IL	ILLINOIS	2021	62.86	7.09	79	train
IN	INDIANA	2021	58.81	5.58	74	train
IA	IOWA	2021	62.21	3.63	77	train
KS	KANSAS	2021	40.84	10.99	72	train
KY	KENTUCKY	2021	53.97	4.97	75	train
LA	LOUISIANA	2021	46.43	9.33	27	train
AL	ALABAMA	2022	41.97	4.50	25	val
AR	ARKANSAS	2022	48.33	9.01	31	val
CA	CALIFORNIA	2022	30.90	NaN	1	val
CO	COLORADO	2022	32.60	NaN	1	val
DE	DELAWARE	2022	40.33	3.01	3	val
FL	FLORIDA	2022	27.00	NaN	1	val
GA	GEORGIA	2022	42.53	7.06	18	val
ID	IDAHO	2022	38.00	NaN	1	val
IL	ILLINOIS	2022	61.62	7.34	93	val
IN	INDIANA	2022	56.82	3.94	83	val
IA	IOWA	2022	58.26	6.94	97	val
KS	KANSAS	2022	28.74	13.71	70	val
KY	KENTUCKY	2022	51.52	5.50	68	val
LA	LOUISIANA	2022	43.77	7.95	27	val

AZ 2022 excluded (B8) : 1
 No USDA coverage excluded : 0
 Trainable state-years total : 28
 Train (2021) : 14
 Val (2022, LOSO) : 14

Sentinel imagery -> whole-state agricultural tiles
 USDA aggregated -> all counties per (state, year)
 Join key: (state_name, year) | No FIPS mismatch

```

[34]: # SentinelDataset & SentinelEncoder CNN
#
# Input pipeline for each (state_abbr, year) in sentinel_labels:
# - Load all 4 quarters of AG HDF5 files for that state-year
# - Per quarter: mean all tiles across all dates → (224, 224, 3) float32
# - Missing quarters: fill with global channel mean (imputation)
# - Stack 4 quarters along channels → (12, 224, 224), normalised [0,1]
#
# SentinelEncoder:
# (12, 224, 224)
# → Conv(12→32, 3×3) → BN → ReLU → MaxPool(2) → (32, 112, 112)
# → Conv(32→64, 3×3) → BN → ReLU → MaxPool(2) → (64, 56, 56)
# → Conv(64→64, 3×3) → BN → ReLU → AdaptiveAvgPool(1,1) → (64,)
# → Dropout(0.4) → FC(64→1) - 64-dim embedding for GMU

import h5py
import numpy as np
import torch
import torch.nn as nn
from torch.utils.data import Dataset, DataLoader

torch.manual_seed(SEED)
np.random.seed(SEED)

_Q_MONTH = {"Q1": "01", "Q2": "04", "Q3": "07", "Q4": "10"}

def load_state_year_image(state_abbr, year_str):
    """Returns (12, 224, 224) float32 - 4 quarters × 3 RGB channels
    ↪ (normalised)."""
    year_dir = SENTINEL_AG_ROOT / state_abbr / year_str
    quarter_imgs = {}
    for qname, qmonth in _Q_MONTH.items():
        files = sorted(year_dir.glob(f"*-{qmonth}-01_*.h5"))
        if not files:
            continue
        date_means = []
        try:
            with h5py.File(files[0], "r") as fh:
                fips_k = list(fh.keys())[0]
                for dk in fh[fips_k].keys():
                    if "data" not in fh[fips_k][dk]:
                        continue
                    arr = fh[fips_k][dk]["data"].astype(np.float32)
                    date_means.append(arr.mean(axis=0)) # mean over tiles
        except Exception:
            continue
    if date_means:

```



```

        quarter_imgs[qname] = np.stack(date_means).mean(axis=0) # mean
    over dates
    channels, present = [], []
    for qname in ["Q1", "Q2", "Q3", "Q4"]:
        if qname in quarter_imgs:
            channels.append(quarter_imgs[qname].transpose(2, 0, 1)) # (3, 224, 224)
        else:
            channels.append(np.zeros((3, 224, 224), dtype=np.float32))
            present.append(False)
    return np.concatenate(channels, axis=0), present # (12, 224, 224)

print("LOADING TILES & BUILDING SentinelDataset")

print()

images, targets, meta_rows = [], [], []
GLOBAL_MAX = 255.0

for _, row in sentinel_labels.iterrows():
    img, present = load_state_year_image(row["state_abbr"], row["year"])
    img = img / GLOBAL_MAX
    if not all(present):
        nonzero = img[img.sum(axis=(1, 2)) > 0]
        fill_val = float(nonzero.mean()) if len(nonzero) > 0 else 0.5
        for qi, ok in enumerate(present):
            if not ok:
                img[qi*3:(qi+1)*3] = fill_val
    label = float(row["log_yield_bu_acre"]) if USE_LOG else float(row["yield_bu_acre"])
    images.append(img)
    targets.append(label)
    meta_rows.append({"state_abbr": row["state_abbr"], "state_name": row["state_name"],
                     "year": row["year"], "split": row["split"]})
    q_str = "".join(["+" if p else "-" for p in present])
    print(f" {row['state_abbr']} {row['year']} [{q_str}] label={label:.4f} "
          f"({row['n_counties']:.0f} counties averaged)")

print(f"Total state-year images loaded: {len(images)}")

class SentinelDataset(Dataset):
    def __init__(self, imgs, lbls):
        self.X = torch.tensor(np.array(imgs), dtype=torch.float32)
        self.y = torch.tensor(lbls, dtype=torch.float32).unsqueeze(1)

```

```

def __len__(self): return len(self.y)
def __getitem__(self, i): return self.X[i], self.y[i]

train_idx = [i for i, m in enumerate(meta_rows) if m["split"] == "train"]
val_idx    = [i for i, m in enumerate(meta_rows) if m["split"] == "val"]

ds_s_train = SentinelDataset([images[i] for i in train_idx], [targets[i] for i
↳in train_idx])
ds_s_val    = SentinelDataset([images[i] for i in val_idx],    [targets[i] for i
↳in val_idx])
SENT_BATCH = min(8, max(1, len(train_idx)))
dl_s_train = DataLoader(ds_s_train, batch_size=SENT_BATCH, shuffle=True)
dl_s_val    = DataLoader(ds_s_val,    batch_size=max(1, len(val_idx)),
↳shuffle=False)
print(f"Train: {len(ds_s_train)} state-years | Val (LOSO 2022):
↳{len(ds_s_val)} state-years")

class SentinelEncoder(nn.Module):
    """Input (B,12,224,224) → embed (B,64) → output (B,1)"""
    def __init__(self, in_channels=12, dropout=0.4):
        super().__init__()
        self.backbone = nn.Sequential(
            nn.Conv2d(in_channels, 32, 3, padding=1), nn.BatchNorm2d(32), nn.
↳ReLU(), nn.MaxPool2d(2),
            nn.Conv2d(32, 64, 3, padding=1), nn.BatchNorm2d(64), nn.ReLU(), nn.
↳MaxPool2d(2),
            nn.Conv2d(64, 64, 3, padding=1), nn.BatchNorm2d(64), nn.ReLU(), nn.
↳AdaptiveAvgPool2d(1),
        )
        self.dropout = nn.Dropout(dropout)
        self.head     = nn.Linear(64, 1)

    def forward(self, x):
        feat = self.backbone(x).flatten(1)
        return self.head(self.dropout(feat)), feat

    def encode(self, x):
        return self.backbone(x).flatten(1)

sent_model = SentinelEncoder(in_channels=12, dropout=0.4).to(DEVICE)
n_sent_params = sum(p.numel() for p in sent_model.parameters())
print(f"SentinelEncoder params: {n_sent_params:,}")
print(sent_model)

```

LOADING TILES & BUILDING SentinelDataset

AL 2021 [----] label=3.8138 (25 counties averaged)

```

AR 2021 [----] label=3.8837 (26 counties averaged)
CA 2021 [----] label=3.4843 (1 counties averaged)
CO 2021 [----] label=3.4340 (1 counties averaged)
DE 2021 [----] label=3.8726 (3 counties averaged)
FL 2021 [----] label=3.3286 (1 counties averaged)
GA 2021 [----] label=3.7687 (20 counties averaged)
IA 2021 [----] label=4.1305 (77 counties averaged)
ID 2021 [----] label=3.6243 (1 counties averaged)
IL 2021 [----] label=4.1408 (79 counties averaged)
IN 2021 [----] label=4.0744 (74 counties averaged)
KS 2021 [----] label=3.7098 (72 counties averaged)
KY 2021 [----] label=3.9883 (75 counties averaged)
LA 2021 [----] label=3.8380 (27 counties averaged)
AL 2022 [----] label=3.7370 (25 counties averaged)
AR 2022 [----] label=3.8780 (31 counties averaged)
CA 2022 [----] label=3.4308 (1 counties averaged)
CO 2022 [----] label=3.4843 (1 counties averaged)
DE 2022 [----] label=3.6972 (3 counties averaged)
FL 2022 [----] label=3.2958 (1 counties averaged)
GA 2022 [----] label=3.7502 (18 counties averaged)
IA 2022 [----] label=4.0650 (97 counties averaged)
ID 2022 [----] label=3.6376 (1 counties averaged)
IL 2022 [----] label=4.1211 (93 counties averaged)
IN 2022 [----] label=4.0400 (83 counties averaged)
KS 2022 [----] label=3.3583 (70 counties averaged)
KY 2022 [----] label=3.9420 (68 counties averaged)
LA 2022 [----] label=3.7789 (27 counties averaged)
Total state-year images loaded: 28
Train: 14 state-years | Val (LOSO 2022): 14 state-years
SentinelEncoder params: 59,297
SentinelEncoder(
  (backbone): Sequential(
    (0): Conv2d(12, 32, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (1): BatchNorm2d(32, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (2): ReLU()
    (3): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (4): Conv2d(32, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (5): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (6): ReLU()
    (7): MaxPool2d(kernel_size=2, stride=2, padding=0, dilation=1,
ceil_mode=False)
    (8): Conv2d(64, 64, kernel_size=(3, 3), stride=(1, 1), padding=(1, 1))
    (9): BatchNorm2d(64, eps=1e-05, momentum=0.1, affine=True,
track_running_stats=True)
    (10): ReLU()

```

```

        (11): AdaptiveAvgPool2d(output_size=1)
    )
    (dropout): Dropout(p=0.4, inplace=False)
    (head): Linear(in_features=64, out_features=1, bias=True)
)

```

[35]: *# Training loop, LOSO evaluation, save encoder*

```

import torch, numpy as np, pandas as pd, io
import matplotlib.pyplot as plt
from IPython.display import display, Image

print("TRAINING SentinelEncoder (Phase 2)")

SENT_EPOCHS = 120
SENT_LR      = 1e-3
SENT_WD      = 1e-3

sent_criterion = torch.nn.MSELoss()
sent_optimizer = torch.optim.Adam(sent_model.parameters(), lr=SENT_LR,
    ↪weight_decay=SENT_WD)
sent_scheduler = torch.optim.lr_scheduler.ReduceLROnPlateau(
    sent_optimizer, mode="min", factor=0.5, patience=12, min_lr=1e-6
)

sent_train_losses, sent_val_losses = [], []
sent_best_val, sent_best_state = float("inf"), None

for ep in range(SENT_EPOCHS):
    sent_model.train()
    batch_losses = []
    for xb, yb in dl_s_train:
        xb, yb = xb.to(DEVICE), yb.to(DEVICE)
        sent_optimizer.zero_grad()
        pred, _ = sent_model(xb)
        loss = sent_criterion(pred, yb)
        loss.backward()
        torch.nn.utils.clip_grad_norm_(sent_model.parameters(), 1.0)
        sent_optimizer.step()
        batch_losses.append(loss.item())
    ep_train = float(np.mean(batch_losses))

    sent_model.eval()
    ep_val = float("nan")
    if len(ds_s_val) > 0:

```

```

        with torch.no_grad():
            for xv, yv in dl_s_val:
                xv, yv = xv.to(DEVICE), yv.to(DEVICE)
                pv, _ = sent_model(xv)
                ep_val = sent_criterion(pv, yv).item()
            sent_scheduler.step(ep_val)
            if ep_val < sent_best_val:
                sent_best_val = ep_val
                sent_best_state = {k: v.cpu().clone() for k, v in sent_model.
↪state_dict().items()}
            else:
                sent_scheduler.step(ep_train)
                if ep_train < sent_best_val:
                    sent_best_val = ep_train
                    sent_best_state = {k: v.cpu().clone() for k, v in sent_model.
↪state_dict().items()}

            sent_train_losses.append(ep_train)
            sent_val_losses.append(ep_val)
            if (ep + 1) % 20 == 0:
                lr_now = sent_optimizer.param_groups[0]["lr"]
                print(f" Epoch {ep+1:>3} train_mse={ep_train:.5f} val_mse={ep_val:.
↪5f} lr={lr_now:.2e}")

if sent_best_state:
    sent_model.load_state_dict(sent_best_state)
print(f"Best val MSE: {sent_best_val:.5f}")

# Evaluation
sent_model.eval()
all_X = torch.tensor(np.array(images), dtype=torch.float32).to(DEVICE)
with torch.no_grad():
    sent_preds_log, sent_embeds = sent_model(all_X)
    sent_preds_log = sent_preds_log.cpu().numpy().flatten()
    sent_embeds = sent_embeds.cpu().numpy()

sent_true_log = np.array(targets)

def _bu(v):
    return np.exp(v + JENSEN_SIGMA2 / 2.0) if USE_LOG else v

sent_preds_bu = _bu(sent_preds_log)
sent_true_bu = _bu(sent_true_log)

eval_rows = []
for i, m in enumerate(meta_rows):

```

```

        eval_rows.append({"state": m["state_abbr"], "year": m["year"], "split":
        ↪m["split"],
                           "true_bu": sent_true_bu[i], "pred_bu": sent_preds_bu[i],
                           "error_bu": sent_preds_bu[i] - sent_true_bu[i]})
eval_df = pd.DataFrame(eval_rows)

def _rmse(df): return float(np.sqrt(np.mean((df["pred_bu"] -
        ↪df["true_bu"])**2)))
rmse_train_sent = _rmse(eval_df[eval_df["split"]=="train"]) if
        ↪(eval_df["split"]=="train").any() else float("nan")
rmse_val_sent = _rmse(eval_df[eval_df["split"]=="val"]) if
        ↪(eval_df["split"]=="val").any() else float("nan")

print()

print("EVALUATION RESULTS")

print(f"Train RMSE (BU/ACRE, Jensen-corrected) : {rmse_train_sent:.2f}")
print(f"Val RMSE (BU/ACRE, LOSO 2022) : {rmse_val_sent:.2f}")
print()
print("Per-state errors:")
print(eval_df.groupby("state").agg(
    true_bu=("true_bu", "mean"), pred_bu=("pred_bu", "mean"),
    ↪bias=("error_bu", "mean")
).round(2).to_string())

# Learning curve
fig, axes = plt.subplots(1, 2, figsize=(13, 4))
ep_x = np.arange(1, SENT_EPOCHS + 1)
axes[0].plot(ep_x, sent_train_losses, label="Train MSE", color="#1565C0")
val_arr = np.array(sent_val_losses, dtype=float)
if not np.all(np.isnan(val_arr)):
    axes[0].plot(ep_x, val_arr, label="Val MSE", color="#D32F2F")
axes[0].set_xlabel("Epoch"); axes[0].set_ylabel("MSE loss")
axes[0].set_title("SentinelEncoder - Training Curve")
axes[0].legend(); axes[0].grid(True, alpha=0.3)

axes[1].scatter(eval_df["true_bu"], eval_df["pred_bu"],
    c=["#D32F2F" if s=="val" else "#1565C0" for s in
    ↪eval_df["split"]],
    s=80, edgecolors="k", linewidths=0.5, zorder=3)
lo = min(eval_df["true_bu"].min(), eval_df["pred_bu"].min()) - 2
hi = max(eval_df["true_bu"].max(), eval_df["pred_bu"].max()) + 2
axes[1].plot([lo, hi], [lo, hi], "k--", lw=1.5, label="Perfect")
axes[1].set_xlabel("True yield (BU/ACRE)"); axes[1].set_ylabel("Predicted (BU/
    ↪ACRE)")

```

```

axes[1].set_title("Predicted vs True(blue=train, red=val 2022)")
axes[1].legend(fontsize=8); axes[1].grid(True, alpha=0.3)
plt.tight_layout()
buf = io.BytesIO(); fig.savefig(buf, format="png", dpi=150, bbox_inches="tight")
buf.seek(0); display(Image(data=buf.read())); plt.close(fig)

# Save artefacts
enc_path = LABELS_DIR / "sentinel_encoder.pt"
ckpt_path2 = LABELS_DIR / "sentinel_phase2_ckpt.pt"

torch.save(sent_model.backbone.state_dict(), enc_path)
torch.save({
    "model_state_dict": sent_model.state_dict(),
    "model_config" : {"class": "SentinelEncoder", "in_channels": 12,
                      "embed_dim": 64, "dropout": 0.4, "n_params": 1},
    "n_sent_params",
    "training_config" : {"n_epochs": SENT_EPOCHS, "lr": SENT_LR, "weight_decay":
    SENT_WD,
                      "criterion": "MSELoss", "split": "LOSO 2022",
                      "label_source": "state-aggregate USDA (all counties
    per state-year)"},
    "evaluation" : {"train_rmse_bu": round(rmse_train_sent, 4),
                    "val_rmse_bu_loso2022": round(rmse_val_sent, 4),
                    "best_val_mse": round(sent_best_val, 6),
                    "n_train": len(ds_s_train), "n_val": len(ds_s_val)},
    "embed_dim" : 64, "jensen_sigma2": JENSEN_SIGMA2,
}, ckpt_path2)

print()

print(f" Encoder backbone → {enc_path}")
print(f" Full checkpoint → {ckpt_path2}")
print()
print(f" Train set : {len(ds_s_train)} state-years (2021 × 15 states, AZ2022
    excluded)")
print(f" Val set : {len(ds_s_val)} state-years (2022 hold-out, LOSO)")
print(f" Val RMSE : {rmse_val_sent:.2f} BU/ACRE")
print(f" Embed dim : 64 - ready for GMU fusion")

```

TRAINING SentinelEncoder (Phase 2)

Epoch	20	train_mse=6.98810	val_mse=8.85530	lr=1.00e-03
Epoch	40	train_mse=3.30752	val_mse=4.21554	lr=1.00e-03
Epoch	60	train_mse=0.70964	val_mse=1.40329	lr=1.00e-03
Epoch	80	train_mse=0.20305	val_mse=0.20764	lr=5.00e-04
Epoch	100	train_mse=0.10888	val_mse=0.11769	lr=2.50e-04
Epoch	120	train_mse=0.26972	val_mse=0.07275	lr=6.25e-05

Best val MSE: 0.06556

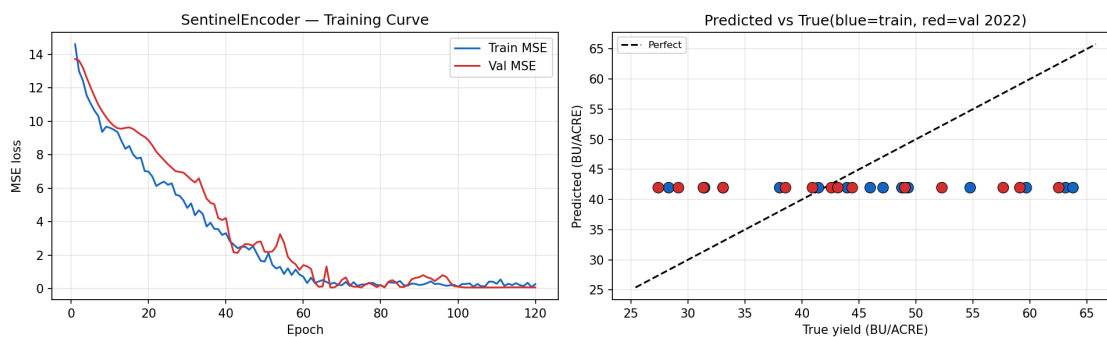
EVALUATION RESULTS

Train RMSE (BU/ACRE, Jensen-corrected) : 11.77

Val RMSE (BU/ACRE, LOSO 2022) : 11.04

Per-state errors:

	true_bu	pred_bu	bias
state			
AL	44.29	41.970001	-2.32
AR	49.17	41.970001	-7.21
CA	32.21	41.970001	9.75
CO	32.27	41.970001	9.70
DE	44.85	41.970001	-2.88
FL	27.85	41.970001	14.11
GA	43.55	41.970001	-1.59
IA	61.12	41.970001	-19.15
ID	38.30	41.970001	3.66
IL	63.15	41.970001	-21.19
IN	58.66	41.970001	-16.70
KS	35.30	41.970001	6.66
KY	53.51	41.970001	-11.55
LA	45.76	41.970001	-3.79



Encoder backbone → e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\sentinel_encoder.pt

Full checkpoint → e:\learnings\University Of Hull AI\assignment-september\MSc project\multimodal_agriculture_prediction-notebook_only\data\CropNet\labels\sentinel_phase2_ckpt.pt

Train set : 14 state-years (2021 × 15 states, AZ2022 excluded)

Val set : 14 state-years (2022 hold-out, LOSO)

Val RMSE : 11.04 BU/ACRE

Embed dim : 64 - ready for GMU fusion